SAFESCAPE V2: AN AI-DRIVEN RESPONSIVE WEB PORTAL FOR FIRE SAFETY INFORMATION DISSEMINATION

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ABSTRACT

This study developed and evaluated SafeScape V2, an AI-driven responsive web portal for disseminating fire-safety information in Sta. Cruz, Laguna. It used a Developmental Research Design and an iterative, incremental development approach. The system was implemented using Laravel 11, React, PostgreSQL, a Gemini API-powered retrieval-augmented generation (RAG) chatbot, Web Speech API-based speech-to-text and text-to-speech functions, two Random Forest adaptive-learning models, and an administrative dashboard. Internal testing showed that all 85 predefined binary test cases passed under controlled conditions. Across 15 controlled fire-safety questions, the chatbot achieved 100.00% response accuracy, 100.00% knowledge retrieval effectiveness, and a 0.00% hallucination/error rate within the test set. The Module Quiz and Final Examination difficulty models achieved accuracies of 91.80% and 98.40%, respectively, using synthetic learner-performance data. The overall evaluation mean scores were 4.23 from 10 IT experts based on ISO/IEC 25010:2023, 4.88 from 15 Bureau of Fire Protection (BFP) personnel, and 4.37 from 50 community end-users. These results indicate favorable technical, operational, and community assessments. Overall, SafeScape V2 demonstrated functionality and technical viability as a localized platform supporting fire-safety information dissemination, education, adaptive assessment, AI-assisted interaction, and administrative monitoring for its intended users.

Keywords: Fire Safety Education, Laravel Architecture, Retrieval-Augmented Generation (RAG), Gemini API, Adaptive Learning, Web Accessibility, ISO/IEC 25010:2023.

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DEFINITION OF TERMS

Operational Terms

The following terms are defined according to their specific meaning and application within SafeScape V2.

Adaptive Learning — In this study, it refers to the computational process that uses learner profile and assessment data to determine appropriate difficulty levels for module quizzes and the final examination. The adaptive learning engine uses Random Forest classification to categorize assessment difficulty as Easy, Medium, or Hard.

Berong SafeScape — It refers to the existing fire safety web portal in Sta. Cruz, Laguna, which is the primary subject of the backend optimization and Laravel migration in this study to resolve performance bottlenecks.

Intelligent Tutoring System (ITS) — In this study, it refers to the educational support framework that uses learner profile and assessment performance to adapt the difficulty of module quizzes and the final examination and to provide learner-performance information for administrative analytics.

Retrieval-Augmented Generation (RAG) — In this study, it refers to the artificial intelligence architecture in which the chatbot retrieves relevant information from a localized 500-record synthetic fire-safety knowledge base using weighted keyword-based

	matching. The retrieved information is supplied as contextual input to the Google Gemini API to support the generation of fire-safety responses.

Technical Terms

The following terms present established computing concepts and technologies as they are applied in the development and evaluation of SafeScape V2.

Artificial Intelligence (AI)	A branch of computer science dealing with the simulation of intelligent human behavior in computers. In this study, it drives the chatbot and adaptive learning features to shift fire safety to a proactive approach.
Gemini API	A cloud-based generative AI service utilized in this study to power the conversational chatbot, configured to perform grounded text generation based strictly on the retrieved fire safety context.
Government Web Hosting Service (GWHS)	In this study, it refers to the structural, security, and hosting-related requirements of the Department of Information and Communications Technology (DICT) that were considered as alignment criteria during the development of SafeScape V2. The study does not claim formal GWHS certification or compliance testing.
ISO/IEC 25010:2023	In this study, it refers to the software product quality model used as the basis for evaluating selected technical quality characteristics of the Enhanced Berong SafeScape Web

	Portal. The selected characteristics were Functional Suitability, Performance Efficiency, Reliability, Security, Maintainability, and Flexibility, which were assessed by ten qualified Information Technology experts. User usability and overall acceptability were evaluated separately through the community end-user evaluation instrument.
Iterative and Incremental Development	An Agile software development methodology utilized in the study where the system is developed through a series of repeated cycles from initial planning to deployment, allowing for flexibility and stakeholder feedback
Laravel	A robust PHP web application framework featuring an expressive syntax and native support for the MVC architectural pattern, utilized as the core backend framework for migrating and modernizing the legacy Berong SafeScape system.
Model-View-Controller (MVC)	An object-oriented architectural pattern implemented in the system that separates data representation and management (Model), user presentation (View), and application request-handling logic (Controller) to support organized, maintainable, and efficient system development.
PostgreSQL	An advanced, enterprise-grade open-source relational database system deployed in the study to securely handle data persistence, user credentials, real-time analytics, and

	adaptive learning tracking.
Weighted Keyword-Based Retrieval	A method used in the study to identify and rank relevant records from the localized BFP knowledge base by assigning different scores to keyword matches in questions, answers, categories, and subcategories.

CHAPTER I

INTRODUCTION AND ITS BACKGROUND

The digital transformation of public safety is now a global priority in order to ensure continuity of public emergency services and to manage disaster risks. In the country, the Department of Information and Communications Technology (DICT), through the E-Government Masterplan (EGMP) 2022, is championing the transformation towards a citizen-centered, One Digitized Government. Moreover, all government agencies are directed by Administrative Order No. 39 to host their government websites in Government Web Hosting Service (GWHS) that shall meet structural and security requirements. Fire safety must be modernized as preventable fires in buildings cause human lives and millions in property damage every year. These tragedies occur mostly in high-density residential areas and are largely preventable through a modern, technology-based fire safety approach that is far from mere reactiveness.

Contrary to the national government's Information and Communications Technology (ICT) initiatives that utilize ICT to enhance public safety and respond to disasters, the Bureau of Fire Protection (BFP) Sta. Cruz, Laguna relies on traditional face-to-face information dissemination on the implementation of the Fire Code of the Philippines (Republic Act No. 9514, 2008). The current ICT initiative of the local BFP is the Berong SafeScape Web Portal which has several performance issues and is not compatible with different types of devices.

Studies further revealed that non-inclusive and lagging e-governance platforms are a cause of digital exclusion which becomes more pronounced during periods of increased inquiries. Users particularly from vulnerable groups are further deprived of

access to important information and services through these non-inclusive online platforms (Jou et al., 2024; Jakaria, 2025).

The existing system required architectural improvements to address the identified issues. This project migrated the backend of the Berong SafeScape platform to the PHP Laravel framework and evaluated the enhanced architecture in terms of performance, scalability, security, and maintainability. The development also incorporated accessibility features and voice-based interaction. Through structured routing, database optimization, caching mechanisms, and other performance enhancements implemented in Laravel, the revamped Berong SafeScape system was designed to process requests efficiently, support different device types, and handle multiple simultaneous user connections while considering the relevant requirements of DICT's Government Web Hosting Service (GWHS) guidelines throughout the development process (Kaptosv, 2025).

The enhanced platform also integrates a Retrieval-Augmented Generation (RAG) chatbot powered by the Gemini API and grounded in a localized fire-safety knowledge base derived from BFP resources. The website also implemented Speech-to-Text (STT) and Text-to-Speech (TTS) technologies to provide alternative methods of interaction and improve accessibility (Shashidhara et al., 2026). During a consultation, Chief FINSP Cesar Morfe, Jr. suggested that the user interface should be designed using high-contrast, simplified, and user-centered elements appropriate for younger users (Sumari et al., 2025). The system featured an adaptive engine that used learner profiles and assessment performance to determine if the difficulty level for children was Easy, Medium, or Hard difficulty levels. Meanwhile, the administrative dashboard allows administrators to monitor user engagement and assessment data.

1.1 Research Problem

The initial assessment of the existing Berong SafeScape Website revealed that the tool had deficiencies which limited its capacity to serve as an effective platform for community-based fire-safety education and information dissemination. These included suboptimal performance due to its legacy architecture, limited channels for automated fire-safety information dissemination, and insufficient administrative oversight of user activities and engagement.

Accordingly, this study aimed to design, develop, and evaluate improvements to the existing Berong SafeScape Web Portal. To guide the research, the following research questions were formulated:

1. How was the existing Berong SafeScape Web Portal restructured and migrated to a PHP Laravel-based architecture to improve performance, scalability, security, and maintainability without compromising necessary adjustments to meet selected characteristics in Department of Information and Communications Technology (DICT) Government Web Hosting Service (GWHS) requirements?
2. How was a Retrieval- Augmented Generation (RAG)-based chatbot, powered by Gemini API and trained using key BFP knowledge resources, developed, integrated, and evaluated based on Response Accuracy, Contextual Relevance, Response Completeness, Knowledge Retrieval Effectiveness, and Hallucination/ Error Rate?
3. How was a voice-enabled accessibility tool utilizing Web Speech API-based Speech-to-Text (STT) and Text-to-Speech (TTS) converters, age appropriate illustrations, and intuitive UI/UX be developed and evaluated based on

usability, accessibility, ease of interaction, and user engagement across user demographics?

4. How was an administrative dashboard and adaptive learning engine be implemented to monitor user activities and assessment performance, as well as provide feedback analytics and allocate difficulty levels for module quizzes and the final exam based on learner profile and assessment results?
5. How was the developed Berong SafeScape Web Portal evaluated in terms of selected quality characteristics under ISO/IEC 25010:2023 by Information Technology (IT) experts, fire-safety information and AI-assisted technologies by Bureau of Fire Protection (BFP) stakeholders, and user experience by community end-users?

1.2 Research Objectives

The objective of this study was to develop and evaluate an enhanced Berong SafeScape Web Portal for the Bureau of Fire Protection (BFP), Santa Cruz, Laguna. The enhanced version was developed using PHP Laravel architecture and incorporated adaptive learning capabilities and voice-enabled accessibility technologies.

To accomplish the study objective, the following specific objectives were designed:

1. To develop and implement an enhanced Berong SafeScape Web Portal by restructuring and migrating the existing portal to use PHP Laravel framework through gradual conversion and evaluation of the revamped system based on performance, scalability, security, and maintainability, as well as meeting selected requirements in DICT GWHS requirements.
2. To develop, integrate, and evaluate a Retrieval-Augmented Generation (RAG)-based chatbot powered by the Gemini API and grounded in BFP knowledge resources in terms of Response Accuracy, Contextual Relevance, Response

Completeness, Knowledge Retrieval Effectiveness, and Hallucination/Error Rate for automated fire safety information dissemination.

3. To design, implement, and evaluate a voice-enabled accessibility framework incorporating Web Speech API-based Speech-to-Text (STT) and Text-to-Speech (TTS), age-appropriate educational content, and user-centered UI/UX design in terms of usability, accessibility, ease of interaction, and user engagement among diverse user groups.
4. To develop an intelligent administrative dashboard with user engagement and assessment analytics, together with an adaptive learning engine using Random Forest classification to determine appropriate Easy, Medium, or Hard difficulty levels for module quizzes and the final examination based on learner profiles and assessment performance.
5. To evaluate the developed Berong SafeScape Web Portal in terms of selected characteristics of ISO/IEC 25010:2023, as assessed by Information Technology experts; validate fire safety information and AI-assisted technologies through Bureau of Fire Protection personnel; and to determine the user experience of the system as assessed by community end-users.

1.3 Research Framework

The research framework of SafeScape V2 is organized into two components. The theoretical framework establishes the scientific and technological foundations that support the study, and the conceptual framework demonstrates how these principles are applied to the design, development, and evaluation of the proposed web portal.

1.3.1 Theoretical Framework

The system's backend and artificial intelligence technical implementation depend on the RAG framework proposed by Lewis et al. (2020), underscoring the need for verified external knowledge access to prevent algorithmic errors. Relying solely on the internal knowledge of generative AI models may result in hallucinations, where the system generates inaccurate, fabricated, or unsupported information. In fire-safety applications, such responses could mislead users and potentially expose them to dangerous situations.

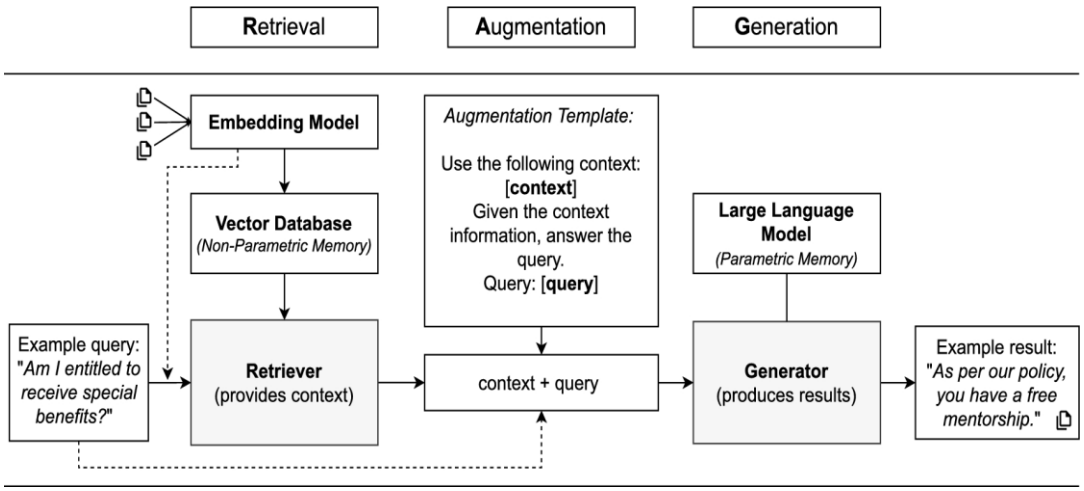


Figure 1. A plain vanilla RAG architecture Adapted from Klesel and Wittmann (2025), based on Lewis et al. (2020).

The RAG framework enforces a strict separation between data storage and generative output to mitigate it. This theoretical architecture is realized through a localized fire-safety knowledge base derived from BFP resources and related validated fire-safety references. When a user submits a query, the retrieval mechanism searches the indexed knowledge base for relevant contextual information before the Gemini API generates a conversational response. By grounding the Large Language Model (LLM) in retrieved fire-safety context, the system reduces the likelihood of unsupported or fabricated responses.

Also, the development of the platform is grounded in established principles from computer science, artificial intelligence, and web architecture. This study is anchored on the recent developments in generative AI and object-oriented software development to support an optimized, cross-device web portal for fire safety dissemination.

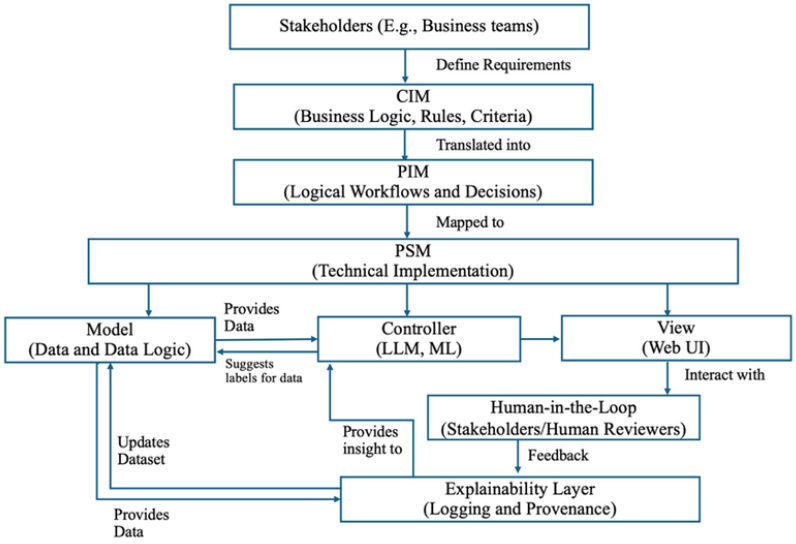


Figure 2. The conceptual architecture integrating MDA, MVC, LLM, human in-the-loop validation and explainability (Gatchalee, 2025)

The cross-device optimization, intelligent data routing, and backend structure of the web portal are anchored on the Model-Driven Architecture integrated with Model-View-Controller and Large Language Models (MDA-MVC-LLM) framework. As demonstrated by Gatchalee (2025), this modernized object-oriented development model not only separates the system's internal data representation from its user interface but explicitly layers this separation within a strict business-rule-driven structure and an artificial intelligence processing environment. By utilizing the PHP Laravel framework to natively enforce this extended MVC architecture, alongside a Retrieval-Augmented Generation (RAG) pipeline, the framework supports a modular, secure, and responsive architecture while allowing the RAG component to reduce the likelihood of unsupported AI responses.

The system's adaptive learning engine is theoretically anchored in the architecture of Intelligent Tutoring Systems (ITS) as described by Thomas and Sahai (2026). As illustrated in Figure 3, an ITS framework replaces static content delivery with a dynamic, continuous feedback loop between three primary interconnected models for adaptive learning.

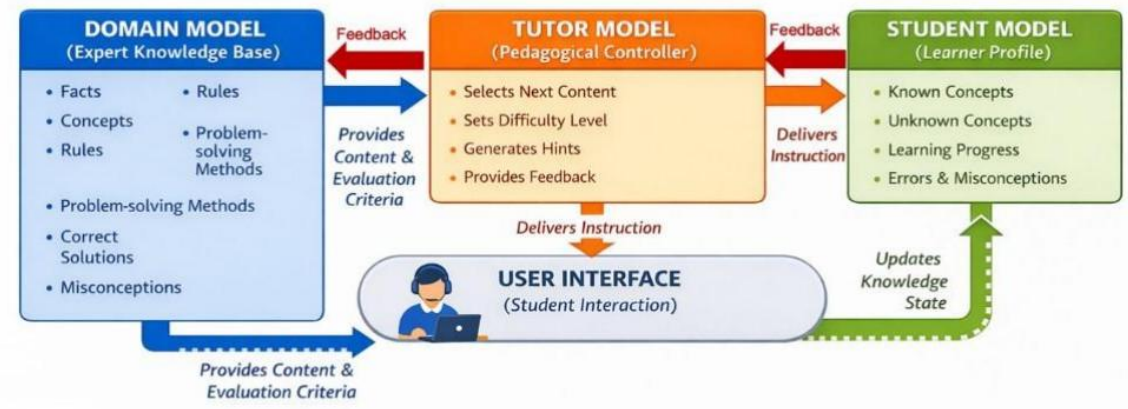


Figure 3. Operation of Intelligent Tutoring System (Thomas & Sahai, 2026)

The Domain model (Expert Knowledge Base) serves as the central repository of facts, concepts, rules, and problem-solving methods. In the context of the platform, the Domain Model is represented by the localized fire-safety knowledge resources used by SafeScape V2, including content derived from BFP resources and related validated fire-safety references. It provides the system with domain-relevant fire-safety information for its educational and intelligent components.

The Student Model (Learner Profile) serves as the diagnostic component, dynamically tracking the user's cognitive state. As the user interacts with the interface, this model updates their knowledge state by recording known and unknown concepts, tracking overall learning progress, and identifying specific errors and misconceptions. In this study, the model is responsible for generating personalized insights.

The Tutor Model (Pedagogical Controller) functions as the central decision-making engine. This model evaluates the feedback from the Student Model against the

rules of the Domain Model. Based on this continuous evaluation, the Tutor Model autonomously selects the next appropriate content, sets the difficulty level, generates hints, and delivers personalized instruction and feedback back to the User Interface. By integrating this architecture, the system supports a responsive and personalized fire safety learning experience that adapts to the needs and pacing of individual learners

The study adopted the learner-modeling and difficulty-adaptation principles of the Intelligent Tutoring System framework within the boundaries of the implemented system. The adaptive learning engine does not autonomously select succeeding modules, reorder educational content, generate instructional hints, or modify the learning pace. Instead, two Random Forest classification models process learner profile and assessment performance data to determine appropriate an Easy, Medium, or Hard difficulty levels for module quizzes and the final examination. Thus, personalization in the present study is operationalized through adaptive assessment difficulty rather than automated content recommendation.

1.3.2 Conceptual Framework

The conceptual framework of this study illustrates the overall architectural flow and development lifecycle of the Website using the Input-Process-Output (IPO) model. This paradigm demonstrates how the theoretical foundations of the study are practically applied to transform safety data into an optimized, accessible digital platform. The framework is mapped in Figure 4, breaking down the system’s requirements, technological processing, and final deliverables.

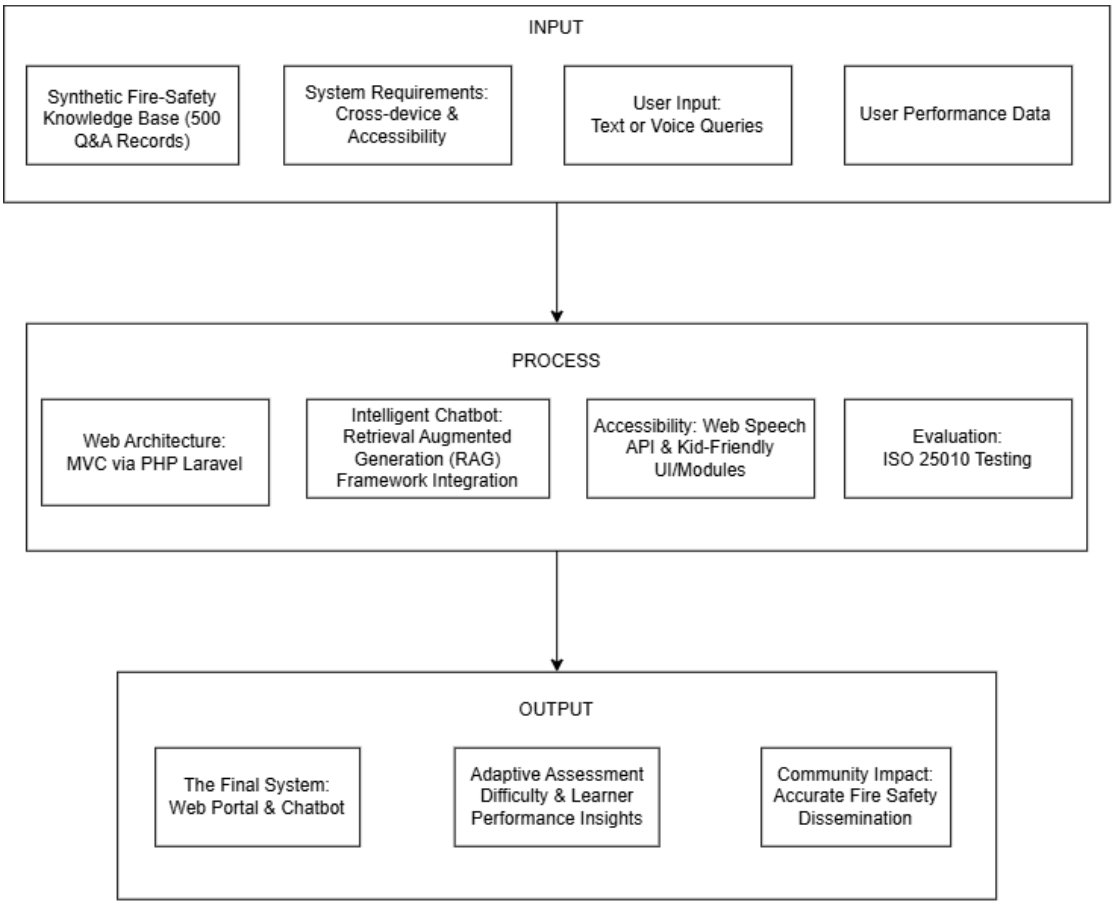


Figure 4. The Conceptual Framework for the Website

The input phase covers the data and requirements needed to develop the system. Its primary knowledge source is the legacy Bureau of Fire Protection (BFP) dataset, which contains 1,200 question-and-answer records. After filtering, cleaning, augmentation, and restructuring, 500 records were selected to form the knowledge base used by the Retrieval-Augmented Generation (RAG) chatbot.

The process phase covers the development, integration, and evaluation of the system. PHP Laravel is used to build the cross-device platform, following the Model-View-Controller (MVC) architecture to organize application logic, manage requests, and handle data securely.

The chatbot uses a Retrieval-Augmented Generation (RAG) approach powered by the Gemini API. It retrieves relevant information from the localized BFP knowledge

base to generate context-based fire safety responses (Klesel & Wittmann, 2025).

The platform also includes an Intelligent Tutoring System (ITS) that uses two Random Forest classification models. The first model assigns an Easy, Medium, or Hard difficulty level to module quizzes based on the learner's Age, Grade Level, and Pre-Assessment Score. The second model determines the difficulty level of the final examination using the same learner information, along with accumulated quiz scores from Modules 1 to 4.

The frontend handles user interactions through an optimized UI/UX design. It also includes specialized learning modules for children and integrates the Web Speech API for Speech-to-Text (STT) and Text-to-Speech (TTS) functions. These features allow users to interact with the system through voice, making it more accessible to individuals who may have difficulty using standard text inputs (Shashidhara et al., 2026).

Finally, the system undergoes software evaluation based on the ISO/IEC 25010:2023 standard. The evaluation covers functional suitability, performance efficiency, reliability, security, maintainability, and flexibility. User usability and overall acceptability are also assessed.

The output phase represents the completed and functional system. Its main objective is to provide accessible and reliable fire safety information to the residents of Sta. Cruz, Laguna. The system combines a Gemini API-powered RAG chatbot with an adaptive learning engine that adjusts assessment difficulty and provides insights into learner performance. Through these features, the platform supports fire safety awareness and disaster preparedness while aligning with the national objectives of the E-Government Masterplan 2022.

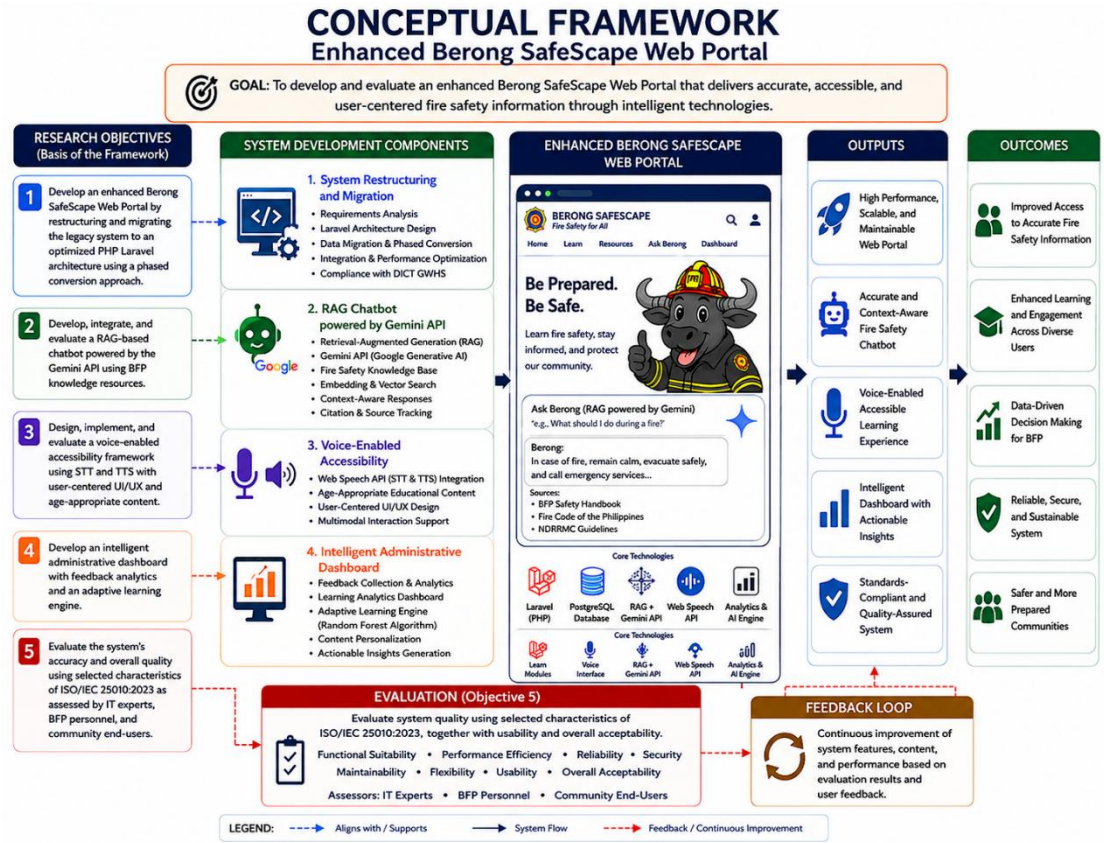


Figure 5. Conceptual Framework of the Enhanced Berong SafeScape Web Portal for Intelligent Fire Safety Information Dissemination and Community Engagement

Note. Developed by the researchers based on the study's research objectives. The conceptual framework was visualized using the image-generation capabilities of ChatGPT by OpenAI (2026).

The conceptual framework maps the architecture and evaluation pipeline of the Enhanced Berong SafeScape Web Portal for the Bureau of Fire Protection (BFP) Santa Cruz, Laguna. Specific research objectives dictate the integration of web technologies, artificial intelligence, accessibility features, and learning analytics.

Research objectives drive the core system. Objective one mandates the restructuring of the legacy portal. Developers migrate the system to an optimized PHP Laravel architecture using a phased conversion approach. The resulting foundation targets specific performance, scalability, security, and maintainability benchmarks. This transition enforces architectural alignment with relevant Department of

Information and Communications Technology (DICT) Government Web Hosting Service (GWHS) requirements.

The second objective was to develop and deploy a conversational chatbot powered by the Gemini API and a Retrieval-Augmented Generation (RAG) architecture. The community gains immediate access to safety guidance. Routine inquiries bypass local fire station personnel completely. The chatbot engine extracts answers exclusively from verified Bureau of Fire Protection records.

Objective three integrates a voice-enabled framework for accessibility. Users with limited digital literacy or reading constraints can interact via alternative auditory and vocal pathways. The application merges age-appropriate educational content and a user-centered interface with Web Speech API-based Speech-to-Text (STT) and Text-to-Speech (TTS) technologies.

The fourth objective requires an intelligent administrative dashboard and an adaptive learning engine. The dashboard centralizes user engagement metrics and assessment data. Administrators access this data stream for direct operational monitoring. So, the adaptive engine processes learner profiles and assessment performance via two Random Forest classification models. The first model classifies module quiz difficulty. The second model predicts the final examination difficulty. Both algorithms output exact difficulty classifications as Easy, Medium, or Hard.

These independent modules consolidate into the final Enhanced Berong SafeScape Web Portal. The production environment combines the optimized Laravel architecture, the Gemini API RAG-based chatbot, the voice-enabled accessibility framework, the adaptive learning engine, and the administrative analytics dashboard.

System evaluation satisfies the fifth research objective. Technical experts, domain specialists, and community end-users execute independent assessment

protocols. Information Technology experts measure specific software quality characteristics against the ISO/IEC 25010:2023 standard. Their evaluation isolates Functional Suitability, Performance Efficiency, Reliability, Security, Maintainability, and Flexibility.

Bureau of Fire Protection personnel execute domain and operational validation. They measure the factual accuracy, contextual relevance, completeness, and appropriateness of all fire safety content and AI-generated responses. Community end-users independently rate usability, accessibility, ease of interaction, and overall acceptability.

A feedback mechanism tells the evaluation results of the development cycle. It extract quantitative findings from system evaluations, and user interactions. Developers then apply these exact metrics to correct interface defects, update content repositories, and retrain intelligent components.

1.4 Scope and Limitations of the Study

This study focused on the design, development, and implementation of a website, an optimized PHP Laravel web portal integrated with a voice-enabled artificial intelligence for fire safety dissemination.

Scope:

- **Geographical Focus:** The system is intended for use in Sta. Cruz, Laguna, covering local BFP stations, community end-users, and schools.
- **Technological Components:** The study covers the restructuring and migration of the existing Berong SafeScape Web Portal to a PHP Laravel architecture. The enhanced architecture incorporates structured routing, database optimization, caching, PostgreSQL data management, and related mechanisms intended to support system performance and concurrent access. Relevant DICT

Government Web Hosting Service (GWHS) requirements are considered as architectural alignment criteria; however, formal GWHS certification or compliance assessment is outside the scope of the study.

- **Key Features:** The scope included the integration of a Retrieval-Augmented Generation (RAG) chatbot powered by the Gemini API and a localized fire-safety knowledge base that is based from BFP resources and related validated fire-safety references, implementation of the Web Speech API for native Speech-to-Text (STT) and Text-to-Speech (TTS) capabilities, and an administrative dashboard featuring user feedback analytics and per-school tracking.
- **Adaptive Learning and Student Performance:** Implements an adaptive learning engine through two Random Forest classification models. The Module Quiz Difficulty Model processes Age, Grade Level, and Pre-Assessment Score. This algorithm assigns an Easy, Medium, or Hard difficulty level for module quizzes. The Final Examination Difficulty Model classifies the final assessment difficulty as Easy, Medium, or Hard using Age, Grade Level, Pre-Assessment Score, and the learner's Module 1 to Module 4 quiz scores. The engine functions strictly as an assessment-difficulty classifier and does not automatically reorder modules or recommend succeeding educational topics.
- **Target Users and Evaluation Participants:** The platform features a kid-friendly and user-centered UI/UX. This design supported younger users and individuals with varying levels of digital literacy. The external evaluation matrix involved ten (10) Information Technology experts, fifteen (15) Bureau of Fire Protection personnel, and fifty (50) community end-users. The IT experts measured selected software quality characteristics defined by ISO/IEC

25010:2023. These characteristics specifically included Functional Suitability, Performance Efficiency, Reliability, Security, Maintainability, and Flexibility. BFP personnel executed domain and operational validation on the fire-safety information, chatbot responses, educational content, and administrative functions. Community end-users assessed usability, accessibility, ease of interaction, and overall acceptability. The researchers/developers solely conducted all internal functional, integration, artificial intelligence, accessibility, adaptive-learning, dashboard, performance, and security testing.

Limitations:

- **Data Dependency and Chatbot Scope:** The Retrieval-Augmented Generation (RAG) chatbot derives its accuracy directly from the quality and completeness of the localized fire-safety knowledge base. This repository relies on BFP resources and related validated fire-safety references. The model restricts responses entirely to fire-safety topics. It may fail to handle inquiries outside this specific domain accurately.
- **Generalizability:** The developers customized the educational and dashboard tracking features specifically for the campaigns of Sta. Cruz, Laguna. Municipalities attempting to deploy the system also require targeted codebase modifications.
- **Adaptive Learning Boundaries:** The adaptive learning engine classifies assessment-difficulty exclusively. The system lacked the ability to generate educational content, reorder learning modules, recommend topics, or modify the learner's pacing. The two Random Forest models restrict their adaptive outputs to Easy, Medium, and Hard difficulty classifications based on predefined learner profile and assessment variables.

- **Synthetic Training Data:** Developers built and evaluated the two Random Forest classification models using two synthetically generated student-performance datasets. The system lacked sufficient real learner records prior to deployment. Each generated dataset contained 5,000 simulated learner records. The researchers allocated 80% or 4,000 records for model training. The remaining 20% or 1,000 records functioned as unseen testing data. Future refinement of the adaptive learning engine depends on real assessment and interaction records collected through continued system use.
- **Legacy-System Baseline Limitation:** The study lacks a controlled quantitative benchmark of the previous Berong SafeScape implementation. The technical performance results of SafeScape V2 describe the measured behavior of the enhanced system under the defined testing conditions. Evaluators must not interpret these measurements as a quantified percentage improvement over the previous portal.
- **Hardware and Network Constraints:** The Laravel migration optimizes database requests for weak mobile networks. The platform still fundamentally requires stable internet connectivity to operate and utilize cloud-based APIs like the Web Speech API.

1.5 Significance of the Study

This research focuses on resolving the performance bottlenecks and accessibility barriers of the current Berong SafeScape portal by introducing an optimized, cross-device platform. The developed system provides the following contributions to its identified beneficiaries:

- **Department of Information and Communications Technology (DICT) & National Government:** The study contributes to the objectives of government

digital modernization by demonstrating the redevelopment of a localized public-safety portal using a structured and maintainable web architecture while considering relevant Government Web Hosting Service (GWHS) requirements as alignment criteria.

- **Bureau of Fire Protection (BFP) Sta. Cruz / Local Authorities: T**he web application features an administrative dashboard containing user feedback analytics and per-school data tracking. BFP administrators extract these specific system metrics to evaluate community engagement and direct upcoming fire-safety educational operations.
- **Schools and Educational Institutions in Sta. Cruz:** Local safety programs can incorporate the platform as a primary digital resource. The adaptive assessment mechanism evaluates baseline learner profiles and ongoing performance. It subsequently restricts the final-examination and module quiz difficulty to Easy, Medium, or Hard parameter assignments. Educators review the resulting analytics alongside BFP personnel to locate precise institutional knowledge deficits.
- **Community End-users (Vulnerable Populations & Children):** The interface combines user-centered UI/UX design with Web Speech API-based Speech-to-Text and Text-to-Speech technologies. These auditory interaction channels bypass traditional text-dependency for critical hazard information. The platform scales the corresponding assessment difficulty against real-time learner profiles.
- **Future Researchers and IT Professionals:** The architectural implementation provides a blueprint for mitigating digital exclusion within municipal e-governance networks. Future research can replicate this specific framework to

analyze database query optimization and Retrieval-Augmented Generation (RAG) chatbots across alternative public sector domains.

CHAPTER II

REVIEW OF RELATED LITERATURE

This chapter reviews related studies that establish the theoretical, technological, and empirical foundations of the enhanced Berong SafeScape Web Portal. The literature is organized into five themes aligned with the research objectives. These studies present relevant approaches, reported findings, and documented implementation limitations. Their findings provide a basis for the design decisions and evaluation procedures adopted in the present study.

2.1 Digital Governance and Public Safety Dissemination

Sabordo (2026) examined the emergency-response challenges experienced by Bureau of Fire Protection (BFP) personnel, particularly those associated with limitations in institutional communication and information and communications technology (ICT). Using a quantitative descriptive-correlational research design, the study identified factors contributing to operational inefficiency, including outdated information and coordination practices. These findings are relevant to Berong SafeScape, particularly in relation to the modernization of its existing system. System improvements should address not only the user interface but also the digital infrastructure responsible for organizing, exchanging, and accessing information. Such improvements can help the system support better coordination and address existing communication limitations.

Pareja (2025) discussed the ongoing digital transformation of Philippine government services and emphasized the potential of artificial intelligence to improve public-service delivery. The study presented AI techniques that can support digital-government strategies and improve user experience. This study is relevant to SafeScape

because artificial intelligence can serve as a supporting component of the platform, with its functions integrated into the delivery of public services.

Saflor and Kudhal (2026) examined digital warning systems in the Philippines and discussed their reliance on technological platforms. Their SEM-ANN analysis identified system reliability as one of the factors that can influence actual preparedness behavior. This finding highlights the importance of users' perceptions of system reliability. It is relevant to SafeScape because the platform must provide clear and dependable information to function effectively as a community fire-safety resource.

Lagrada and Santiago (2025) identified an imbalance between reactive response and preventive activities in disaster management in Metro Manila. Although post-disaster operations generally received favorable assessments, prevention and mitigation were consistently identified as weaker areas. Their findings indicate that public agencies need stronger interventions before emergencies occur, including preventive programs and technologies that can maintain public awareness over time. This evidence supports the role of SafeScape as more than an emergency-response aid.

Longa and Perena (2024) examined the emergency-response capabilities of BFP personnel and the operational stress associated with firefighting demands, outdated communication equipment, and the continued use of manual reporting. Although strong firefighting skills are essential, repetitive administrative tasks and poor information flow can create additional operational difficulties. These challenges may contribute to fatigue and reduce the time available for actual operational work. The findings support the value of self-service digital information channels in the public-safety environment.

Daminar and Cabonce (2026) examined the implementation of the Disaster Risk Reduction and Management Act (RA 10121) in educational institutions and identified continuing weaknesses in the sustainability of risk-reduction information networks.

Their study showed that strong administrative leadership does not automatically guarantee continuous safety programming when schools lack a permanent mechanism for maintaining information exchange with public-safety institutions. This observation is relevant to the educational role of SafeScape. A persistent digital connection between schools and safety agencies can provide a more stable channel for distributing fire-safety materials, supporting educational activities, and sustaining community engagement beyond one-time seminars or physical campaigns.

A practical Philippine example of web-supported emergency coordination is provided by Parallag et al. (2023), who developed and evaluated RescueLink for emergency agencies. Their implementation combined Laravel with a relational PostgreSQL database and was associated with improved coordination and strong usability results, including a 73% strong-agreement rating for usefulness. The study demonstrated how replacing congested or fragmented communication practices with a responsive web application can make reporting and inter-agency interaction more manageable.

De Castro and De Castro (2022) investigated e-government initiatives among local government units and highlighted the dependence of many local institutions on the Department of Information and Communications Technology (DICT) for long-term digital infrastructure and technical support. Their findings suggest that local offices with limited internal IT capacity may adopt fragmented or temporary systems when broader standards and support mechanisms are absent. This situation is directly relevant to a locally deployed public-safety portal such as SafeScape. Sustainable implementation requires attention not only to current functionality but also to maintainability, technical standards, and the possibility of future institutional support.

Alignment with national guidance therefore helps reduce the risk that the platform becomes an isolated short-term software solution.

Araojo et al. (2024) prepared an Information Systems Strategic Plan for a municipal BFP unit and reported a low level of IT maturity characterized by siloed information and the absence of modern analytics. Their assessment showed that fragmented data can prevent fire-service managers from producing timely, evidence-based decisions or developing a clear picture of community safety conditions. This local public-safety context supports the inclusion of administrative analytics in SafeScape. By consolidating engagement and assessment information into a centralized dashboard, the portal can provide BFP personnel with a clearer view of user activity and learning patterns.

Bokingkito et al. (2025) audited Philippine e-government websites and identified a gap between nominal accessibility compliance and the availability of meaningful connected digital services. Although basic accessibility compliance reached 82.5%, the study reported a substantially lower 53.43% level for functional web presence. The authors observed that standards such as WCAG and government website design guidance can be treated as checklist requirements without ensuring that citizens, particularly vulnerable users, can complete useful interactions. This distinction is important for SafeScape because accessibility should be demonstrated through actual tasks and user experience, not inferred solely from the presence of technical accessibility features. The portal's evaluation should therefore include practical interaction with its learning, chatbot, and voice-enabled functions.

Ferrer (2025) reviewed Philippine Disaster Risk Reduction and Management frameworks and identified a continuing shortage of grassroots-level technological applications despite widespread awareness of disaster preparedness concepts. The

review suggested that knowledge of hazards does not automatically develop into practical resilience when communities lack interactive tools that allow them to engage with safety information in usable ways. This gap is relevant to SafeScape because the portal is intended to move beyond passive dissemination. Interactive learning functions, guided information access, and community-oriented digital features can help convert general awareness into repeated engagement with fire-safety practices.

Barrios and Moreno (2024) examined e-government initiatives in Zamboanga City and found that faster administrative services can coexist with a persistent digital divide. Their mixed-methods analysis showed that citizens with limited technical skills may still be excluded when digital platforms are designed primarily for technically confident users. This finding demonstrates that the success of government digitalization cannot be judged only by the number of services placed online or by processing speed. For SafeScape, meaningful accessibility requires that community members with different levels of digital literacy can understand the interface and complete important tasks.

Rivas (2026) developed a Barangay Information Management System with automated messaging and showed that centralized digital communication can distribute public-safety information more quickly than manual dispatch practices. The study reported that replacing repetitive administrative communication with automated mechanisms improved the speed at which urgent advisories could reach intended recipients. This finding supports the use of digital channels for routine and time-sensitive safety communication. In the context of SafeScape, centralizing fire-safety resources and allowing users to retrieve information independently can reduce dependence on case-by-case dissemination.

Lagura et al. (2026) evaluated accessibility across Philippine institutional websites and reported that none of the reviewed sites achieved full compliance with WCAG 2.1. Their audit identified structural and semantic barriers that can restrict access to digital services for people with disabilities, demonstrating that institutional websites may remain exclusionary even when online services are available. The findings are relevant to SafeScape because a public-safety platform must account for users who interact with information in different ways. Standards-based interface design should therefore be combined with practical accessibility functions, including simplified layouts and voice-enabled interaction.

Dandan et al. (2026) applied the ISO/IEC 25010:2023 quality model to evaluate a localized barangay management system and demonstrated the usefulness of standardized software-quality criteria in a Philippine municipal setting. Their descriptive-developmental study used defined quality characteristics to assess areas such as functional suitability, usability, and performance efficiency rather than relying on general impressions of system quality. This evaluation approach is relevant to SafeScape because the portal also requires a structured method for professional assessment. Using selected ISO/IEC 25010:2023 characteristics enables the study to examine specific dimensions of technical quality through measurable criteria, while other stakeholder groups can separately evaluate domain relevance, accessibility, and overall user acceptability.

Carsola and Bagtaso (2026) analyzed BFP response performance in Pangasinan and associated operational inefficiencies with analog communication practices and limited digital tracking. Their findings indicate that personnel shortages can be intensified when information must be handled manually and when administrative records are dispersed across separate processes. The researchers consequently

recommended more centralized digital tracking to reduce operational friction. This evidence is relevant to SafeScape because the portal's administrative and monitoring functions can help consolidate information that would otherwise remain fragmented.

Roque et al. (2024) demonstrated the use of a bilingual Natural Language Processing chatbot within a Philippine local information system. Their system showed that conversational agents can answer routine citizen inquiries and reduce repetitive communication demands on government personnel. The study provides a local example of how AI-assisted interaction can expand access to public information without requiring a staff member to respond to every basic question. For SafeScape, this supports the inclusion of a conversational interface as a supplementary information channel. The present study extends that concept by grounding chatbot responses through retrieval from domain-specific fire-safety resources and by subjecting the resulting answers to controlled technical testing and BFP validation.

Taiwiland et al. (2026) developed a web-based management information system using Laravel and reported improvements in administrative processing after moving away from document-intensive manual workflows. The study linked the MVC-oriented framework and relational data structure with more organized operations, reducing a reported eight-hour processing task to approximately 25 minutes. Although the application context differs from fire safety, the result demonstrates how a structured framework can support data-intensive public-service work.

Amil (2024) examined collaborative governance in disaster risk reduction and emphasized that community resilience cannot be built through top-down instructions alone. The analysis showed that preparedness improves when citizens have opportunities to participate, learn, and interact with public authorities rather than receiving safety messages only as passive recipients. This participatory perspective is

important to SafeScape because its educational component is intended to encourage repeated interaction with fire-safety information. The inclusion of assessments, learning resources, and other interactive functions supports a model in which community members actively engage with preparedness content.

Esbieto et al. (2026) implemented the iPlan Gov decision-support system and demonstrated the value of real-time analytics for local-government planning. Their evaluation indicated that automated reporting and visual presentation of data can reduce workflow inconsistencies and help administrators base decisions on current information rather than fragmented manual records. This finding supports the administrative-dashboard component of SafeScape. By summarizing user engagement, assessment results, and related portal activity, the dashboard can convert routine system records into information that is easier for BFP administrators to interpret. The study therefore provides a practical basis for treating analytics not merely as a display feature but as a mechanism for supporting evidence-informed planning and follow-up activities.

2.2 Web Architecture and Performance Optimization

Saputri et al. (2024) examined Laravel applications deployed on cloud infrastructure to determine how caching and cloud scaling affect response speed during heavy use. Their load tests, which reached 1,000 concurrent users, showed that Laravel caching, Eloquent ORM optimization, and cloud load balancing reduced average latency from 4.5 seconds to 2.2 seconds while supporting 99.9% availability through auto-scaling. These findings show that performance depends on both application-level optimization and deployment strategy. For SafeScape, the study supports the use of caching and workload-aware configuration, but it also indicates that these mechanisms should be verified through actual response-time and concurrent-user testing rather than assumed to be effective.

Ahmed et al. (2024) compared PHP framework performance under simulated stress and specifically examined Laravel optimization commands. Their benchmark found that route caching, view caching, and Eloquent eager loading reduced response latency from 119 milliseconds to 94 milliseconds in the tested single-iteration scenario. Although the numerical improvement was measured in a controlled environment, the result demonstrates that framework configuration can materially influence execution time without requiring a complete change of technology stack. This is relevant to SafeScape because performance tuning can be performed within Laravel itself. The study therefore supports evaluating route, view, and query-related optimizations as part of backend performance improvement.

Ariyanto et al. (2024) compared native PHP and Laravel RESTful API implementations to examine the balance among execution speed, code organization, and routing structure. Native PHP achieved the faster baseline in their test, at 18.3 ms compared with 344.52 ms for Laravel, but Laravel provided stronger organization through MVC and Eloquent ORM while reducing repetitive manual SQL code. The results illustrate an important architectural trade-off: the fastest baseline framework is not always the most maintainable or scalable choice for a long-lived application. For SafeScape, Laravel is justified by structure and maintainability, while its additional processing overhead must still be measured and optimized in the actual system environment.

Putra et al. (2025) evaluated Laravel, CodeIgniter, and Symfony using a 10,000-record PostgreSQL database under load and stress conditions. Their tests showed that lightweight frameworks performed well in simpler workloads, while Laravel improved more noticeably under heavier demand after eager loading and database indexing were applied. The reported optimization produced a 35.1% reduction in latency. This finding

is relevant to a public-facing system such as SafeScape because user demand may not remain constant. A framework should therefore be assessed not only during ordinary single-user operation but also when traffic increases, making concurrency testing and database optimization important parts of the portal's performance evaluation.

Kundan and Makwana (2025) analyzed common Laravel and PostgreSQL bottlenecks by measuring eager loading, composite indexing, and cursor pagination under large concurrent workloads. Their experiments simulated 10,000 API requests at 1,000 requests per second and showed that resolving the N+1 query problem with eager loading reduced query overhead and improved response time from 1,200 ms to 300 ms, while indexing helped avoid full table scans. The study demonstrates that application performance can be degraded by query design even when the underlying framework is capable. For SafeScape, these results support profiling database access, applying indexing where appropriate, and controlling repeated ORM queries instead of treating backend optimization as a purely server-level concern.

Urnikienė et al. (2026) compared PostgreSQL 17.5 and MongoDB 7.0.14 for filtering and analytical queries in a constrained single-host environment. Their results showed that PostgreSQL's normalized relational design performed substantially better for several complex, non-indexed analytical operations, with execution improvements ranging from 1.6 to 15.1 times and shorter processing times of roughly 65% to 80%. The comparison is relevant because SafeScape stores highly structured user, learning, assessment, and administrative records. However, the study does not establish that PostgreSQL is automatically be faster for every SafeScape workload. It instead supports selecting PostgreSQL for relational analysis while still benchmarking the portal's own query patterns.

Salunke and Ouda (2024) tested PostgreSQL 14.6 and MySQL 8.0.20 under simultaneous read and write operations using a continuous-authentication workload. During parallel INSERT and SELECT activity, MySQL latency increased to approximately 13 ms, whereas PostgreSQL maintained read latency of about 0.7 to 0.9 ms through its Multi-Version Concurrency Control architecture. The result illustrates the importance of database behavior when multiple users are reading and updating data at the same time. This is relevant to SafeScape because the portal must serve content and analytics while also recording user activity, assessments, and other transactions. Stable concurrency therefore forms part of the rationale for PostgreSQL and for direct system-level performance testing.

Wodyk and Skublewska-Paszkowska (2020) compared SQL Server, MySQL, and PostgreSQL through a Laravel web application and found that database engines respond differently depending on the operation being executed. Their results caution against selecting a database solely from general reputation or isolated benchmarks because application behavior changes with query type and workload. For SafeScape, the study provides support for evaluating PostgreSQL within a Laravel environment while emphasizing that the database should be judged using the portal's actual usage patterns. This includes content retrieval, assessment transactions, dashboard queries, and concurrent activity rather than relying only on performance claims obtained from unrelated applications.

Permana and Triandini (2021) evaluated Laravel's Eloquent ORM and Query Builder in a crowdfunding information system to compare development convenience with execution efficiency. Their findings showed that the two data-access approaches differ in performance, while Eloquent offers a more structured object-oriented model for application development. The comparison suggests that a single database-access

method does not need to be used for every operation. Within SafeScape, Eloquent can support maintainable model-based development, while more direct query approaches may be appropriate when profiling identifies a specific performance-sensitive path. The study therefore supports selective optimization rather than replacing the ORM across the entire application.

Revaliana et al. (2025) focused on the memory and execution overhead created when Laravel Eloquent instantiates large numbers of model objects. Using 10,000 synthetic records, the study reported an average retrieval time of 0.0251 seconds for Raw SQL compared with 0.0949 seconds for Eloquent ORM, with the difference largely attributed to object hydration and additional memory allocation. The result does not imply that Eloquent should be avoided altogether, but it demonstrates that convenience can introduce measurable cost for high-volume retrieval. For SafeScape, large or frequently accessed datasets should therefore be profiled individually so that performance-critical queries can be optimized without sacrificing maintainable ORM-based development elsewhere.

Szewczyk and Skublewska-Paszkowska (2025) compared REST API implementations across several development environments, including Laravel, to examine latency and resource consumption. Their results indicated that feature-rich frameworks can require more memory than minimalist alternatives, yet they offer compact code, established routing, and stable behavior during demanding workloads. The study illustrates that framework assessment should consider more than raw resource use. In SafeScape, a modest runtime overhead may be acceptable when it is accompanied by clearer organization, stronger routing support, and easier maintenance. Nevertheless, these benefits do not remove the need to measure actual response time and resource behavior in the implemented Laravel application.

Pamungkas and Suranegara (2026) evaluated a Laravel application using MySQL Master-Slave replication in a Docker environment to determine how separating read and write activity affects scalability. With simulated workloads from 100 to 500 concurrent users, their configuration maintained high throughput and low error levels by distributing database operations according to function. Although SafeScape uses PostgreSQL and does not replicate the exact architecture, the study demonstrates a general principle: backend scalability can improve when database workloads are organized instead of competing through a single processing path. The findings therefore provide a useful reference for future scaling while SafeScape's present deployment is evaluated using its own database structure and concurrent-user conditions.

Ramadhani, Nindyasari, and Murti (2025) examined the use of Model-View-Controller architecture in a Laravel-based management system and reported benefits in code organization and development efficiency. Separating business logic from presentation reduced coupling between application components and made the system easier to modify as functionality expanded. Their results support MVC as a maintainability strategy rather than merely a coding convention. This is directly relevant to SafeScape, where learning modules, administrative analytics, authentication, and AI integrations must coexist within one application. A structured separation of responsibilities makes it easier to maintain these components and reduces the risk that changes in one part of the portal unintentionally affect unrelated functions.

Olipas et al. (2022) developed a Laravel-based health-record management system and evaluated its quality using ISO 9126. The study reported improvements in data accuracy and waiting time after information was centralized and retrieval processes were organized through the web framework and relational database. Although the health-record context differs from fire safety, both applications require dependable

handling of structured records and frequent user interaction. The findings therefore demonstrate how centralized data management can support operational quality in information-intensive systems. For SafeScape, the study strengthens the rationale for consolidating user, content, learning, and administrative data within a structured application rather than relying on disconnected data-handling processes.

Surahmat and Darmawan (2024) investigated Laravel in the development of an organizational information system and highlighted its routing, session-management, and user-interaction capabilities. Their work showed that framework-provided mechanisms can support stable operation while allowing the application to accommodate continuing user interaction and changing requirements. This is relevant to SafeScape because the portal serves multiple user roles and combines public information, learning, chatbot, and administrative functions within the same environment. The study supports Laravel as an extensible foundation, particularly where authentication, sessions, and route organization must remain manageable as the system evolves. Actual performance, however, still depends on the way these framework features are implemented in the final application.

Abutaleb, Tamimi, and Alrawashdeh (2021) benchmarked widely used PHP frameworks to compare efficiency and scalability characteristics. Their study found that mature frameworks are capable of supporting substantial database activity and varied request workloads, while offering built-in features that simplify application development. These general findings support the selection of an established PHP framework for a system expected to serve continuing user interaction. At the same time, the results are not a substitute for application-specific evidence. SafeScape includes AI requests, database transactions, dashboards, and accessibility functions that may create a different workload from benchmark applications. The framework choice therefore

requires validation through response-time, concurrency, and functional testing in the actual portal.

Hamadou et al. (2026) compared index behavior in PostgreSQL and MongoDB across several data workloads and found that properly indexed PostgreSQL performed strongly for complex structured queries. Their experiments emphasized that the benefit of indexing depends on the way data are organized and queried rather than on simply adding indexes indiscriminately. For SafeScape, this is important because its PostgreSQL database contains relationships among users, assessments, learning records, content, and administrative information. Efficient retrieval therefore requires indexes that reflect actual access patterns. The study supports deliberate relational indexing as part of backend optimization while reinforcing the need to inspect query behavior before deciding which fields and combinations should be indexed.

Zhu et al. (2021) studied cardinality estimation and query optimization in PostgreSQL using extensive benchmark workloads and machine-learning-assisted estimation methods. Their results showed that improved query planning can substantially reduce execution time, with reported speedups reaching 5.69 times for complex multi-join operations compared with less optimized configurations. The findings demonstrate that database performance is influenced not only by hardware and indexing but also by the optimizer's ability to estimate query cost and choose an efficient execution plan. For SafeScape, this supports examining query plans for complex administrative or analytical operations when performance issues appear, rather than assuming that all latency originates from the application framework itself.

Erdelt (2024) evaluated PostgreSQL benchmarking in cloud-native and multi-tenant settings, focusing on parallel activity, contention, and throughput. The study found that PostgreSQL can maintain strong performance as concurrent transactions and

isolated tenants increase when the environment is configured appropriately. This work is relevant primarily as a scalability reference for future deployment rather than as direct evidence of SafeScape's current performance. If the portal is later hosted in a shared or more elastic environment, contention and parallel processing will become increasingly important. The study therefore supports considering database concurrency during architectural planning, while the present research still relies on measured behavior under the specific loads and hosting conditions defined for SafeScape V2.

Pratama and Raharja (2023) evaluated server-side framework performance and resource use during demanding processing tasks, with particular attention to architectures that follow MVC principles. Their findings associated structured framework design with advantages in security, maintainability, and integration with external APIs during sustained workloads. These qualities are important for a public-sector application expected to remain operational and maintainable beyond initial development. In SafeScape, backend performance cannot be considered separately from authorization, maintainability, and service integration because the portal connects database functions with external AI and accessibility services. The study therefore supports balancing execution speed with architectural organization and security rather than optimizing one performance measure at the expense of long-term system quality.

2.3 Artificial Intelligence and the RAG Framework

Chae et al. (2025) investigated retrieval and question-answering performance in a fire-safety legal domain using 4,437 statutory documents drawn from 117 fire-safety codes. Their system combined lexical retrieval with Weighted keyword matching through weighted Reciprocal Rank Fusion (wRRF) to handle complex regulatory hierarchies. Although continual pre-training improved general comprehension, the study found that answer quality deteriorated when the retrieved context was incomplete

or noisy, with model accuracy falling to 39.44%. The researchers also observed that domain-specific parameter adaptation could increase the production of confident but incorrect answers when adequate supporting context was unavailable. These findings are directly relevant to SafeScape because they show that model capability alone cannot guarantee reliable answers in regulated or safety-sensitive domains. The quality of the retrieved evidence must therefore be assessed separately from the quality of the generated response.

Li et al. (2026) proposed DisastRAG, a multi-source disaster-information system designed to combine heterogeneous data within a retrieval-augmented generation environment. Rather than relying on a single retrieval method, the framework included a query-understanding and routing mechanism that directed requests toward the most suitable source. Depending on the type of inquiry, information could be obtained through semantic document search, structured relational SQL queries, or external web-based services. The architecture demonstrates that retrieval can be organized around the nature of the information being requested instead of forcing all data into one search mechanism. For disaster-information systems, this approach is especially useful because operational data may exist in several formats and repositories. In the context of SafeScape, the study reinforces the value of preserving clear provenance and structured retrieval before information reaches the generative component.

Wong and Wong (2026) examined the reliability of retrieval-supported reasoning through the Multi-Evidence Clinical Reasoning with Retrieval-Augmented Generation (MECR-RAG) framework. Their system used DeepSeek-V3 together with two independent evidence sources: a procedural-rule database based on emergency guidelines and an experiential database containing 3,000 historical triage cases. In a

retrospective evaluation involving 236 real-world triage encounters, the dual-source architecture achieved a quadratic weighted kappa of 0.902 and an accuracy of 0.802. These results were higher than those obtained by the ungrounded baseline model, which produced a QWK of 0.801 and an accuracy of 0.542. The system also reduced overtriage from 28.8% to 12.7%. The study shows that generated decisions in high-risk environments can become more dependable when multiple forms of supporting evidence are consulted instead of relying solely on the language model's internal knowledge.

Alhusain (2026) introduced Triage Retrieval-Augmented Generation (TRAG), which translated standardized triage criteria into formal if-then rules and stored them as dense vector representations in a FAISS index. When a user submitted a query, Weighted relevance scoring was used to retrieve the most relevant rules, which were then inserted into a structured prompt before response generation. Testing on 100 curated triage cases showed that rule-supported prompting improved GPT-3.5 classification accuracy from 0.45 to 0.68. The corresponding quadratic weighted kappa increased from 0.67 to 0.82 under the Top-15 retrieval configuration. These results illustrate how explicit procedural rules can reduce variability in model behavior by constraining generation to relevant evidence. For fire-safety applications, this principle is important because many user questions depend on matching the inquiry to a specific protocol rather than producing a broadly plausible answer.

Granata et al. (2025) developed ELERAG, an Entity Linking Enhanced Retrieval-Augmented Generation architecture intended to improve educational question answering where technical terminology may create ambiguity. The system added a Wikidata-based entity-linking stage to conventional vector retrieval and then combined entity matches with semantic similarity through Reciprocal Rank Fusion. In

evaluation, ELERAG produced a 29-percentage-point improvement in Recall@10 for low-frequency technical entities compared with standard vector-only retrieval and more computationally demanding cross-encoder rerankers. The findings demonstrate that semantic similarity by itself may fail when related technical terms appear in similar contexts. Entity-aware retrieval can therefore strengthen precision by helping the system distinguish among closely related concepts. This is relevant to fire-safety information, where several terms may sound similar but refer to different procedures, materials, hazards, or equipment.

Escribano Arias et al. (2026) evaluated large language models in maritime accident-report analysis and compared factors such as Keyword-based retrieval, generation configuration, and prompt-injection vulnerability. Their study emphasized that retrieval settings and generation settings should be examined independently because weaknesses in one stage can affect the reliability of the entire pipeline. The researchers also highlighted the risk that malicious or misleading retrieved context can influence otherwise capable language models. This observation is applicable to SafeScape because the chatbot's performance depends not only on whether Gemini can generate coherent answers, but also on whether the retrieval process supplies appropriate and trustworthy fire-safety information. Consequently, the evaluation of the SafeScape chatbot separates knowledge retrieval from the accuracy and quality of the generated response.

Pujari et al. (2026) developed LawGPT, a legal-assistance platform that implemented a complete retrieval-augmented generation pipeline over statutory documents. The legal texts were divided into overlapping chunks of 512 tokens with an overlap of 50 tokens. These segments were transformed into high-dimensional embeddings and stored in a FAISS index for nearest-neighbor retrieval using Weighted

relevance scoring. The study reported improvements in ROUGE-L and BLEU after retrieved statutory content was supplied to the generation stage, while factual hallucinations were reduced to near zero when outputs were required to reference specific legal provisions. LawGPT demonstrates how a domain-specific assistant can combine retrieval, generation, and source traceability to compensate for the knowledge limitations of standalone language models. The same general principle supports SafeScape's use of retrieved fire-safety context before Gemini produces a response.

Kang et al. (2026) explored whether RAG could remain effective under strict device limitations through a client-side first-aid assistant operating within a 2.0 GB memory ceiling. Their design used a two-stage retrieval procedure in which a fast lexical filter narrowed the search space before dense semantic reranking was applied. Additional optimizations included selective context compression, batched decoding, and quantization of the key-value cache. Despite using a compact 0.6-billion-parameter Qwen model, the system achieved 94.5% accuracy for physical first-aid questions and 97.0% for psychological first-aid questions. Time-to-First-Token was also reduced from 14.2 seconds to 3.7 seconds. Although SafeScape depends on the cloud-based Gemini API rather than an entirely local language model, this study provides evidence that retrieval efficiency, context management, and runtime optimization strongly influence the responsiveness of safety-oriented AI systems.

Xie et al. (2025) developed a RAG-supported multi-agent system for natural-hazard resilience and adaptation. Instead of using one general-purpose agent, the framework distributed responsibilities among specialized components such as evacuation mapping and hazard assessment. Each agent was connected to retrieval pipelines that could access localized geospatial information, meteorological data, sensor observations, and municipal regulations. Across ten expert-validated case

studies, the multi-agent platform outperformed ungrounded language-model baselines by maintaining context-aware reasoning and producing safety recommendations without inventing unsupported spatial or risk information. SafeScape does not implement this type of multi-agent architecture, but the study demonstrates a broader principle that is relevant to the project: domain-grounded retrieval can constrain generative systems when they operate on complex safety information.

Adhipradhana et al. (2026) investigated protocol-grounded artificial intelligence for emergency dispatch through a multimodal response framework. Dashcam footage was first converted into textual descriptions, after which the system retrieved relevant emergency procedures from a curated collection of Japanese traffic-law and emergency-medical-service guidelines. A language-model reasoning component then used the retrieved information to assess severity and formulate a dispatch recommendation from three perspectives. Comparative testing showed that separating perception from protocol-based reasoning was more reliable than relying directly on end-to-end visual-language models. The proposed architecture reduced critical visual hallucinations and produced recommendations that could be traced to rule-level evidence. This approach closely reflects the design principle adopted in SafeScape, where a user query is not answered directly from unrestricted model knowledge but is first grounded in relevant fire-safety information retrieved from the knowledge base.

Ndum et al. (2026) examined the use of retrieval-augmented artificial intelligence in nuclear engineering, where incorrect information can produce serious consequences. Their framework followed a local-first and model-independent design and connected the language model to a metadata-rich knowledge base capable of tracking information at page and figure level. Two measures, Context of Provenance

(CoP) and Verification of Response (ViR), were used to assess whether generated outputs remained traceable to their supporting documentation. Expert-curated evaluations produced compliance levels ranging from 85% to 98%, with the retrieval-supported system exhibiting lower hallucination rates than general-purpose ungrounded commercial models. Although the domain differs from fire safety, the underlying requirement is comparable: generated information must remain reviewable and connected to trusted evidence whenever it may affect safety-related decisions.

Balasubramanian (2024) proposed a real-time RAG intelligence framework for disaster-response logistics that focused on combining information from several rapidly changing sources. The retrieval component dynamically prioritized high-relevance evidence under strict time constraints and merged structured inputs, including weather and resource data, with unstructured information from social media and news reports. The study reported reduced decision latency, improved resource allocation, and more effective public-safety communication. These findings demonstrate the potential of RAG to organize and synthesize large volumes of operational information. However, they also show that speed and reliability need to be assessed in the actual application environment. For SafeScape, the study supports the use of retrieval for rapid information access while reinforcing the need to measure chatbot response time and retrieval effectiveness directly.

Zhang et al. (2026) integrated named entity recognition, a domain-specific knowledge graph called EmergencyKG, and a knowledge-graph RAG pipeline to process unstructured emergency reports. Weighted keyword-based retrieval was used to identify multi-hop subgraphs containing information such as disaster parameters, locations, and resource requirements. These structured relationships were then added to the language model's prompt context. When evaluated using the EM-InstructNER test

set, the hybrid approach achieved a macro F1-score of 0.9205 and produced a 36.6% relative improvement over strong ungrounded zero-shot baselines. The study demonstrates that raw emergency narratives can be transformed into structured relationships that are easier for retrieval systems to use. Although SafeScape does not implement a knowledge graph, the findings support careful preparation and structuring of source material before it enters the retrieval pipeline.

Cho et al. (2026) identified an important reliability problem that can emerge when RAG chatbots are used in long conversations. The researchers repeatedly issued the same medication-dosing question while varying the amount of dialogue history from zero to 50 question-and-answer exchanges. At the start of a conversation, the recorded hallucination rate was 5%, but it increased to 40% after only ten previous exchanges. Analysis showed that accumulated conversation history introduced irrelevant contextual information and gradually redirected the retriever away from the user's current question. As a result, less relevant document segments were returned to the generator. This finding demonstrates that retrieval-supported systems are not automatically stable across extended interactions. It also supports evaluating chatbots using varied and repeated questions rather than judging reliability from a small number of isolated examples.

Huang (2026) conducted a systematic review of hallucination in medical large language models, examining its causes, risks, benchmarks, detection approaches, and mitigation methods. The review argued that hallucination cannot be eliminated merely by increasing model size because uncertainty remains inherent in autoregressive next-token generation. To support safer use, the study introduced the CR^2 framework, representing Capability, Reliability, Cost, and Clinical Risk. Huang further emphasized the value of combining retrieval grounding, automated verification, controlled

generation, and human oversight rather than relying on any single safeguard. For SafeScape, this perspective supports evaluating the chatbot through several complementary measures. Response Accuracy alone cannot fully describe reliability; Contextual Relevance, Response Completeness, Knowledge Retrieval Effectiveness, and Hallucination/Error Rate are also needed to determine whether the chatbot's answers are both supported and useful.

Pawlik and Deniziak (2026) investigated how prompt design influences factual behavior in clinical RAG systems using the MedQA USMLE benchmark with 500 cases. Three prompting strategies were compared: Neutral, Expert Chain-of-Thought, and StrictCitations. Unexpectedly, the Expert-CoT condition produced an Unsupported Sentence Ratio of approximately 95% to 100%, indicating that detailed reasoning prompts encouraged the models to rely heavily on their internal clinical knowledge instead of remaining within the retrieved context. In contrast, StrictCitations produced the strongest verifiable grounding. The study also identified a significant relationship between verbosity and uncertainty, with a reported correlation of $r = 0.81$ and $p < 0.001$. These results show that prompt engineering can improve or weaken grounding depending on how instructions are constructed. In SafeScape, generation control is therefore treated as one layer of reliability rather than as a complete solution to hallucination.

Siddique and Arain (2026) examined practical factors affecting the reliability of retrieval-augmented systems through qualitative interpretation of industry cases and semi-structured interviews. Their analysis identified retrieval success, the quality of retrieved documents, and effective incorporation of relevant context into the model's generation window as central factors in response credibility. The study further suggested that properly cleaned and structured source information may contribute more

to reliability than simply increasing model size. This is particularly relevant to SafeScape because the effectiveness of its RAG chatbot depends on the quality of the underlying fire-safety knowledge base. Preprocessing, organization, and context selection therefore remain important even when a capable generative model such as Gemini is used.

Ronco et al. (2026) demonstrated the use of RAG and large language models for transforming unstructured disaster information into structured data. Their automated pipeline processed more than 3,000 disaster events reported globally between 2014 and 2024. News narratives were converted into structured, FAIR-compliant representations and knowledge graphs that captured relationships among hazards and risks. The researchers reported that this automated structuring process reduced the effort required to organize historical disaster information and accelerated subsequent analysis. The work illustrates another important function of retrieval-oriented AI: it can help organize large amounts of unstructured evidence into forms that are easier to search and reuse. For SafeScape, this reinforces the value of converting loosely structured fire-safety information into a consistent knowledge base before retrieval and generation occur.

Amazou et al. (2025) evaluated RAG in a legal-information setting where non-expert users needed understandable access to complex statutory material. Their system divided large legal PDF documents into approximately 4,000-character segments, generated vector representations for those segments, and retrieved the three most relevant pieces of information for each query. The selected content was then used to construct the prompt supplied to the generative model. Results indicated that grounding responses in actual legal provisions helped users work with complex contractual information while reducing hallucinated or misleading legal statements. The study demonstrates that RAG can make highly technical rules more accessible when

generation remains tied to source documents. This principle is applicable to SafeScape because fire-safety procedures can also contain specialized terminology that community users may need to understand in simpler conversational form.

George and Samuel (2025) proposed Retrieval-Augmented Governance (RAGov) as a framework for increasing the accountability of AI systems operating in regulated environments. Their architecture combined curated policy knowledge stores with detailed logging of query and retrieval activity and incorporated human verification into the response process. By preserving the exact context used to support each generated output, the system created an auditable trail that could be reviewed after generation. The researchers argued that traceability is essential when AI is introduced into governance contexts because decision makers must be able to determine how and from where a response was derived. This perspective is highly relevant to SafeScape because the chatbot provides fire-safety information within a public-service setting. The study supports treating source control, traceability, and professional validation as integral design considerations rather than optional additions.

Taken together, the reviewed studies show that Retrieval-Augmented Generation is most reliable when retrieval quality, source preparation, generation control, and verification are treated as interconnected but separately measurable components. Research across legal, medical, disaster-response, educational, and regulatory environments consistently demonstrates that grounding language models in external evidence can improve factual reliability, yet retrieval errors, poor context selection, long conversational histories, or inappropriate prompt strategies can still produce unsupported responses. These findings directly inform the SafeScape V2 chatbot architecture, which retrieves relevant context from a localized fire-safety knowledge base before providing that context to the Gemini API. They also support the

use of separate evaluation measures for Response Accuracy, Contextual Relevance, Response Completeness, Knowledge Retrieval Effectiveness, and Hallucination/Error Rate rather than relying on a single overall measure of chatbot performance.

2.4 Inclusive UX/UI Design and Digital Accessibility

Zallio et al. (2025) examined the relationship between artificial intelligence and inclusive design through the development and validation of the IDEAS framework, representing Inclusion, Diversity, Equity, Accessibility, and Safety. Using a mixed-methods approach, the researchers emphasized that AI-enabled systems can improve participation and accessibility when they are designed with diverse users in mind. At the same time, poorly designed AI services may reproduce or intensify existing inequalities when accessibility, safety, and representation are treated as secondary concerns. The framework therefore extends the concept of accessibility beyond technical compliance and presents it as a broader design responsibility involving equitable participation and protection of users. For SafeScape, this perspective supports the development of accessibility features as part of the system's overall design rather than as isolated additions introduced after development.

Hussain et al. (2024) reviewed User-Centered Design principles within emergency evacuation applications and identified recurring interface problems that can interfere with decision-making during urgent situations. Their analysis found that overly complex navigation structures, excessive information, and inadequate consideration of disability or demographic differences can increase cognitive burden at the exact moment when users require clear guidance. The authors emphasized simplified interface structures, high-contrast presentation, and human-centered interaction as important design characteristics for emergency applications. These findings are directly relevant to SafeScape because fire-safety information must remain

understandable to users with different levels of technical experience and literacy. Clear visual hierarchy, reduced interaction complexity, and task-oriented controls can therefore help users access important information without unnecessary cognitive effort.

Ackerman et al. (2025) investigated how human-centered design and behavioral science can improve health literacy among vulnerable and historically underserved populations. Their mixed-methods work incorporated user-centered development into information-delivery interventions and used pre-test and post-test measures to assess changes in participant behavior and confidence. The results showed a statistically significant increase in confidence regarding health-seeking activities, with $p < 0.01$, together with a reduction in perceived barriers to information. The study illustrates that information becomes more useful when it is presented according to the capabilities and circumstances of intended users rather than simply being made digitally available. For SafeScape, the same principle applies to fire-safety education: the interface should make information easier to understand, navigate, and act upon for children, vulnerable users, and community members with varying levels of digital literacy.

Sermet and Demir (2021) explored the technical integration of intelligent web functionality through Instant Expert, an open-source semantic framework based on Web Components and Shadow DOM. Their work demonstrated that complex natural-language processing functions can be embedded within a browser-based platform while remaining isolated from the surrounding Cascading Style Sheets and JavaScript environment. This form of modular integration reduces the likelihood that an intelligent component will interfere with existing interface behavior. Although SafeScape uses a different technology stack, the study provides an architectural lesson that remains applicable: advanced AI or accessibility functions should be incorporated without destabilizing the principal web application. The portal's chatbot, voice tools, and

learning features therefore need to operate as integrated components while preserving consistent navigation, responsiveness, and interface behavior.

Faiz et al. (2025) examined whether contextual and immersive presentation can improve fire-safety learning compared with more conventional instructional media. Using a pre-test and post-test experimental design, the researchers compared participants exposed to non-contextual fire-safety material with those who interacted with immersive virtual-reality simulations. Participants using contextual spatial environments showed stronger improvement in identifying complex fire hazards, as well as better recall and motivation. The findings suggest that fire-safety education can become more memorable when users actively interact with meaningful scenarios rather than only reading or viewing static information. SafeScape does not rely exclusively on virtual reality, but the principle supports the use of age-appropriate, interactive content such as educational activities, simulations, videos, and assessments to encourage active participation among younger users.

Igarashi et al. (2024) studied conversational artificial intelligence through a comparison between human interviewers and a three-dimensional AI avatar equipped with speech-recognition and speech-synthesis capabilities. Their qualitative evaluation indicated that participants were able to engage with the AI avatar during semi-structured interactions without substantial technological anxiety. The results demonstrate that conversational interfaces can sustain user attention when the interaction is sufficiently understandable and responsive. However, conversational AI should not be considered a replacement for accessible visual design. In SafeScape, the BFP Assistant can provide an additional mode of interaction, particularly for users who prefer conversational information retrieval, while the main portal must still maintain clear text, navigation, visual hierarchy, and conventional interface controls.

El Fakir et al. (2025) examined Speech-to-Text and Text-to-Speech technologies through Universal Design for Learning and Cognitive Load Theory. Their analysis considered how auditory and textual channels can be combined to support students who experience difficulty with conventional reading-based instruction. The case studies reported significant improvement in reading comprehension among underperforming learners when Text-to-Speech was incorporated into instructional activities. According to the authors, distributing information across visual and auditory channels can reduce demands on working memory and provide alternative routes for understanding. This finding is relevant to SafeScape's voice-enabled accessibility functions because fire-safety information should not require strong reading proficiency as the only means of access. Combining spoken and written information can make educational content more accessible to younger learners and other users who benefit from auditory support.

Baker et al. (2022) evaluated automatic speech-recognition technology in pediatric speech-language pathology, where speech patterns can differ substantially from those of typical adult users. The study examined the transcription of children's narrative speech, including participants with communication disabilities, and found that modern ASR systems can achieve high recognition performance under appropriate conditions. In some cases, automated transcription approached or exceeded human transcription performance. The study is particularly relevant to systems intended for younger users because recognition results obtained from adult speech cannot automatically be assumed to apply to children. SafeScape's Speech-to-Text feature should therefore be tested using realistic utterances and the characteristics of its intended users rather than relying solely on general claims regarding speech-recognition accuracy.

Sezgin et al. (2022) investigated the practical acceptability of voice interaction among caregivers of children with special healthcare needs. Using the Technology Acceptance Model, the researchers evaluated whether hands-free voice functions could support users performing physically demanding care-related activities. Approximately 80% of the participants agreed that automatic-speech-recognition-based voice notes helped them perform tasks when their hands were occupied. The study illustrates an important practical advantage of voice interaction: it can reduce the physical effort required to operate a device when conventional typing or touch input is inconvenient. Within SafeScape, voice input may similarly provide an alternative interaction method when users cannot easily type, when younger users find spoken questions easier to formulate, or when accessibility needs make standard text entry less convenient.

Blanco et al. (2023) evaluated Voice User Interfaces within a web-based Geographic Information System and reported generally positive user acceptance of spoken spatial navigation. However, the study also identified weaknesses in commercially available voice technologies when users pronounced context-sensitive geographic names and other location-specific terminology. These errors demonstrate that acceptable general speech-recognition performance does not guarantee reliable recognition of domain-specific words. For SafeScape, the finding is relevant because emergency and fire-safety interactions may contain specialized terminology, local place names, and uncommon vocabulary. Testing should therefore include realistic fire-safety phrases and representative spoken inputs so that recognition performance is evaluated in the same context in which users will actually operate the system.

Alqarni and Alhramelah (2025) examined the use of AI-powered voice assistants in elementary education and reported improvements in learner engagement, communication skills, and retention when voice technology was incorporated into

instructional activities. Compared with more conventional non-auditory methods, students interacting with voice-enabled learning tools demonstrated more active participation. These findings support the potential educational value of speech-based interfaces when they are incorporated into age-appropriate learning activities rather than treated only as accessibility utilities. In SafeScape, Speech-to-Text and Text-to-Speech may therefore contribute not only to accessibility but also to engagement by giving younger learners additional ways to interact with fire-safety information, educational content, and the chatbot.

Calahorra-Candao and Martín-de Hoyos (2024) investigated the factors influencing adoption of AI-based voice assistants by combining the Technology Acceptance Model with Uses and Gratifications Theory. Their findings showed that perceived usefulness is not the only determinant of user willingness to adopt voice interfaces. Feelings of safety, security, emotional comfort, and trust were also important in determining whether users were willing to rely on the technology. This is particularly relevant to public-safety applications, where incorrect or confusing information can have greater consequences than in ordinary entertainment or productivity systems. SafeScape must therefore consider not only whether its voice functions operate correctly but also whether the broader interaction remains predictable, understandable, and trustworthy to intended users.

Zooli and Chakraoui (2026) reviewed artificial-intelligence-based assistive technologies in inclusive educational environments and described the continuing development of speech-enabled systems beyond simple text reading. The review highlighted technologies capable of adjusting speech characteristics, emphasizing context, and supporting increasingly natural interaction for users with cognitive or physical impairments. Such developments demonstrate that accessibility technologies

can provide more personalized assistance than traditional one-directional Text-to-Speech systems. Although SafeScape implements a more focused Web Speech API approach, the review supports the broader rationale for multimodal interaction. Offering both spoken and visual information creates alternative access routes and helps reduce dependence on a single method of interacting with educational content.

Goller, Fraser, and Graßl (2026) developed MeowCrophone, a voice-controlled interface for children with motor disabilities using the Scratch programming environment. Because default browser speech recognition did not consistently interpret children's commands, the researchers added multiple matching techniques, including regular expressions, phonetic comparison, and fuzzy matching. These mechanisms helped compensate for recognition errors that could otherwise prevent users from completing intended tasks. The project demonstrates that browser-based speech recognition can be practical for children but may require interface-level strategies to handle imperfect transcription. This is relevant to SafeScape because recognition failures should be treated as an interaction-design concern rather than solely as a speech-engine limitation. Clear retry mechanisms, understandable error handling, and flexible interaction options can reduce frustration when a spoken query is not recognized correctly.

Svensson et al. (2021) investigated the effects of Speech-to-Text and Text-to-Speech support on individuals with reading disabilities through a controlled intervention involving 149 participants. The research found that participants using multimodal assistive technologies improved their ability to process and communicate information and also displayed greater motivation during learning activities. These findings provide evidence that speech-based assistance can support both comprehension and participation, particularly when conventional text presents a barrier.

The relevance to SafeScape lies in providing users with alternatives to exclusively reading or typing fire-safety information. Incorporating speech output and spoken input can support learners who require additional assistance while preserving conventional text interaction for users who prefer it.

Défossez et al. (2024) developed Moshi, a full-duplex speech-to-text and speech-to-speech model designed for responsive conversational interaction. Experimental evaluation reported approximately 100 milliseconds of speech-to-text latency and conversational response latency between 600 and 800 milliseconds. These figures demonstrate how rapidly modern speech systems can support spoken interaction under optimized conditions. However, Moshi uses a different architecture and runtime environment from the browser-based Web Speech API incorporated into SafeScape. Its reported latency should therefore be interpreted as a technical reference rather than as an expected SafeScape performance result. The present system must independently evaluate its own speech-recognition behavior, browser compatibility, and interaction responsiveness using the actual technologies and devices defined in the study.

Song et al. (2022) examined Voice User Interface adoption among older adults and found that accessibility benefits can be weakened by poorly configured interaction timing. Although hands-free operation was generally helpful, short listening windows and rigid recognition behavior created usability difficulties for participants who spoke more slowly or used distinctive speech patterns. The study demonstrates that recognition accuracy is only one component of a usable voice interface. Listening duration, pacing, error recovery, and the opportunity to repeat or correct an utterance also influence whether users can successfully interact with the system. This is relevant to SafeScape because its intended users may differ considerably in age, confidence, speaking style, and technical ability. Voice-enabled accessibility should therefore be

evaluated as an interaction process rather than solely through raw transcription accuracy.

Prado et al. (2023) reviewed digital accessibility and examined how the Web Content Accessibility Guidelines address communication and methodological barriers experienced by users in digital environments. Their analysis emphasized that accessibility in websites and software differs fundamentally from accessibility in physical environments because many barriers originate directly from program structure, content presentation, and interaction design. Consequently, digital inclusion requires accessibility principles to be implemented within the software itself. The review is applicable to SafeScape because features such as Speech-to-Text and Text-to-Speech cannot compensate for an interface that remains difficult to navigate. Accessibility therefore needs to be considered across forms, navigation, contrast, feedback, controls, and alternative modes of interaction, and its effectiveness should ultimately be verified through user-oriented evaluation.

Ren et al. (2023) conducted a systematic review of digital game-based interventions for children and examined the influence of User Experience and User Interface design on participation and learning. The review identified recurring problems in systems whose interfaces were not adequately adapted to children's cognitive and developmental characteristics. Complex navigation, inappropriate visual presentation, and interaction demands can reduce engagement even when the educational material itself is valuable. These findings support designing child-oriented digital platforms around simpler navigation, understandable feedback, appropriate typography, and manageable interaction complexity. In SafeScape, this principle applies to the learning modules, games, assessments, and other educational components intended for younger

users. A kid-friendly interface should therefore reflect children's interaction capabilities rather than simply applying a visually colorful theme to an adult-oriented layout.

Eragamreddy (2025) systematically reviewed the educational role of Voice User Interfaces, including conversational assistants such as Google Assistant, and examined their contribution to learner autonomy and skill development. The review found that voice-based AI can provide immediate feedback, customized interaction, and opportunities for learners to obtain information without depending continuously on an instructor. These characteristics can support more independent learning, especially when the interface remains understandable and easy to control. For SafeScape, voice-enabled conversational interaction complements the platform's fire-safety learning environment by allowing users to ask questions and receive information using an additional interaction mode. However, the educational value of voice technology depends on appropriate integration with the system's content and visual interface rather than replacing conventional educational materials.

Collectively, the reviewed literature indicates that inclusive digital systems require a combination of human-centered interface design, multimodal interaction, age-appropriate presentation, and practical accessibility evaluation. Speech-to-Text and Text-to-Speech technologies can reduce barriers associated with reading, typing, physical interaction, or situational constraints, but their effectiveness depends on recognition quality, pacing, error handling, browser behavior, and the characteristics of intended users. Likewise, child-friendly design requires more than visual styling; navigation complexity, typography, feedback, content presentation, and interaction demands must correspond to the abilities of younger learners. These findings support SafeScape V2's integration of voice-enabled functions, responsive interface design, simplified user interaction, and age-appropriate educational resources while also

justifying direct technical and end-user evaluation of accessibility rather than assuming that the presence of these features alone guarantees inclusive use.

2.5 Adaptive Learning and Intelligent Tutoring System (ITS)

Endla et al. (2024) presented an adaptive-learning framework designed to support intelligent instruction in environments where computational resources may be limited. Their approach used lightweight multimodal artificial-intelligence techniques to process real-world student interaction data without depending entirely on large, centralized cloud systems. The framework incorporated Natural Language Processing and Computer Vision to interpret both structured and unstructured learning materials and used these inputs to support learner modeling and instructional adaptation. The researchers reported improvements in student engagement and knowledge retention, demonstrating that adaptive-learning systems can remain useful even when deployed under constrained technical conditions. For SafeScape, the study is relevant because it shows that personalization does not necessarily require extremely complex infrastructure. A structured learner model can still provide meaningful adaptation when the variables used are clearly defined and appropriately linked to the instructional task.

Abu-Issa, Butmeh, and Tumar (2024) examined the problem of adapting instruction for new learners who have little or no previous interaction history. Their Intelligent Tutoring System addressed this cold-start problem through an initial learner-profiling process that combined VARK learning-preference questionnaires with time-limited knowledge assessments. By collecting explicit information at the beginning of the learning experience, the Student Model could immediately establish an initial representation of the learner instead of waiting for a long history of interaction data. The study reported a 90% user-satisfaction rate together with improvements in early instructional personalization. These findings indicate that useful adaptation can begin

from relatively simple learner-profile and assessment information. This is relevant to SafeScape because its adaptive-learning component similarly relies on available learner-profile variables and assessment scores rather than requiring extensive behavioral history before assigning assessment difficulty.

Deng and Yuan (2026) investigated the combination of adaptive education and knowledge grounding through a framework that automatically generated multimodal knowledge graphs from instructional videos and textbook material. The resulting knowledge structure was connected to a Retrieval-Augmented Generation component that supplied verified information to the tutoring system. Their cross-modal approach reduced hallucination-related problems by constraining generated educational responses to content obtained from trusted instructional sources. This study is important to SafeScape because the system combines two distinct intelligent functions: adaptive assessment and a retrieval-grounded chatbot. Although the adaptive-learning engine does not generate instructional content, any educational information presented alongside adaptive assessments should still remain consistent with verified fire-safety knowledge. The study therefore supports keeping learner adaptation and factual content validation as separate but complementary responsibilities.

Sarkar (2025) developed the Dynamic Academic Response and Tutoring System, or DARTS, as a mobile-oriented intelligent tutoring approach that delivers formative assessments and AI-supported explanations through SMS. The design avoided requiring users to install a specialized application or maintain high-bandwidth internet access. The study reported that this low-barrier delivery model increased learner participation, particularly among users in resource-constrained environments. The findings demonstrate that adaptive-learning systems should consider accessibility not only in terms of instructional intelligence but also in terms of how easily learners

can access the educational service. SafeScape uses a web-based approach rather than SMS, yet the same principle remains relevant: an intelligent learning function provides limited value if users cannot conveniently access it. Responsive web design and straightforward interaction therefore complement the adaptive-assessment mechanism.

Pérez-Ortiz et al. (2024) developed a conversational Intelligent Tutoring System that integrated speech recognition, large language models, and Retrieval-Augmented Generation into a multimodal learning environment. The system processed spoken learner interactions, interpreted the conversation, and generated instructional responses while maintaining continuity across the session. By combining speech input with retrieval-supported generation, the framework demonstrated that accessibility functions can operate together with intelligent tutoring logic rather than existing as unrelated system features. This perspective is relevant to SafeScape because its voice-enabled functions and adaptive-learning components are both part of the same educational portal. Although SafeScape does not use speech information as an input to its Random Forest models, the study demonstrates that multiple interaction modalities can coexist within an adaptive educational system without requiring them to perform the same computational role.

López-Goyez, González-Briones, and Demazeau (2026) investigated a more autonomous approach to adaptive tutoring through a multi-agent architecture implemented in a Moodle-based environment. Their system used tabular Q-learning to coordinate tutorial strategies through a Meta-Agent Orchestrator. The architecture separated pedagogical decision-making from generative content production, enabling the system to update its routing strategies according to inferred learner cognitive and emotional states. This design illustrates how learner interpretation can be separated from the mechanisms responsible for producing instructional material. SafeScape

follows a substantially simpler form of adaptation, but the architectural principle remains relevant. Its Random Forest models perform a clearly bounded classification task based on learner-profile and assessment information, while fire-safety content and chatbot generation are handled by separate components. This separation makes the adaptive behavior easier to measure and evaluate.

Mehta et al. (2022) introduced the LEARNER framework for training emergency responders through a human-centered adaptive environment. Their system monitored psychoneurobiological indicators and user-interaction behavior in real time and used the resulting learner state to modify the difficulty of emergency-training scenarios. The researchers found that static training environments were less responsive to differences in individual performance and that adaptation based on real-time interaction could accelerate the development of responder competency. The study is especially relevant because it places adaptive difficulty within an emergency-preparedness context. SafeScape does not monitor physiological variables or dynamically alter complex emergency scenarios, but it applies the same general concept of performance-sensitive difficulty. Learner information is used to determine whether an Easy, Medium, or Hard assessment is appropriate rather than presenting an identical level to every learner.

Chakraborty et al. (2026) developed PAL, or Personal Adaptive Learner, to introduce adaptation directly into video-based instruction. The system used Hybrid Reinforcement Learning to insert timestamped questions into lecture videos and adjust their difficulty according to learner accuracy and performance streaks. In addition to adapting question difficulty, PAL generated lesson summaries and attempted to maintain the learner within an appropriate level of challenge. The researchers reported higher engagement and better concept retention than those observed with static video

instruction. Their findings illustrate how timing and performance-sensitive interventions can influence learning even when the main instructional material remains unchanged. SafeScape does not dynamically interrupt videos or control instructional pacing, but the study is relevant because it supports the use of assessment difficulty as one mechanism for adapting the learning experience without necessarily changing the underlying educational content.

Abrar et al. (2025) examined AI-supported personalized learning pathways that used machine-learning models to interpret repeated formative-assessment results. Based on learner performance, the system could accelerate the progression of stronger students or provide additional resources to users who required remediation. Their quasi-experimental evaluation reported that learners using the adaptive pathways completed tasks approximately 25% faster and demonstrated a 25-percentage-point improvement in knowledge acquisition compared with learners in static instructional conditions. The results provide evidence that assessment information can serve as a meaningful input for educational adaptation. SafeScape applies this idea within a narrower scope. Rather than constructing complete individualized learning pathways, it uses learner-profile and assessment-performance data to classify the difficulty of module quizzes and the final examination.

As'ad et al. (2024) explored the use of generative artificial intelligence in a healthcare-oriented tutoring application and incorporated a rule-based validation layer to provide additional oversight over generated instructional information. The proof of concept demonstrated the value of separating content generation from validation when educational material is used in a safety-sensitive field. This principle is applicable to SafeScape because its subject matter concerns fire safety, where inaccurate information may create risks beyond ordinary educational error. Although the adaptive-learning

models themselves do not generate fire-safety content, the wider system includes a generative chatbot. The study therefore supports the broader SafeScape architecture in which intelligent adaptation, generated information, and domain validation are treated as separate functions rather than being controlled by a single unrestricted AI component.

Ivanova (2026) proposed an ontology-based layered hybrid artificial-intelligence knowledge model for organizing complex educational domains. The architecture separated lightweight taxonomic structures from more computationally intensive reasoning mechanisms and rule engines. According to the study, dividing these responsibilities reduced computational overhead and improved the transparency of pedagogical decision-making. This separation is relevant to adaptive-learning systems because educational knowledge and learner information often serve different purposes. Within SafeScape, the fire-safety knowledge base provides the domain information used by the RAG chatbot, while learner-profile and assessment variables are processed separately by the Random Forest difficulty classifiers. Maintaining these functions as distinct components helps ensure that the decision to assign a particular assessment difficulty can be evaluated independently from the generation or retrieval of fire-safety information.

Ho and Lam (2026) developed EduMSRA, an educational agent that incorporated hierarchical Retrieval-Augmented Generation together with a Conflict-Aware Fusion Module. The framework retrieved material from multiple educational sources and compared the available evidence before generating instructional responses. This process was intended to reduce contradictory or unsupported information and was reported to operate with low latency in instructional environments. The study illustrates how retrieval and conflict checking can strengthen intelligent educational systems by

ensuring that adaptive or conversational behavior remains connected to reliable content. SafeScape does not implement the complete EduMSRA architecture, but its separation of a retrieval-grounded chatbot from the adaptive-assessment classifier reflects a similar concern. Adaptation should influence assessment difficulty without compromising the factual integrity of the fire-safety material presented to learners.

Kurniawan et al. (2024) designed an Intelligent Tutoring System framework for the Indonesian National Armed Forces that incorporated continuous learner profiling. Their model considered visual, auditory, and kinesthetic learning characteristics together with emotional and concentration indicators to modify instructional delivery. The study demonstrates that structured training environments can benefit from continuous assessment of learner condition and behavior. However, it also represents a substantially more complex learner model than the one adopted in SafeScape. The present study does not monitor affective state, learning style, or concentration. Instead, it deliberately uses a limited set of explicit variables, including Age, Grade Level, Pre-Assessment Score, and relevant quiz performance. The comparison shows that learner adaptation can range from highly multidimensional profiling to simpler performance-based classification depending on the requirements and scope of the system.

Nikolaidis et al. (2026) investigated real-time affective adaptation using an ensemble of convolutional and transformer-based neural networks. Their system continuously monitored indicators associated with learner frustration and cognitive overload and could adjust task difficulty at intervals as short as 0.01 seconds. This form of automatic adaptation demonstrates the potential of using physiological or affective signals to make highly responsive decisions without requiring learners to provide explicit feedback. At the same time, such systems introduce greater computational and implementation complexity. SafeScape intentionally adopts a simpler strategy based on

observable learner-profile and assessment information. The study is therefore useful not because SafeScape duplicates its approach, but because it clarifies a boundary of the current research. The adaptive engine focuses on interpretable assessment variables rather than continuous affective sensing.

Naga and Perez (2026) explored learner profiling through natural-language interaction by applying K-means clustering to student dialogue in LLM-supported tutoring sessions. Their analysis identified distinct patterns of engagement, including profiles described as Passive Reactors and Reflective Performers. The findings suggested that conversational behavior can provide additional evidence about how learners interact with educational systems and can be used to update the Student Model dynamically. SafeScape's BFP Assistant records conversational interactions as part of the wider platform, but these conversations are not used as predictive variables in the adaptive-learning classifiers. The adaptive engine instead depends on explicitly defined assessment and profile data. The study nevertheless demonstrates how future intelligent tutoring systems could incorporate richer behavioral information after sufficient real-world interaction data become available.

Chrysafiadi and Virvou (2024) developed PerFuSIT, a fuzzy-logic-based module for selecting instructional interventions when learner behavior contains uncertainty. The system considered factors such as learner error patterns and task-completion time and used fuzzy inference to decide whether the learner should receive supplemental material or advance to another topic. The work illustrates an adaptive strategy in which educational decisions are made from variables that do not always fit into sharply defined categories. SafeScape follows a different computational approach. Its two Random Forest classifiers generate discrete Easy, Medium, or Hard assessment-difficulty classes rather than selecting among several tutoring interventions. This

distinction is important because the study does not claim that SafeScape implements a complete fuzzy tutoring environment; instead, it applies supervised classification to a clearly bounded adaptation task.

Huang, Sun, and Yu (2025) introduced SP-TeachLLM, an adaptive educational framework that combined Cognitive Load Theory, Bloom's Taxonomy, and reinforcement learning. The system decomposed complex educational objectives into smaller tasks and continuously refined its instructional strategies according to learner success rates. This architecture represents a relatively autonomous form of adaptive tutoring in which the system can revise its own instructional decisions over time. SafeScape does not implement reinforcement learning or automatic decomposition of educational goals. Its adaptive behavior is deliberately more constrained and interpretable. The Random Forest models classify assessment difficulty based on predefined inputs and fixed output classes. The comparison therefore demonstrates that adaptive-learning systems can differ greatly in autonomy, and that a simpler classification-based mechanism may be more appropriate when the research objective requires measurable and clearly defined outputs.

Mustapha (2025) examined the role of explainability in adaptive-learning systems through ERD-PRO, a mixed-initiative Intelligent Tutoring System that incorporated Explainable Artificial Intelligence. The system was designed to communicate the reasons behind pedagogical decisions instead of presenting instructional changes without explanation. The study reported approximately 90% user satisfaction and found that users responded positively when they could understand why a particular intervention had been selected. This result highlights the relationship between transparency and trust in adaptive systems. For SafeScape, the adaptive-learning engine does not implement a full XAI explanation module, but the principle

remains relevant. Difficulty assignments should be understandable within the context of learner-profile and assessment data, and administrators should be able to interpret the learner-performance information presented through the system rather than treating classification results as unexplained outputs.

Lai and Lin (2025) investigated staged instructional support through ITS-CAL, an LLM-based tutoring system that provided assistance progressively. Rather than immediately supplying complete answers, the system initially presented broader conceptual guidance and then offered more specific debugging support when necessary. Their mixed-method evaluation reported a 72.22% pass rate among students who interacted with the scaffolded support, indicating that gradual assistance can encourage learners to engage more actively with a problem. Although SafeScape does not automatically generate hierarchical hints or debugging instructions, the study demonstrates that educational systems benefit from adjusting the level of support according to learner needs. The principle may inform future learning activities, but it should not be interpreted as a current function of the SafeScape adaptive engine.

Villegas-Ch et al. (2025) proposed a modular Intelligent Tutoring System for STEM education that relied on continuous Natural Language Processing to adjust learner trajectories dynamically. Their evaluation reported precision values above 78%, demonstrating that continuously collected interaction information can be used to improve personalization decisions. The work highlights the potential of large volumes of learner behavior to support increasingly responsive tutoring systems. SafeScape currently operates with a more limited set of explicit learner and assessment variables and evaluates its Random Forest models using Accuracy, Precision, Recall, F1-Score, per-class measures, and confusion matrices. The comparison emphasizes that the performance of an adaptive system should be measured using metrics appropriate to its

prediction task rather than simply assuming personalization is effective because adaptation is present.

Across the reviewed Intelligent Tutoring System literature, adaptive learning ranges from simple learner profiling to highly autonomous systems capable of content sequencing, reinforcement learning, affective monitoring, conversational analysis, adaptive feedback, and continuous modification of instructional pathways. These studies establish that learner information can support educational personalization, but they also demonstrate that the term adaptive learning can describe substantially different levels of system intelligence. For this reason, the adaptive-learning component of SafeScape V2 is deliberately bounded to a specific and measurable function.

SafeScape V2 does not automatically choose the next learning module, rearrange the sequence of educational materials, generate instructional hints, modify video pacing, or create complete personalized learning pathways. Instead, the implemented engine uses learner-profile and assessment-performance data to assign assessment difficulty. The Module Quiz Difficulty Model uses Age, Grade Level, and Pre-Assessment Score to classify the appropriate module-quiz difficulty as Easy, Medium, or Hard. The Final Examination Difficulty Model additionally incorporates the learner's Module 1 to Module 4 quiz scores and produces the same three difficulty classes for the final examination. In this study, adaptive learning is therefore operationalized as supervised assessment-difficulty classification rather than as autonomous content recommendation or instructional sequencing. This bounded approach provides a clearly defined prediction problem that can be evaluated using established classification metrics while still introducing performance-sensitive personalization into the fire-safety learning environment.

Synthesis

The reviewed literature demonstrates that the modernization of a public-safety information platform requires the coordinated development of several technological and user-centered components rather than the addition of isolated digital features. Research on digital governance and disaster preparedness consistently identifies reliable information delivery, continuous public engagement, institutional coordination, accessibility, and data-supported administration as important characteristics of effective public-service platforms. These findings support the modernization of Berong SafeScape as an information and educational environment rather than merely a redesign of its existing interface.

Studies concerning web architecture further show that application performance depends on the interaction among framework design, database structure, query behavior, caching, concurrency handling, and deployment configuration. Laravel provides organized routing, Model-View-Controller separation, authentication mechanisms, and maintainable application structure, while PostgreSQL offers relational data management appropriate for user, content, assessment, chatbot, and administrative information. However, the literature also demonstrates that framework and database selection alone cannot guarantee performance. Eloquent query behavior, indexing, caching, query planning, simultaneous requests, and application workload must still be measured in the implemented system. This supports the study's decision to evaluate SafeScape V2 through actual response-time, concurrent-access, security, maintainability, and related technical measures rather than claiming improvement based solely on the technologies selected.

The literature on Retrieval-Augmented Generation provides a strong basis for the BFP Assistant while also identifying important limitations that must be considered

during implementation and evaluation. Studies from legal, medical, emergency-response, disaster-management, and regulated environments show that connecting a generative model to trusted external evidence can improve factual reliability and reduce unsupported outputs. At the same time, RAG systems remain vulnerable to poor retrieval, irrelevant context, long conversational histories, inadequate source preparation, and inappropriate prompting strategies. These findings support SafeScape's architecture in which relevant fire-safety information is retrieved from a localized knowledge base before being supplied to the Gemini API. They also justify evaluating Response Accuracy, Contextual Relevance, Response Completeness, Knowledge Retrieval Effectiveness, and Hallucination/Error Rate as separate measures instead of treating the chatbot as reliable merely because it uses retrieval.

Research on digital accessibility and inclusive interface design demonstrates that effective access extends beyond technical standards or the existence of assistive features. User-centered layouts, simplified navigation, readable visual presentation, multimodal interaction, appropriate pacing, understandable feedback, and realistic error-recovery mechanisms influence whether users can successfully interact with a system. Studies involving children, users with disabilities, older adults, and other populations also show that Speech-to-Text and Text-to-Speech can reduce barriers associated with typing, reading, physical interaction, and literacy. However, speech technology can still be affected by accents, pediatric voices, domain-specific terminology, browser behavior, and recognition timing. Consequently, SafeScape's voice-enabled functions require direct testing of recognition accuracy and functional behavior together with end-user evaluation of accessibility and ease of interaction.

The literature on adaptive learning and Intelligent Tutoring Systems establishes that learner information can be used to modify educational experiences in many ways,

including difficulty adjustment, content sequencing, feedback generation, reinforcement-learning strategies, affective monitoring, and personalized learning pathways. Nevertheless, the reviewed systems vary considerably in complexity and purpose. The present research does not attempt to reproduce the full functionality of an autonomous Intelligent Tutoring System. Instead, SafeScape V2 implements a controlled form of personalization in which two Random Forest classifiers use learner-profile and assessment data to determine Easy, Medium, or Hard difficulty levels for module quizzes and the final examination. This design retains the central concept of performance-sensitive adaptation while defining a prediction task that can be evaluated objectively using classification metrics.

Taken together, the reviewed studies identify a research and implementation gap for a localized fire-safety platform that combines a maintainable web architecture, structured public-safety information, retrieval-grounded conversational artificial intelligence, voice-enabled accessibility, adaptive assessment difficulty, and administrative analytics within one system. Existing literature frequently evaluates these technologies separately or applies them within domains such as healthcare, education, legal services, disaster management, and general e-government. SafeScape V2 brings these elements together within the localized context of fire-safety information dissemination and education in Sta. Cruz, Laguna.

The literature therefore informs both the architecture and the evaluation strategy of the present study. Laravel and PostgreSQL support the web and data-management layer; Retrieval-Augmented Generation and the Gemini API support context-grounded conversational access to fire-safety information; the Web Speech API provides alternative Speech-to-Text and Text-to-Speech interaction; Random Forest classification supports assessment-difficulty adaptation; and the administrative

dashboard provides user-engagement and assessment information for monitoring purposes. Their effectiveness is examined through researcher-conducted technical testing, IT-expert software-quality evaluation, BFP domain and operational validation, and community end-user assessment. In this way, the literature establishes the technical basis for SafeScape V2 while also demonstrating why the system's performance, reliability, accessibility, AI accuracy, adaptive-learning behavior, and user acceptability must be measured directly rather than inferred from the capabilities reported by previous studies.

CHAPTER III

RESEARCH METHODOLOGY

This chapter presents the methodology used to achieve the objectives of the study. It describes the research design, locale of the study, applied concepts and techniques, data collection and preparation procedures, data model generation, system development methodology, software tools, system architecture, and software testing and evaluation procedures implemented in the study.

3.1 Research Design

This study employed a Developmental Research Design, also referred to as a Research and Development (R&D) approach, with a primarily quantitative evaluation component. This design was appropriate because the primary purpose of the study was to develop, integrate, and evaluate an AI-driven public-safety web portal rather than to conduct a controlled experiment comparing different algorithms or treatments.

The developmental phase was guided by the Design Science Research Methodology (DSRM), which provided a systematic approach for designing and developing an information-technology artifact to address the identified problems. The methodology guided the identification of system requirements, architectural design, implementation of the Laravel-based platform, integration of the RAG chatbot and adaptive-learning components, technical testing, and iterative refinement of SafeScape V2.

Following development, the completed system underwent quantitative evaluation using selected characteristics of ISO/IEC 25010:2023, specifically Functional Suitability, Performance Efficiency, Reliability, Security, Maintainability, and Flexibility, together with complementary measures of community-user usability, accessibility, ease of interaction, and overall acceptability. Technical measurements

were also used to evaluate the adaptive-learning models, RAG chatbot, Speech-to-Text and Text-to-Speech functions, response time, concurrent access, security, and other system components.

Internal technical testing was conducted by the researchers/developers, while external evaluation involved ten (10) Information Technology experts, fifteen (15) BFP personnel, and fifty (50) community end-users. Open-ended questions were included as supplementary feedback for identifying possible areas for improvement and were not treated as a separate qualitative research component.

3.2 Locale of the Study

This study was conducted within Sta. Cruz, Laguna, which served as the primary deployment and evaluation environment of SafeScape V2. The locale was selected because the study focused on enhancing the existing Berong SafeScape initiative of BFP Santa Cruz and supporting localized fire-safety information dissemination and education.

Requirements gathering and consultation with BFP personnel identified reported performance, accessibility, information-dissemination, and educational limitations in the previous portal. The localized fire-safety information used for the enhanced system originated from legacy BFP-derived records and related fire-safety resources. These materials were filtered, augmented, cleaned, and restructured into a 500-record knowledge base used by the RAG chatbot.

The adaptive-learning component used two separate synthetic student-performance datasets containing 5,000 records each. The Module Quiz Difficulty Model used Age, Grade Level, and Pre-Assessment Score to classify module quiz difficulty as Easy, Medium, or Hard. The Final Examination Difficulty Model used Age, Grade Level, Pre-Assessment Score, and Module 1 to Module 4 quiz scores to

classify final-examination difficulty. These models adapted **assessment difficulty only** and did not automatically reorder learning modules or recommend succeeding educational content.

3.3 Applied Concepts and Techniques

The development of SafeScape V2 integrated several computing concepts selected according to the requirements of the research objectives. These included web-system modernization, machine-learning classification, retrieval-augmented generation, voice-enabled accessibility, and data-driven administrative functions.

3.3.1 Legacy System Modernization

The existing Berong SafeScape Web Portal underwent systematic modernization based on the performance, accessibility, and functional limitations identified during requirements gathering and consultation with BFP personnel. The modernization involved restructuring the existing portal and implementing the enhanced system using PHP Laravel 11 as the primary backend framework. The redevelopment also introduced a responsive user interface, PostgreSQL-based data management, artificial-intelligence integrations, accessibility functions, and administrative analytics. The exact technology stack of the previous implementation was not treated as a variable in the study; therefore, the modernization is evaluated based on the architecture and measured technical behavior of the developed SafeScape V2.

3.3.2 Laravel-Based Web Application Architecture

The enhanced platform utilizes the PHP Laravel 11 framework as its primary application development framework to provide an organized structure for server-side routing, database interaction, authentication, and application modularity. The system

applies the Model-View-Controller (MVC) architectural pattern. The Model manages data related to users, fire-safety information, learning modules, chatbot records, and other core database entities. The View provides the responsive user interface for community users, learners, and BFP administrators. Finally, the Controller processes user requests and coordinates application logic between the Model and the view, ensuring optimized data routing.

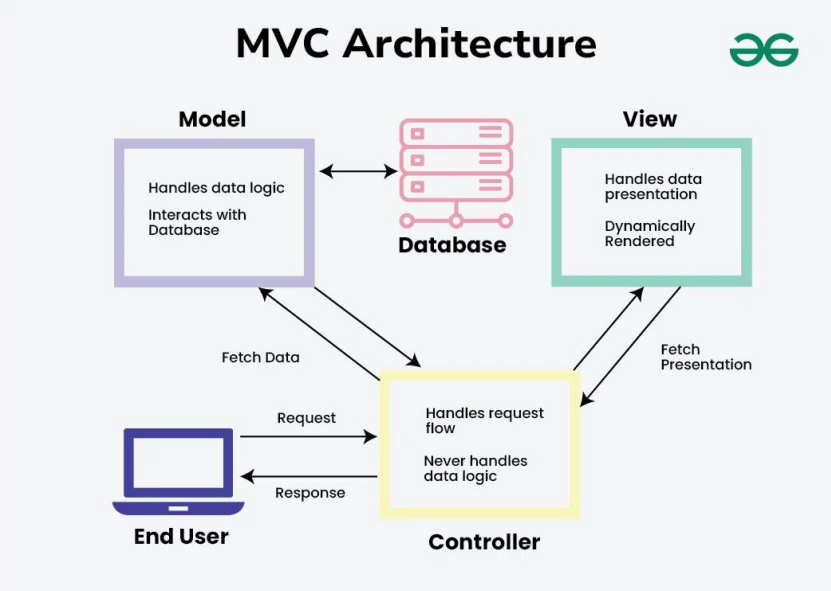


Figure 6. Model-View-Controller (MVC) Architecture (geekforgeeks, 2025)

3.3.3 Responsive and User-Centered Web Design

To accommodate vulnerable demographics and younger learners, a user-centered design approach was applied. The frontend utilizes Tailwind CSS to implement a mobile-first, responsive web design that adapts its layout, typography, and interactive elements to different screen sizes. For the educational modules, cognitive load was considered through the use of high-contrast visual hierarchies, simplified navigation paths, and child-friendly graphical elements. These design features were intended to make important fire-safety information easier to read and understand.

3.3.4 Retrieval-Augmented Generation (RAG)

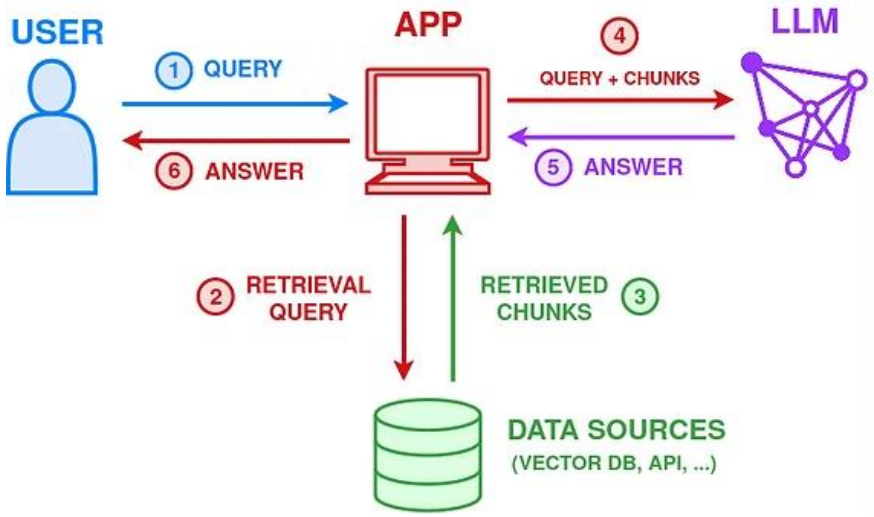


Figure 7. Architecture of a basic RAG system (Araya, 2024)

Weighted Keyword-Based Retrieval

The chatbot uses a weighted keyword-based retrieval approach to identify relevant information from the BFP knowledge base. The knowledge base contains fire-safety questions, answers, categories, and related information stored in a CSV file. When a user submits a question, the system processes the input and compares its keywords with the available records.

To determine the relevance of each record, the retrieval mechanism applies different weights to matching fields. Matches in the question field receive three points, matches in the category or subcategory receive two points, and matches in the answer field receive one point. The system uses the resulting scores to identify the most relevant records from the knowledge base.

The retrieved information is then used as context for generating a response through the Gemini API. This approach allows the chatbot to provide responses based

on the available BFP fire-safety records rather than relying solely on the generative model's internal knowledge.

3.3.5 Natural Language Processing (NLP)

Natural Language Processing (NLP) techniques were applied to process and interpret the unstructured, conversational queries submitted by community users. Before a search is executed, the user's input undergoes text normalization and preprocessing to handle varied sentence structures and localized terminology. This NLP layer helps the system process both formal questions and fragmented user inputs and identify the intended meaning of the submitted query.

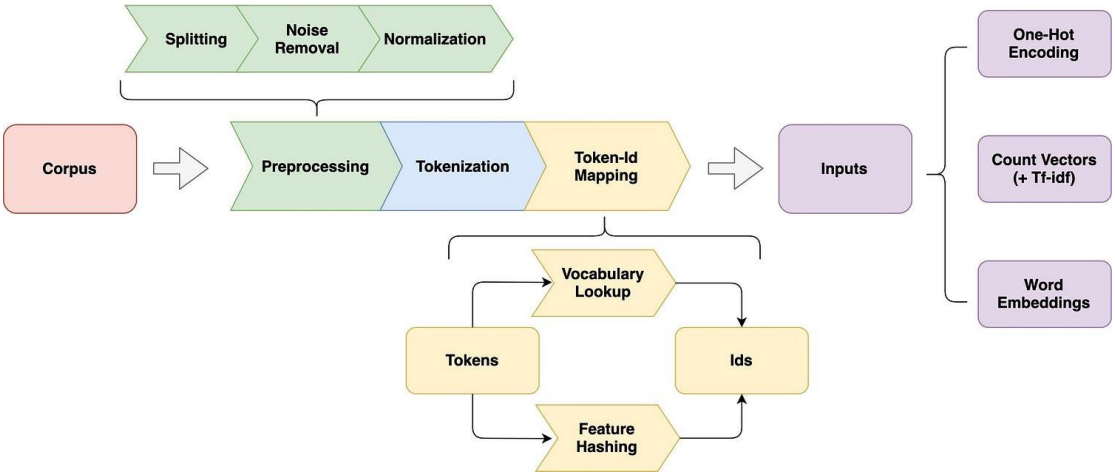


Figure 8. Natural Language Processing Pipeline (Sudani, 2023)

3.3.6 Weighted Keyword-Based Information Retrieval

The platform uses a weighted keyword-based information retrieval approach to identify relevant fire-safety information from the localized knowledge base. When a user submits a question, the system first converts the input into lowercase text, removes punctuation, and filters common stop words to extract the remaining keywords. These keywords are then compared against the questions, answers, categories, and subcategories stored in the CSV dataset. A weighted scoring mechanism assigns three points for matches in questions, one point for matches in answers, and two points for

matches in categories or subcategories. The retrieved records are subsequently ranked according to their accumulated scores, allowing the system to select the highest-scoring entries as contextual information for the chatbot. This approach enables the platform to identify relevant fire-safety records based on keyword occurrences in the user's inquiry.

3.3.7 Prompt and Response Control

Advanced prompt engineering techniques were applied within the Laravel backend to govern the operational behavior of the Gemini API. System-level instruction templates explicitly define the chatbot's persona, setting a calm, authoritative, and helpful tone. Furthermore, strict response control mechanisms are programmed to execute fallback protocols; if the weighted keyword-based retrieval mechanism fails to find a relevant BFP document with a high confidence score, the system prompt forces the Gemini API to state its limitations rather than guessing the answer.

3.3.8 Voice-Enabled Accessibility

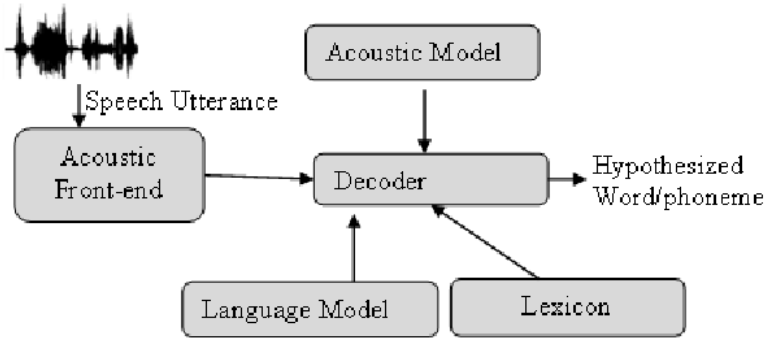


Figure 9. Standard Architecture of a Speech-to-Text System (Ahmed & Ghabayen, 2017)

To eliminate digital literacy constraints, particularly for young children or physically occupied users, the platform applies multimodal voice interaction. Utilizing the browser-native Web Speech API, the system implements Speech-to-Text (STT)

capabilities, allowing users to query the AI chatbot hands-free by simply speaking into their device's microphone. Conversely, Text-to-Speech (TTS) functionalities automatically read system responses and educational modules aloud, ensuring that reading proficiency does not act as a barrier to critical safety knowledge.

3.3.9 Accessibility and Inclusive Design

Beyond voice tools, broad digital accessibility principles are integrated directly into the web application's structure. The user interface is designed to accommodate diverse abilities by ensuring sufficient color contrast ratios, clear error state indicators, and accessible form controls. By prioritizing programmatic inclusivity, the platform ensures that marginalized or vulnerable populations maintain equitable access to the municipality's e-governance safety resources.

3.3.10 Adaptive Learning

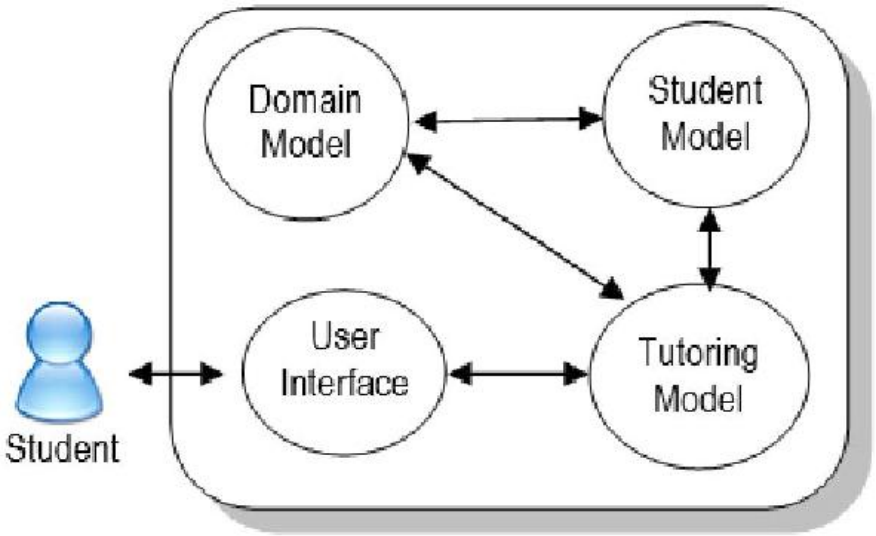


Figure 10. Intelligent Tutoring Systems framework (Thai-Nghe & Schmidt-Thieme, 2015)

The platform applies an adaptive assessment mechanism through two supervised Random Forest classification models. The first classifier, referred to as the Module Quiz Difficulty Model, uses the learner's Age, Grade Level, and Pre-

Assessment Score to determine an appropriate Easy, Medium, or Hard difficulty level for module quizzes. The second classifier, referred to as the Final Examination Difficulty Model, uses Age, Grade Level, Pre-Assessment Score, and the learner's accumulated Module 1 to Module 4 quiz scores to determine an appropriate Easy, Medium, or Hard difficulty level for the final examination. In this study, adaptive learning is therefore operationalized through performance-based assessment-difficulty classification rather than automatic module recommendation or content sequencing.

3.3.11 Administrative Analytics and Decision Support

Data aggregation and visualization techniques were applied to transform raw database interactions into an intelligent administrative dashboard. The system actively tracks and compiles metrics such as total registered users, module completion rates, and aggregate quiz performances. This provides local BFP authorities with a real-time, data-driven decision-support tool, allowing them to instantly identify widespread community knowledge gaps and tailor their physical fire safety campaigns accordingly.

3.3.12 Role-Based Access Control (RBAC)

To maintain system integrity, Role-Based Access Control (RBAC) is implemented using Laravel’s native middleware. The system defines strict access boundaries based on user profiles: Community Users can access learning modules and the chatbot, while BFP Personnel and System Administrators are granted elevated privileges to manage content, view analytical dashboards, and monitor user feedback. This ensures that sensitive administrative functions are entirely isolated from public-facing interfaces.

3.3.13 Database Management and Data Integrity

The platform utilizes PostgreSQL to execute advanced database management principles. Relational data normalization techniques (e.g., establishing primary and

foreign keys) were applied to securely link user profiles to their specific learning progress, quiz scores, and chatbot session histories. Strict database constraints and transactions are enforced to prevent data duplication, orphan records, and maintain data integrity across the educational platform.

3.3.14 Security by Design

Security principles were integrated into the development lifecycle to support alignment with relevant DICT Government Web Hosting Service (GWHS) security considerations. The Laravel backend applies automated CSRF (Cross-Site Request Forgery) protection, secure password hashing algorithms (Bcrypt), and utilizes Eloquent ORM to neutralize SQL injection vulnerabilities. Furthermore, API keys for the Gemini service are securely isolated within server-side environment variables to prevent unauthorized cloud access.

3.3.15 Privacy-Aware Data Processing

The system adheres to strict data minimization and privacy-aware processing rules. Only data essential for educational tracking and system optimization are collected. During conversational interactions, the application layer acts as a sanitizer, ensuring that no unnecessary Personally Identifiable Information (PII) from the user's profile is embedded into the prompt payload before it is transmitted to the external cloud-based Gemini API.

3.3.16 API and Service Integration

The platform architecture heavily relies on Application Programming Interface (API) integrations to establish seamless interoperability between discrete technology stacks. Rather than embedding heavy computational logic directly into the web application, Laravel's HTTP client is utilized to orchestrate secure communication between the core PHP web portal and external services. This includes routing data to

Python-hosted machine learning endpoints (for Random Forest adaptive learning predictions) and the Google Gemini endpoints (for chatbot generation). This microservice-inspired integration allows computationally intensive AI tasks to be offloaded from the main web server, reducing application-side processing overhead and supporting responsive user experiences across the public-facing platform.

3.3.17 Software Testing and Quality Assurance

Software quality assurance procedures were applied iteratively throughout the development of SafeScape V2. Internal technical testing was conducted by the researchers/developers and included functional testing, integration testing, artificial intelligence chatbot verification, Speech-to-Text and Text-to-Speech testing, accessibility and user-task testing, adaptive-learning verification, administrative-dashboard validation, performance testing, and security testing. Controlled test accounts, test records, and non-production, anonymized, or aggregate data were used whenever applicable to protect the personal information of existing users. Following internal technical testing and necessary corrective actions, ten (10) Information Technology experts independently assessed selected software quality characteristics based on ISO/IEC 25010:2023, specifically Functional Suitability, Performance Efficiency, Reliability, Security, Maintainability, and Flexibility. Community-user usability and acceptability and BFP domain validation were assessed separately through their respective evaluation instruments.

3.3.18 Iterative Development and Refinement

The project relies on an Agile iterative development methodology, allowing for continuous integration and refinement. The lifecycle (Requirement → Design → Development → Testing → Evaluation → Refinement) ensures that the software is built in modular increments. This allows the developers to repeatedly test the AI's

response accuracy, adjust adaptive learning thresholds, and refine the kid-friendly UI components based on continuous empirical feedback before final deployment.

3.4 Algorithm Analysis

Algorithm analysis was conducted to systematically evaluate the computational procedures implemented in the enhanced Berong SafeScape Web Portal and to determine whether they satisfy the fundamental characteristics of a valid algorithm while demonstrating appropriate computational efficiency. Each algorithm is examined based on its defined inputs, processing steps, outputs, termination conditions, time complexity, space complexity, and performance metrics.

3.4.1 Retrieval-Augmented Generation (RAG) Chatbot Algorithm

The chatbot utilizes a Retrieval-Augmented Generation (RAG) algorithm powered by the Google Gemini API. The algorithm accepts a user's query, either through text input or transcribed audio, as its primary input. The system first preprocesses the query by converting it into lowercase text, removing punctuation, and filtering common stop words.

After preprocessing, the remaining keywords are compared against the questions, answers, categories, and subcategories of the localized 500-record BFP knowledge base stored in a CSV file. A weighted scoring mechanism is applied, assigning three points for matching keywords in questions, one point for matches in answers, and two points for matches in categories or subcategories. The records with positive scores are sorted in descending order, and the highest-ranking entries are selected according to the configured retrieval limit.

The retrieved question-and-answer records are then combined with the user's original query, system instructions, and available conversation history before being transmitted to the Gemini API for natural language generation. The chatbot returns the generated response to the user. When the Gemini API is unavailable, the system attempts to retrieve the highest-scoring dataset entry and displays the corresponding information as an offline response. If no relevant record is found, an appropriate fallback message is provided.

3.4.2 Speech-to-Text (STT) Algorithm

The voice-enabled accessibility component uses the browser-native Web Speech API to convert spoken user input into machine-readable text. The algorithm accepts the user's speech as input. It begins by activating the microphone interface to capture audio, which is then processed by the browser's Web Speech Recognition engine. The engine converts the recognized speech into text. The process ends when the system either produces a transcribed query or prompts the user to repeat the input when recognition is unsuccessful. The Speech-to-Text component was evaluated using Recognition Accuracy and Word Error Rate (WER).

Because speech recognition is handled by the browser's speech-recognition engine, the application performs primarily interface-level operations. The study did not independently measure the computational time and space complexity of the underlying browser engine. The overall effectiveness and performance of the STT algorithm are evaluated using Recognition Accuracy percentages and Word Error Rate (WER).

3.4.3 Text-to-Speech (TTS) Algorithm

The Text-to-Speech algorithm converts system-generated text into audible speech using the browser-native Web Speech Synthesis API. The algorithm accepts a system-generated text string as input. It then initializes the speech synthesis engine,

converts the text into a speech utterance, configures the available voice parameters, and generates audio playback. The process ends when the synthesized speech has finished playing. The component was evaluated through functional test cases that examined successful playback and speech intelligibility.

3.4.4 Adaptive Learning Classification Algorithm (Random Forest)

The adaptive learning engine employs two supervised Random Forest classification models to personalize the difficulty of fire safety assessments. The first classifier, referred to as the Module Quiz Difficulty Model, uses the learner's Age, Grade Level, and Pre-Assessment Score as input features to predict the appropriate ModuleQuizDifficulty class. The predicted output is categorized as Easy, Medium, or Hard and is used to determine the difficulty level presented for the module quizzes.

The second classifier, referred to as the Final Examination Difficulty Model, uses Age, Grade Level, Pre-Assessment Score, and the learner's quiz scores from Modules 1 to 4 as input features. These features are processed by a separate Random Forest classifier to predict the FinalExamDifficulty class, also categorized as Easy, Medium, or Hard. This enables the difficulty of the culminating assessment to reflect the learner's accumulated assessment performance throughout the educational modules.

Both Random Forest models use an ensemble of 100 decision trees and aggregate the individual tree predictions through majority voting. For each classifier, the approximate inference time complexity is $O(T \times D)$, where T represents the number of trees and D represents the maximum traversal depth. Model storage depends on the total number of nodes constructed across the ensemble and may be represented as $O(T \times M)$, where M denotes the average number of nodes per tree. The predictive performance of each classifier is evaluated independently using Accuracy, Precision, Recall, F1-Score, and a confusion matrix.

3.5 Data Collection Method

The data collection process for this study encompassed the acquisition of verified domain knowledge for the artificial intelligence retrieval engine, the preparation of synthetic training data for the adaptive-learning system, the curation of held-out test datasets, the collection of internal technical testing records generated by the researchers/developers, and the collection of evaluation responses from external participants. The researchers employed a multi-phase approach to ensure that the data integrated into the platform and the resulting evaluation data were appropriately structured for their respective computational, technical-testing, and stakeholder-evaluation purposes.

3.5.1 Consultation and Problem Identification

The project originated from direct consultation with the Bureau of Fire Protection (BFP) in Sta. Cruz, Laguna. Initial discussions with key BFP personnel revealed limitations in the existing Berong SafeScape web portal and challenges associated with traditional community outreach, including performance bottlenecks, difficulties in engaging younger residents, and accessibility barriers for populations with varying levels of digital literacy. These consultations served as part of the requirements-gathering and problem-identification process and helped establish the project's core direction: to modernize the legacy platform into an inclusive and proactive digital tool for continuous fire safety information dissemination and education.

3.5.2 Requirements Gathering and Benchmarking

Following the initial problem identification, requirements consultations and iterative meetings were conducted with BFP personnel to determine the functional and operational requirements of the enhanced system. During these consultations, the need

for an accessible educational component suited to diverse user groups was identified, leading to the integration of Web Speech API-based Speech-to-Text and Text-to-Speech capabilities. Stakeholder requirements also supported the integration of a Retrieval-Augmented Generation (RAG) chatbot for automated fire safety inquiries and an adaptive learning engine capable of adjusting learning difficulty according to learner performance. The gathered requirements were used as development inputs and were not treated as a separate qualitative research dataset.

3.5.3 Technical Data Sourcing

The study initially obtained approximately 1,200 existing BFP-related question-and-answer records as the source of the chatbot's knowledge base. Before integrating the records into the system, the researchers performed data cleaning and preparation to improve the consistency and usability of the available information.

The cleaning process involved reviewing the records, identifying and removing unsuitable entries, and retaining the records that met the requirements of the study. After this process, a total of 500 records were retained for integration into the system's CSV-based knowledge base.

The prepared dataset served as the primary information source for the chatbot's weighted keyword-based retrieval mechanism. The retained records contained fire-safety questions and corresponding answers that the system used to identify relevant information when responding to user inquiries.

The preparation process helped ensure that the chatbot operated using a more organized and relevant collection of BFP-related information.

3.6 Dataset Development and Preparation

To support algorithmic integrity and prevent data leakage, the collected data were divided into development datasets, which were used to build the system's components, and evaluation datasets, which were reserved for testing their performance.

3.6.1 Synthetically Generated Knowledge Base (RAG Development Data)

To populate the chatbot's localized context with highly structured, retrieval-friendly data, the researchers synthetically generated a 500-record knowledge base. This synthetic dataset was carefully modeled after the legacy BFP queries to ensure a realistic and comprehensive coverage of localized fire safety topics. Rather than being used for local model fine-tuning, the synthetic dataset was converted into Weighted keyword matching to populate the CSV-based BFP knowledge base . This dataset serves as the localized knowledge corpus used by the Gemini API Retrieval-Augmented Generation (RAG) architecture to ground generated responses in the fire-safety information represented in the knowledge base.

Table 1. Attributes of the Q&A Chat

ATTRIBUTES	DESCRIPTION	VALUES
id	Unique record identifier	Unique ID
Category	Main fire safety topic	String
Subcategory	Specific focus within category	String
Region	Philippine geographic region	String
Language	pair	String
user_question	Varied ways users ask the same question	Text
bot_answer	Standardized safety	Text

Table 1 continued…

ATTRIBUTES	DESCRIPTION	VALUES
	response from BFP	
source	Origin of the guidance	String
tags	Keywords for search/filtering	String
created_at	Record creation timestamp	Date

3.6.2 Chatbot Evaluation Dataset (Held-Out Test Data)

To evaluate the effectiveness of the Gemini API and RAG pipeline without introducing data leakage, a separate held-out testing dataset was collected. This dataset consists of curated, simulated fire safety questions paired explicitly with BFP-validated reference answers. By keeping this evaluation data completely separate from the synthetic vector knowledge base, the researchers can empirically test the chatbot's retrieval precision and response faithfulness against unseen community queries.

3.6.3 Adaptive Learning Training Data

To initialize the supervised machine learning models that drive the adaptive learning component of the Intelligent Tutoring System (ITS), two synthetic student-performance datasets were generated due to the absence of sufficient live learner interaction records prior to system deployment. Each dataset consisted of 5,000 simulated learner records containing variables representative of learner profiles and assessment performance.

The two synthetic datasets correspond to the two adaptive-learning prediction tasks. The first dataset, presented in Table 2, contains Age, Grade Level, and Pre-Assessment Score as predictor variables, with ModuleQuizDifficulty as the target variable. This dataset is used to train the Module Quiz Difficulty Random Forest Model,

which predicts whether the appropriate module quiz difficulty should be classified as Easy, Medium, or Hard based on the learner's baseline profile and pre-assessment performance. The second dataset, presented in Table 3, contains Age, Grade Level, Pre-Assessment Score, and Module 1 to Module 4 quiz scores, with FinalExamDifficulty as the target variable. This dataset is used to train the Final Examination Difficulty Random Forest Model, which determines an appropriate Easy, Medium, or Hard difficulty classification for the final examination based on the learner's accumulated assessment performance. The target difficulty labels were encoded into numerical representations before model training. Each modeling dataset was independently divided using an 80/20 train-test split with a fixed random state of 42, where 4,000 records (80%) were used for model training and 1,000 records (20%) were reserved as unseen testing data. This separation allows the predictive performance of the two Random Forest models to be evaluated independently.

Table 2. Attributes of the Pre-Assessment Score Dataset

ATTRIBUTES	DESCRIPTION	VALUES
Age	Age of the student user	Integer
GradeLevel	Academic grade level of the student user	Integer
PreTestScore	Score achieved by the student.	Integer
ModuleQuizDifficulty	Target difficulty level assigned by the algorithm for learning modules based on the student's baseline performance	String (Categorical)

Table 3. Attributes of the Synthetic Final Exam Score Dataset

ATTRIBUTES	DESCRIPTION	VALUES
Age	Age of the student	Integer
GradeLevel	Academic grade level of the student user	Integer
PreTestScore	Score achieved by the	Integer

Table 3 continued…

ATTRIBUTES	DESCRIPTION	VALUES
	student.	
module1Score	Raw score by the student on the quiz for module 1	Integer
module2Score	Raw score by the student on the quiz for module 2	Integer
module3Score	Raw score by the student on the quiz for module 3	Integer
module4Score	Raw score by the student on the quiz for module 4	Integer
FinalExamDifficulty	Target difficulty level assigned to the final exam by the adaptive learning algorithm based on accumulated learner assessment performance	String (Categorical)

3.7 Data Model Generation

This study aimed to design, develop, and evaluate an AI-driven responsive web portal equipped with an adaptive-learning engine and an intelligent chatbot. To achieve these objectives, the researchers followed a structured Machine Learning (ML) process covering data collection, data preparation, model development, model evaluation, and deployment. For the adaptive-learning component, two Random Forest classification models were developed: the Module Quiz Difficulty Model, which predicts the appropriate Easy, Medium, or Hard difficulty level for module quizzes based on learner profiles and pre-assessment performance; and the Final Examination Difficulty Model, which predicts final-examination difficulty based on the learner's cumulative quiz performance. In parallel, a Retrieval-Augmented Generation (RAG) architecture

powered by the Gemini API was configured and evaluated for the conversational chatbot.

3.7.1 Data Collection and Loading

The development required the ingestion of distinct datasets corresponding to the system's artificial intelligence components. For the conversational AI, the augmented 500-record Bureau of Fire Protection (BFP) knowledge base was loaded into the Python processing environment. For the adaptive learning engine, two synthetic student-performance datasets containing 5,000 records each were prepared using the Pandas data manipulation library: one for predicting module quiz difficulty from baseline learner profile and pre-assessment performance, and another for predicting final-examination difficulty from baseline learner information and cumulative module assessment performance.

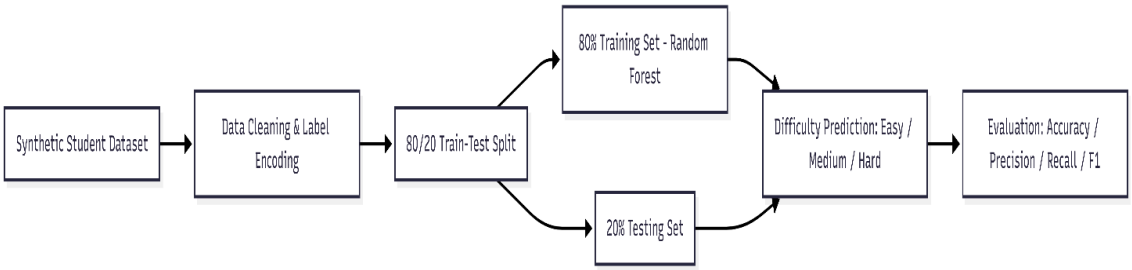


Figure 11. Adaptive Learning Pipeline (Classification)

Figure 11 illustrates the general supervised machine learning workflow applied to both Random Forest classifiers of the adaptive learning engine. The workflow begins with the synthetic student-performance data, which undergo data cleaning and label encoding to prepare the Easy, Medium, and Hard target classes. The prepared data for each prediction task are independently divided using an 80/20 train-test split with a fixed random state of 42. The Module Difficulty Model is trained using baseline learner variables to predict ModuleQuizDifficulty, while the Final Examination Difficulty Model is trained using baseline and accumulated module assessment scores to predict

FinalExamDifficulty. For each model, 80% of the corresponding data is used for training and 20% is reserved as unseen testing data. The resulting predictions are evaluated independently using Accuracy, Precision, Recall, and F1-Score.

Figure 12 illustrates the artificial intelligence architecture designed for the portal's conversational agent. The pipeline initiates by ingesting the foundational data source: the 500-record synthetic fire-safety knowledge base derived from BFP resources. To ensure factual grounding, the retrieval-augmented generation (RAG) framework processes this data through text chunking and dense vector embedding, storing the high-dimensional outputs within a specialized CSV-based BFP knowledge base. During live execution, the pipeline receives the user's query and performs semantic retrieval via Weighted relevance scoring to extract the most precise factual context. This retrieved context is then programmatically fed directly into the cloud-based Gemini API. This structure constrains the Gemini API to generate responses using the retrieved BFP context. Ultimately, the RAG chatbot is evaluated internally by the researchers/developers using a validated set of fire-safety questions and reference answers. The evaluation measures Response Accuracy, Contextual Relevance, Response Completeness, Knowledge Retrieval Effectiveness, and Hallucination/Error Rate to determine the reliability and grounding of the generated information.

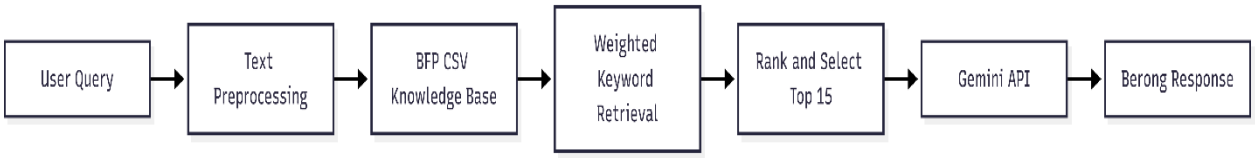


Figure 12. The Chatbot Pipeline (Retrieval-Augmented Generation)

3.7.2 Transformations and Splitting

To prepare the data for machine learning, distinct transformations were applied to the two artificial intelligence components. For the two Random Forest classification models, the student-performance data underwent cleaning and label encoding to convert

the target learning difficulty classes of Easy, Medium, and Hard into numerical representations. Each corresponding modeling dataset was then independently divided using an 80/20 train-test split through the Scikit-Learn library with a fixed random state of 42, with 4,000 records allocated for model training and 1,000 records reserved as unseen testing data for each model. For the Retrieval-Augmented Generation (RAG) pipeline, the textual data underwent semantic chunking and was converted into Weighted keyword matching before being stored in the CSV-based BFP knowledge base. Retrieved text chunks were then formatted into a structured prompt template and supplied to the Gemini API as contextual information for response generation.

3.7.3 Model Selection

Following data preprocessing, specific algorithms and models were selected to drive the platform's AI components based on their computational efficiency and task suitability. A tree-based classification architecture, specifically the Random Forest algorithm, was selected for the adaptive learning component because of its suitability for structured tabular data, its ability to model nonlinear relationships among learner variables, and its capability to perform multiclass classification of the Easy, Medium, and Hard assessment-difficulty categories. For the conversational AI component, the 500-record synthetic fire-safety knowledge base derived from validated BFP resources was prepared and stored in a CSV file. Unlike the adaptive learning component, which requires data transformation and splitting for Random Forest classification, the chatbot's knowledge base is processed through a weighted keyword-based retrieval mechanism. The system reads the dataset, extracts the question, answer, category, and subcategory fields, and uses these fields to identify relevant records based on the user's query. The retrieved records are subsequently supplied as contextual information to the Gemini API for response generation.

3.7.4 Model Making and Training

This section breaks down how the researchers configured and trained the two core AI engines for the enhanced Berong SafeScape Web Portal. A Retrieval-Augmented Generation (RAG) pipeline powered by the cloud-based Gemini API was configured for the intelligent conversational chatbot, while two Random Forest classifiers were trained for the adaptive learning engine. The data preprocessing and Random Forest model development were conducted in Google Colab, which provided a reproducible Python-based computational environment for data preparation, classification, and model evaluation.

3.7.4.1 Retrieval-Augmented Generation (RAG) Architecture Setup

The retrieval subsystem was implemented to serve as the localized knowledge provider for the conversational agent. It uses a 500-record synthetic fire-safety knowledge base derived from validated Bureau of Fire Protection (BFP) safety resources and stored in a CSV file. The system accesses the dataset through the Laravel backend to identify relevant information based on the user's inquiry.

During retrieval, the user's query undergoes text preprocessing, including lowercase conversion, punctuation removal, and stop-word filtering. The remaining keywords are compared against the question, answer, category, and subcategory fields of the dataset using a weighted keyword-matching approach. The scoring mechanism assigns three points for question matches, one point for answer matches, and two points for category or subcategory matches. The matching records are sorted according to their accumulated scores in descending order.

For online chatbot interactions, the system selects up to fifteen of the highest-ranking question-and-answer records and incorporates them into the prompt supplied

to the Gemini API. The prompt also includes system instructions and available recent conversation history to provide additional context for response generation.

When the Gemini API is unavailable, the system switches to its offline retrieval mechanism, which selects the highest-ranking matching record from the local dataset and displays its corresponding answer to the user. This allows the chatbot to provide locally stored fire-safety information even when the external generative service cannot be accessed.

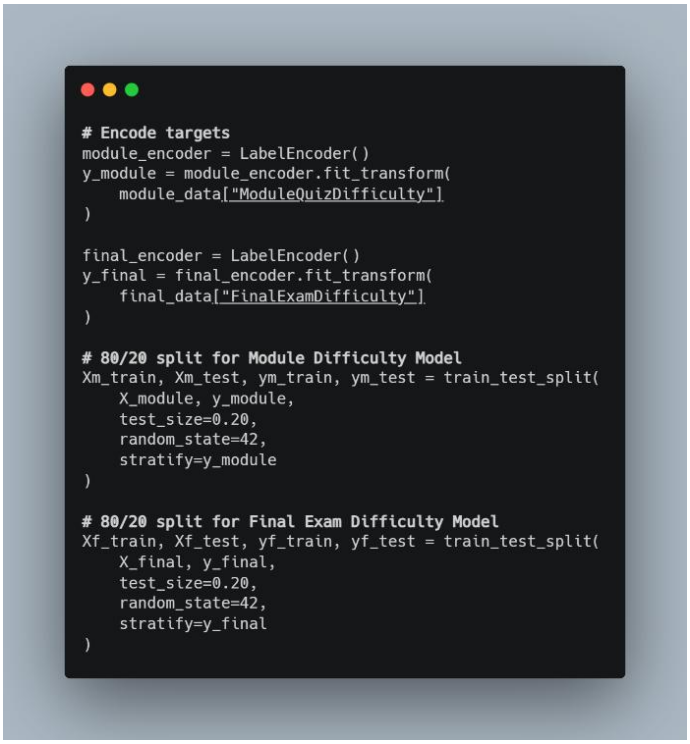
3.7.4.2 Gemini API Configuration

To ensure the model strictly adhered to the fire safety domain, advanced prompt engineering techniques were applied within the Laravel backend. The retrieved contextual chunks are injected into a structured instructional prompt that directs the Gemini API to prioritize the supplied BFP information when generating a response. The generation temperature was treated as a controlled configuration parameter. A low-temperature setting was selected during system configuration to reduce output variability, and the selected setting remained constant throughout the formal chatbot evaluation. The factual correctness of the resulting responses was independently evaluated using Response Accuracy, Contextual Relevance, Response Completeness, Knowledge Retrieval Effectiveness, and Hallucination/Error Rate.

3.7.4.3 Adaptive Learning Engine Data Preprocessing

The data preparation subsystem was implemented to clean and transform the student assessment records prior to classification. Separate feature sets and target variables were prepared for the two adaptive-learning models. For the Module Quiz Difficulty Model, Age, Grade Level, and Pre-Assessment Score were used as predictor variables, while ModuleQuizDifficulty served as the target variable. For the Final Examination Difficulty Model, Age, Grade Level, Pre-Assessment Score, and Module 1 to Module

4 quiz scores were used as predictor variables, while FinalExamDifficulty served as the target variable. Using the Scikit-Learn preprocessing framework, LabelEncoder was applied independently to the target variables of both datasets to convert the Easy, Medium, and Hard difficulty classes into numerical labels. Each dataset was subsequently divided using an 80/20 train-test split with a fixed random state of 42. For each 5,000-record dataset, 4,000 records were allocated for training, while the remaining 1,000 records were reserved as unseen testing data.



```
# Encode targets
module_encoder = LabelEncoder()
y_module = module_encoder.fit_transform(
    module_data["ModuleQuizDifficulty"]
)

final_encoder = LabelEncoder()
y_final = final_encoder.fit_transform(
    final_data["FinalExamDifficulty"]
)

# 80/20 split for Module Difficulty Model
Xm_train, Xm_test, ym_train, ym_test = train_test_split(
    X_module, y_module,
    test_size=0.20,
    random_state=42,
    stratify=y_module
)

# 80/20 split for Final Exam Difficulty Model
Xf_train, Xf_test, yf_train, yf_test = train_test_split(
    X_final, y_final,
    test_size=0.20,
    random_state=42,
    stratify=y_final
)
```

Figure 13. Code snippet of the label encoding and train-test splitting configuration

3.7.4.4 Random Forest Models for Adaptive Learning Engine Training

Two Random Forest classifiers were implemented using the Scikit-Learn ensemble learning framework. The Module Quiz Difficulty Model was trained using the baseline learner feature set to predict ModuleQuizDifficulty, whereas the Final Examination Difficulty Model was trained using the cumulative learner-performance feature set to predict FinalExamDifficulty.



```
# Initialize the two Random Forest models
module_model = RandomForestClassifier(
    n_estimators=100,
    max_depth=10,
    random_state=42
)

final_exam_model = RandomForestClassifier(
    n_estimators=100,
    max_depth=10,
    random_state=42
)

# Train each model
module_model.fit(Xm_train, ym_train)
final_exam_model.fit(Xf_train, yf_train)

# Generate predictions
module_pred = module_model.predict(Xm_test)
final_exam_pred = final_exam_model.predict(Xf_test)
```

Figure 14. Code snippet of the two Random Forest classifiers' initialization and model fitting routines.

Both classifiers were configured using the same baseline hyperparameters: 100 decision trees (n_estimators=100), a maximum tree depth of 10 (max_depth=10), and a fixed random state of 42 (random_state=42). The .fit() function was applied independently to the corresponding training partitions of each model, after which the .predict() function generated difficulty classifications for their respective unseen testing partitions. Limiting the maximum tree depth helps control model complexity and reduce the risk of overfitting to the synthetic training data.

Table 4. Hyperparameter configuration for the Random Forest classification models.

HYPERPARAMETER	VALUE	DESCRIPTION
n_estimators	100	Determines the number of independent decision trees generated in the ensemble to ensure stable majority-voting.

Table 4 continued…

HYPERPARAMETER	VALUE	DESCRIPTION
max_depth	10	Restricts the maximum structural depth of each tree to help reduce the risk of overfitting by limiting excessive tree growth.
random_state	42	Establishes a fixed random seed to support reproducibility across model training executions.

3.7.5 Model Evaluation

The two artificial intelligence components of the system were evaluated according to their respective computational tasks and performance requirements. For the supervised-learning-based Random Forest models, the evaluation focused on multiclass classification performance using Accuracy, Precision, Recall, F1-Score, and confusion matrices. For the Gemini API Retrieval-Augmented Generation (RAG) chatbot, the evaluation used controlled researcher/developer verification against validated fire-safety reference answers and retrieved knowledge resources to assess Response Accuracy, Contextual Relevance, Response Completeness, Knowledge Retrieval Effectiveness, and Hallucination/Error Rate.

3.7.5.1 Gemini API RAG Chatbot Evaluation Metrics

Because the reliability of a fire-safety chatbot cannot be adequately established solely through automated linguistic similarity measures, the Gemini API Retrieval-Augmented Generation (RAG) chatbot underwent a dedicated internal performance evaluation conducted by the researchers/developers. A curated test set containing fire-safety questions and corresponding validated reference answers was used as the basis for verification.

For each test query, the generated chatbot response was compared with the corresponding validated reference answer. Response Accuracy measured whether the substantive fire-safety information agreed with the validated reference answer. Contextual Relevance measured whether the response directly addressed the submitted question without introducing unrelated information. Response Completeness determined whether the response contained the essential information required by the reference answer. Knowledge Retrieval Effectiveness determined whether the knowledge segment retrieved by the RAG pipeline was relevant to the submitted query and supported the generated response. Hallucination/Error Rate measured the proportion of responses containing unsupported, fabricated, misleading, or factually incorrect fire-safety information.

The internal RAG performance measures were computed from the completed researcher/developer test records. The separate BFP personnel evaluation provided complementary professional domain validation of the factual accuracy, contextual relevance, completeness, and operational appropriateness of the chatbot's fire-safety information.

3.7.5.2 Random Forest (Adaptive Learning Engine) Evaluation Metrics

The two Random Forest classification models were evaluated independently according to their respective prediction tasks. The Module Quiz Difficulty Model was evaluated based on its ability to correctly classify ModuleQuizDifficulty, while the Final Examination Difficulty Model was evaluated based on its ability to correctly classify FinalExamDifficulty. Accuracy, Precision, Recall, and F1-Score were computed using the corresponding unseen testing partitions. Precision, Recall, and F1-Score were reported for the Easy, Medium, and Hard classes together with macro-averaged and weighted-averaged results.

Accuracy: measures the proportion of test instances that are correctly classified by the Random Forest model. A higher accuracy indicates a greater proportion of correct learning difficulty predictions across the testing dataset.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Equation 1. Accuracy

Where TP, TN, FP, and FN represent the number of True Positive, True Negative, False Positive, and False Negative classifications, respectively.

Precision: Measures the proportion of instances predicted as a particular difficulty class that actually belong to that class.

$$Precision = \frac{TP}{TP + FP}$$

Equation 2. Precision

Recall: Measures the proportion of instances belonging to a particular difficulty class that are correctly identified by the model.

$$Recall = \frac{TP}{TP + FN}$$

Equation 3. Recall

F1-Score: Represents the harmonic mean of Precision and Recall and provides a balanced measure when both false-positive and false-negative classifications are important.

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

Equation 4. F1-Score

3.7.6 Model Deployment

The final phase of the machine-learning process involved integrating the evaluated artificial-intelligence components into the developed web application. The Random Forest classification engine was deployed as a local Python-based backend service communicating with the PHP Laravel 11 architecture. The conversational chatbot was integrated through secure API requests to the cloud-based Gemini API, with the Laravel backend coordinating Weighted keyword-based retrieval and prompt construction. During operation, the application sends the required learner variables to the adaptive-learning service and forwards user fire-safety queries together with retrieved contextual information to the Gemini API. The integrated services return assessment-difficulty classifications and context-grounded fire-safety responses to the application interface. Actual response latency and system behavior were subsequently measured through the technical evaluation. (Parallag et al., 2023).

3.8 System Development Methodology

The developed Berong SafeScape Web Portal underwent systematic internal technical testing and external stakeholder evaluation to determine whether the implemented components satisfied the identified functional, technical, artificial-intelligence, accessibility, software-quality, domain-specific, and user-oriented requirements. Internal system testing was conducted solely by the researchers/developers using controlled test accounts, controlled records, and non-production, anonymized, or aggregate data whenever applicable. No external evaluator or Information Technology expert executed the internal System Testing and Verification Tool. Following internal verification and corrective action, the system underwent separate ISO/IEC 25010:2023 software-quality evaluation by Information

Technology experts, domain and operational validation by Bureau of Fire Protection personnel, and usability and acceptability evaluation by community end-users.

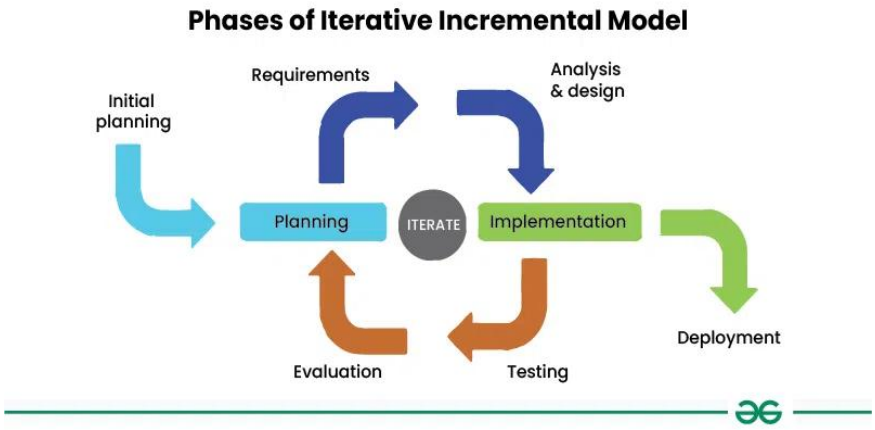


Figure 15. The Phases of the Iterative and Incremental Model. (Iterative Incremental Model in Designing System, 2024)

The application of this methodology followed structured phases during each development cycle. The process began with initial planning, during which the overall project vision was established. Each subsequent iteration involved a recurring sequence of the following phases:

1. **Requirements Analysis:** Each development cycle began with an analysis of the core functionalities required for the next system increment. This phase involved defining the technical requirements for migrating the legacy infrastructure to an optimized PHP Laravel architecture to address the identified performance limitations. It also involved establishing the dataset requirements for the Retrieval-Augmented Generation (RAG) chatbot and considering relevant DICT Government Web Hosting Service (GWHS) requirements when defining the architectural, security, and deployment-related requirements of the enhanced system.

2. **System Design:** Based on the established requirements, the architectural designs and user interfaces were formulated before implementation. This phase involved structuring the relational database schema, configuring the Model-View-Controller (MVC) routing logic, and designing the user interfaces. Attention was given to an accessible, child-friendly UI/UX design to help intended users navigate the educational safety modules.
3. **System Development:** During this phase, the proposed designs were translated into executable code. The researchers developed the PHP Laravel backend, integrated the Web Speech API to provide Speech-to-Text (STT) and Text-to-Speech (TTS) capabilities, and implemented the computational logic for the administrative dashboard and adaptive-learning engine.
4. **Integration:** After the individual system components were developed, they were combined into a unified environment. This phase focused on establishing secure and efficient communication among the different technological components. Specifically, the PHP Laravel web architecture was connected to the Python-based artificial intelligence services to allow data and API requests to be processed through the integrated system.
5. **Testing:** The developed system components underwent testing to identify defects and assess their performance. Functional testing and load testing were conducted to examine whether the Laravel architecture could process database queries under the specified test conditions. The RAG chatbot powered by the Gemini API was also evaluated using controlled fire-safety questions to determine whether its responses were supported by the knowledge resources used in the study.

6. **Deployment:** Following internal system testing and correction of identified defects by the researchers/developers, the completed software increment was deployed within the defined scope of the study and subjected to stakeholder evaluation. Ten (10) Information Technology experts assess the selected ISO/IEC 25010:2023 software quality characteristics through the IT Expert System Evaluation Form. Bureau of Fire Protection personnel perform domain and operational validation of the fire-safety information and AI-assisted functions, while community end-users assess usability, accessibility, ease of interaction, and overall acceptability. The resulting evaluation findings and supplementary comments are used to identify areas for continued system refinement.

3.9 Software Tools Used

The development and training of the artificial intelligence engines for the adaptive learning system and conversational chatbot were conducted primarily in a Python-based environment. Google Colab was selected as the computational environment because it provides a convenient and reproducible Python-based platform for data preprocessing, model development, and evaluation.

Table 5. Software Tools for AI Model Development

SOFTWARE	DESCRIPTION
Google Colab	Used as the computational environment for data engineering, algorithm experimentation, and training the Random Forest model.
Python	Used to script data preprocessing, dataset preparation, algorithm experimentation, and the development and evaluation of the Random Forest adaptive learning models.

Table 5 continued…

SOFTWARE	DESCRIPTION
Scikit-Learn	Used for label encoding, dataset splitting, classification, model training, and evaluation of the Random Forest adaptive learning model.
Pandas	Used for dataset loading, cleaning, transformation, and preparation of the structured student-performance tabular data.
Google Gemini API	Serves as the core generative engine for the chatbot, synthesizing responses based strictly on the context provided by the retrieval pipeline.
Laravel Cache	Used to temporarily store the parsed CSV knowledge base, reducing repeated file-reading operations during chatbot retrieval.

The main web platform, which serves as the user-facing hub for the fire safety portal, was developed using a modern, server-side rendered architecture. The tools listed in Table 6 were chosen to build a highly responsive, scalable frontend and a robust backend capable of handling advanced administrative dashboards, user analytics, and API integrations with the AI engines.

Table 6. Software Tools for Web Development

SOFTWARE	DESCRIPTION
Antigravity IDE	Coding, debugging, project file management, and maintenance of PHP, Laravel, React, JavaScript, and configuration files.
PHP	Development of server-side logic, controllers, models, validation, authentication, middleware, and API communication.
Laravel 11	Migration of the legacy system to an MVC architecture; handling routing,

Table 6 continued…

SOFTWARE	DESCRIPTION
	authorization, database operations, and backend services.
React	Development of reusable, interactive interfaces for the portal, chatbot, and administrative dashboard. Its implementation directly supports the user-centered and kid-friendly UI/UX requirement.
Redis & Laravel Octane	These tools were integrated to optimize application performance. Redis served as an advanced caching mechanism, while Laravel Octane supercharged the application server, ensuring the platform remained fast and responsive even with concurrent users accessing the AI features.
Web Speech API	Implementation of Speech-to-Text for spoken queries and Text-to-Speech for reading information aloud. This explicitly supports the study's objective of improving accessibility and reducing
Inertia.js	Integration of Laravel routes with React components, managing page navigation and server-to-client data delivery without full page reloads.
Tailwind CSS	Development of responsive, accessible, and user-centered layouts across mobile and desktop interfaces, supporting the kid-friendly design requirements.
PostgreSQL	Secure storage and management of users, content, learning records, assessments, feedback, chatbot interactions, and administrative data.
Git	Tracking source-code changes, managing branches, and supporting iterative development during the legacy migration.
Github	Hosting the source code, documenting iterative changes, and maintaining the version history of the evolving system.

3.10 System Architecture

The system architecture of the AI-driven responsive web portal is structurally anchored on the Model-View-Controller (MVC) design pattern, augmented by specialized artificial intelligence microservices and external cloud APIs. This architecture strictly separates the application's presentation layer, business logic, and data management, ensuring a highly scalable, secure, and maintainable platform for fire safety dissemination. The system is fundamentally divided into distinct, interconnected operational layers: the User Interaction Layer, the Web Application Framework, the Database Layer, the AI Service Architecture, and External Support Services.

The Presentation Layer, or the View, is the user-facing environment developed using React integrated with Inertia.js. This layer is responsible for delivering a dynamic, single-page application experience across various device form factors without the latency of traditional page reloads. It renders the specialized user interfaces for local residents, Bureau of Fire Protection (BFP) personnel accessing specialized training videos and content, and the kid-friendly educational dashboard. Furthermore, this layer handles client-side input validation and integrates the Web Speech API to provide native Text-to-Speech and Speech-to-Text accessibility features for younger learners navigating the interactive modules.

The Application Layer functions as the Controller and is driven by the Laravel 11 framework. Operating on the server side, this layer dictates the core business logic of the portal. It intercepts all HTTP requests from the React frontend, manages secure user authentication, handles session states, and coordinates the routing of data. Crucially, the Laravel backend acts as the central communication bridge, receiving user inputs and operational triggers from the frontend and securely passing them to the appropriate database or external artificial intelligence processing pipelines.

The Data Management Layer constitutes the Model aspect of the architecture and utilizes a dual-database strategy to handle diverse data types efficiently. A PostgreSQL relational database is employed to manage structured system data, including encrypted user credentials, community analytics, and the continuous tracking of student performance metrics across the educational modules. Parallel to this, a specialized CSV-based BFP knowledge base is deployed to store the high-dimensional Weighted keyword matching generated from the augmented BFP knowledge base, optimized specifically for rapid similarity search operations.

The Artificial Intelligence and External Services Layer operates as an integrated environment where complex data processing and generation occur. When a student interacts with the educational component, the Laravel backend transmits the required learner profile and assessment performance variables to the Python-based adaptive learning classification engine. The engine returns an Easy, Medium, or Hard difficulty classification used to determine the appropriate module quiz difficulty or final-examination difficulty according to the applicable prediction task. Concurrently, when a user queries the platform’s conversational agent, the backend routes the prompt to the Retrieval-Augmented Generation (RAG) pipeline. The chatbot uses a localized CSV-based BFP knowledge base containing 500 synthetic fire-safety question-and-answer records. The Laravel backend accesses the dataset and applies weighted keyword matching to retrieve relevant information based on the user's inquiry. The retrieved

records are then supplied to the Gemini API as contextual information for response generation.

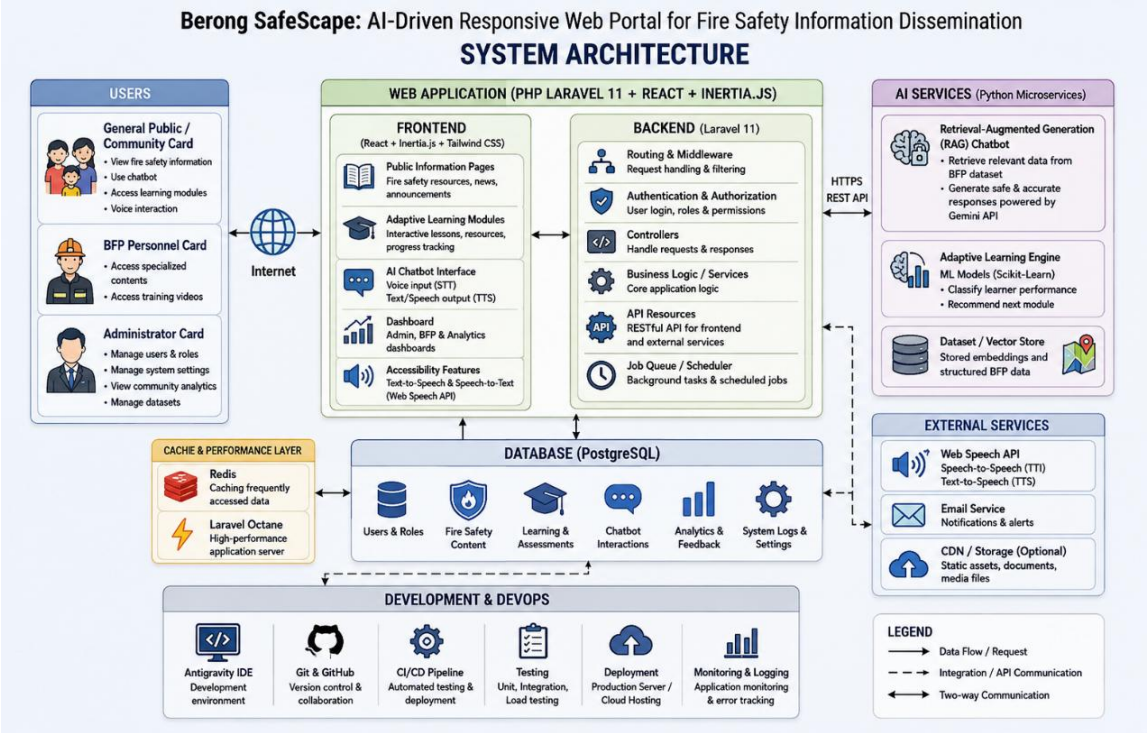


Figure 16. The overall system architecture of the AI-driven responsive web portal.

Note. Developed by the researchers based on the study's system architecture. The system architecture was visualized using the image-generation capabilities of Gemini by Google (2026).

3.11 Software Testing

The developed Berong SafeScape Web Portal underwent systematic internal technical testing and external stakeholder evaluation to determine whether the implemented components satisfied the identified functional, technical, artificial-intelligence, accessibility, software-quality, domain-specific, and user-oriented requirements. Internal system testing was conducted solely by the researchers/developers using controlled test accounts, controlled records, and non-production, anonymized, or aggregate data whenever applicable. No external evaluator

or Information Technology expert executed the internal System Testing and Verification Tool. Following internal verification and corrective action, the system underwent separate ISO/IEC 25010:2023 software-quality evaluation by Information Technology experts, domain and operational validation by Bureau of Fire Protection personnel, and usability and acceptability evaluation by community end-users.

3.11.1 System Testing and Evaluation Participants

The testing and evaluation process distinguished between internal technical testing conducted by the researchers/developers and external stakeholder evaluation conducted by selected participants. The researchers/developers were solely responsible for executing the Individual Researcher/Developer System Testing and Verification Tool, recording results, identifying defects or deviations, applying corrective actions, and conducting retesting when necessary.

A total of 75 external participants took part in the stakeholder evaluation, consisting of ten (10) Information Technology experts, fifteen (15) Bureau of Fire Protection personnel, and fifty (50) community end-users. The participants were selected purposively according to the requirements of their respective evaluation roles.

The ten IT experts completed only the ISO/IEC 25010:2023 Software Quality Evaluation. The fifteen BFP personnel performed domain and operational validation of the fire-safety information, chatbot responses, educational content, and administrative functions. The fifty community end-users evaluated usability, accessibility, ease of navigation and interaction, perceived usefulness, satisfaction, and overall acceptability. No external participant executed the researchers' internal technical test cases.

This separation of responsibilities ensures that internal technical testing remains under the control of the researchers/developers while external participants provide independent software-quality, domain-specific, and end-user assessments.

3.11.2 Artificial Intelligence (AI) Chatbot Testing

The Gemini API Retrieval-Augmented Generation (RAG) chatbot underwent dedicated internal testing conducted by the researchers/developers using a validated set of fire-safety questions derived from BFP-approved or otherwise authoritative fire-safety knowledge resources. For each test query, the generated response was compared with its corresponding validated reference answer.

Response Accuracy was calculated as the proportion of factually correct responses relative to the total number of test questions. Contextual Relevance and Response Completeness were rated by the researchers/developers through direct comparison of each generated response with the corresponding validated reference answer. Knowledge Retrieval Effectiveness was determined by identifying whether the knowledge segment retrieved by the RAG pipeline was relevant to the submitted query and adequately supported the generated answer. Hallucination/Error Rate was calculated as the proportion of responses containing unsupported, fabricated, misleading, or factually incorrect information. The resulting internal RAG test measures were reported separately from the BFP personnel evaluation. BFP ratings served as complementary professional domain validation rather than as a substitute for the controlled researcher/developer RAG test.

3.11.3 Voice-Enabled Accessibility Testing (STT & TTS)

The Web Speech API integration underwent internal technical testing conducted by the researchers/developers. Speech-to-Text (STT) was tested using predefined spoken utterances representing realistic fire-safety and emergency-interaction scenarios. Recognition Accuracy was calculated by dividing the number of correctly transcribed utterances by the total number of test utterances and multiplying the result by 100. Word Error Rate (WER) was calculated using the number of substitutions,

deletions, and insertions relative to the total number of words in the reference transcription.

Text-to-Speech (TTS) was evaluated through researcher/developer functional test cases that assessed whether readable content was converted into understandable speech, long passages were read without inappropriate interruption, relevant fire-safety terminology was pronounced reasonably, playback controls functioned correctly, and TTS operated appropriately in the supported browser and device environments defined by the study.

These internal technical results were reported separately from community end-user ratings of perceived accessibility, usability, and ease of interaction.

3.11.4 Adaptive Learning and Dashboard Testing

The two Random Forest classifiers of the adaptive learning engine were evaluated independently using their unseen testing partitions and controlled system test scenarios. The Module Quiz Difficulty Model was evaluated based on its ability to correctly classify Easy, Medium, or Hard module quiz difficulty, while the Final Examination Difficulty Model was evaluated based on its ability to correctly classify Easy, Medium, or Hard final-examination difficulty. Accuracy, Precision, Recall, F1-Score, per-class results, and confusion matrices were generated from the model predictions.

In addition to model-level evaluation, the researchers/developers executed controlled adaptive-learning system test scenarios using designated test accounts to verify that the predicted difficulty class was correctly reflected by the assessment presented by the portal.

The administrative dashboard was likewise tested internally by the researchers/developers. Displayed analytics were cross-referenced against authorized

aggregate counts or controlled test records to verify data consistency without exposing personally identifiable information belonging to actual users.

3.11.5 Functional, Performance, and Security Testing

The researchers/developers conducted the internal functional, integration, performance, and security testing of the Laravel 11 web application. Functional testing verified whether the major system features operated according to their defined requirements using valid, invalid, boundary, and favorable test conditions. Integration testing verified the proper exchange of data among Laravel, PostgreSQL, the Gemini API/RAG service, and the Random Forest classification service.

Performance testing measured actual system response times and error rates under the researcher-defined test loads and acceptance criteria specified in the testing instrument. This included single-user requests, chatbot response time, dashboard access, database retrieval, simulated concurrent access, and mobile-access conditions. Security testing verified authentication, authorization, input handling, protected routes, session behavior, and other specified security conditions through controlled manual test scenarios.

All internal system tests used designated test accounts and test records. Existing users' private records were not used as test data except where authorized aggregate information was required for non-identifying dashboard verification.

3.11.6 ISO/IEC 25010:2023 Software Quality Evaluation

Following completion of the internal technical testing and correction of identified critical defects, the developed system underwent an external software quality evaluation involving ten (10) qualified Information Technology experts. The IT experts completed the IT Expert System Evaluation Form aligned with selected characteristics of the ISO/IEC 25010:2023 software product quality model.

The selected characteristics were Functional Suitability, Performance Efficiency, Reliability, Security, Maintainability, and Flexibility. The IT experts provided their professional ratings based on their technical assessment and interaction with the developed portal. Their role was limited to the ISO/IEC 25010:2023 software quality evaluation and did not include execution of the internal System Testing and Verification Tool.

BFP domain and operational validation and community end-user evaluation were conducted separately through their respective evaluation instruments.

3.11.7 Testing and Validation Sequence

The testing and evaluation process followed a sequential procedure. It began with Internal System Testing, during which the researchers/developers executed the functional, integration, AI chatbot, Speech-to-Text, Text-to-Speech, accessibility and user-task, adaptive-learning, administrative-dashboard, performance, and security test cases using controlled test accounts and records. Any identified defect or deviation was documented, corrected, and subjected to retesting when necessary.

Following satisfactory internal technical verification, the system proceeded to Software Quality Evaluation, wherein ten (10) Information Technology experts assessed the selected ISO/IEC 25010:2023 characteristics using the IT Expert System Evaluation Form.

The succeeding stage consisted of Domain and Operational Validation, wherein fifteen (15) BFP personnel evaluated the factual accuracy, contextual relevance, completeness, operational appropriateness, and domain relevance of the fire-safety information and AI-assisted functions.

Finally, Community End-User Evaluation was conducted with fifty (50) community end-users to assess usability, accessibility, ease of interaction, perceived usefulness, satisfaction, and overall acceptability.

3.11.8 Data Analysis

Quantitative data obtained from the external evaluation instruments completed by the ten (10) Information Technology experts, fifteen (15) BFP personnel, and fifty (50) community end-users were organized, tabulated, and summarized using appropriate descriptive statistics. Frequency and percentage distributions were used where applicable to describe participant and response distributions, while mean scores and standard deviations were used to summarize the corresponding evaluation ratings.

Internal technical testing records generated by the researchers/developers were analyzed separately from the external stakeholder evaluation results. For applicable functional, integration, accessibility, dashboard, Text-to-Speech, and security test cases, pass/fail results and corresponding pass rates were summarized. Performance testing reported actual response time, error rate, and behavior under the specified test loads.

Speech-to-Text performance was analyzed using Recognition Accuracy and Word Error Rate (WER). The RAG chatbot was analyzed using Response Accuracy, Contextual Relevance, Response Completeness, Knowledge Retrieval Effectiveness, and Hallucination/Error Rate based on the controlled researcher/developer verification records. The two Random Forest adaptive-learning models were evaluated independently using Accuracy, Precision, Recall, F1-Score, per-class, macro-averaged, weighted-averaged results, and confusion matrices for the Easy, Medium, and Hard classes.

Open-ended comments and recommendations obtained from the IT expert ISO evaluation, BFP personnel evaluation, and community end-user evaluation were reviewed and summarized to identify commonly mentioned recommendations. These responses remained supplementary to the primary quantitative and technical findings and did not constitute a separate qualitative or thematic analysis.

Overall, the methodology combined researcher-conducted internal technical testing, ISO/IEC 25010:2023 software-quality evaluation by ten (10) IT experts, BFP domain and operational validation, and community end-user evaluation to determine whether the developed Berong SafeScape Web Portal satisfied its functional, technical, software-quality, domain-specific, accessibility, and user-acceptability requirements.

CHAPTER IV

RESULTS AND DISCUSSION

This chapter presents and discusses the results of the development, internal technical testing, and external evaluation of SafeScape V2: An AI-Driven Responsive Web Portal for Fire Safety Information Dissemination. The findings are organized according to the research objectives and the evaluation procedures established in Chapter III.

Internal technical testing was conducted by the researchers/developers using controlled test accounts, test records, and the prescribed technical measures for the developed system. External evaluation involved 75 participants, consisting of 50 community end-users, 15 Bureau of Fire Protection (BFP) personnel, and 10 Information Technology (IT) experts. The community end-users assessed the usability, accessibility, ease of interaction, and overall acceptability of the system. The BFP personnel evaluated the factual accuracy, contextual relevance, completeness, and operational relevance of the fire-safety information and AI-assisted functions, while the ten IT experts assessed the selected software quality characteristics based on ISO/IEC 25010:2023. The chapter also presents the technical results of the RAG chatbot, voice-enabled functions, adaptive learning models, administrative dashboard, performance, security, and other internal system tests.

4.1 Results of Research Objective 1 – Laravel Migration and System Performance

Research Objective 1 focused on restructuring and migrating the existing Berong SafeScape Web Portal to an optimized PHP Laravel architecture and evaluating the resulting system in terms of performance, scalability, security, maintainability, and alignment with relevant Department of Information and Communications Technology (DICT) Government Web Hosting Service (GWHS) requirements.

The enhanced SafeScape V2 was implemented using Laravel 11 for backend routing, authorization, database operations, and application services. The user interface was developed using React and integrated with Laravel through Inertia.js, while PostgreSQL was used for persistent system data. Redis and Laravel Octane were also incorporated as performance-related mechanisms for caching and application-server processing.

4.1.1 Implementation of the Enhanced Laravel Architecture

The restructuring of SafeScape V2 established a Model-View-Controller-based application architecture in which presentation, application logic, and data-management responsibilities were separated into organized components. Laravel 11 provides the primary backend framework, while React supports reusable interface components and Inertia.js provides communication between Laravel routes and the React-based interface without requiring conventional full-page reloads. PostgreSQL manages user, educational, assessment, chatbot, feedback, and administrative records.

Redis and Laravel Octane were incorporated into the developed architecture to support caching and application processing under concurrent access. These technologies form part of the optimization mechanisms implemented in the enhanced portal rather than constituting independent measures of performance improvement over the previous implementation.

Figure 17 presents the implemented user-facing interface of SafeScape V2. The portal provides access to the system's fire-safety educational environment through a responsive interface containing centralized navigation, user controls, fire-safety information, and access to the system's learning and administrative functions.

The use of a responsive and user-centered interface is consistent with the accessibility and digital-service considerations discussed by Barrios and Moreno

(2024), Jou et al. (2024), and Bokingkito et al. (2025). These studies provide context for designing public-facing digital services that consider usability, accessibility, and differences in users' access to technology. The figure presents the implemented interface of SafeScape V2 rather than an independent measurement of user satisfaction.

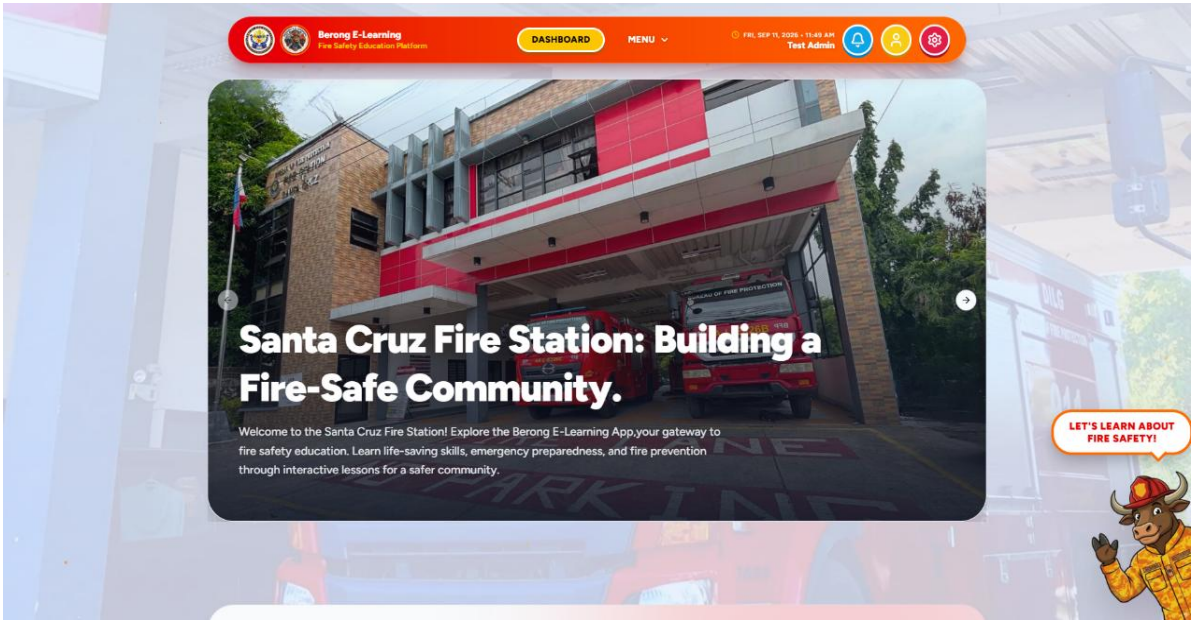


Figure 17. Implemented SafeScape V2 Web Portal

The completed portal demonstrates the implementation of the enhanced architecture described in Chapter III. However, because the study did not conduct a controlled quantitative benchmark of the previous Berong SafeScape implementation, the succeeding technical results describe the measured performance of SafeScape V2 itself and should not be interpreted as a quantified percentage improvement over the previous portal.

4.1.2 Performance and Scalability Results

The performance of SafeScape V2 was evaluated through ten predefined scenarios covering the major functions of the web portal. As shown in Table 7, all ten scenarios met their corresponding acceptance criteria, with recorded response times

ranging from 0.15 seconds to 1.40 seconds and an error rate of 0% across the tested scenarios. Search and simulated mobile access recorded the lowest response times at 0.15 seconds, while the administrative dashboard recorded the highest at 1.40 seconds. The chatbot recorded a response time of 1.39 seconds, which remained within its predefined seven-second acceptance criterion.

Table 7. Summary of Performance and Scalability Results Relevant to Research Objective 1

PERFORMANCE MEASURE	RESULT
Total Performance Test Scenarios	10
Scenarios Meeting Acceptance Criteria	10/10
Recorded Response-Time Range	0.15–1.40 seconds
Error Rate Across Recorded Scenarios	0%
Simulated Concurrent Users	50 users
Average Response Time at 50-User Load	0.54 seconds
Error Rate at 50-User Load	0%

The system's scalability was further examined through a controlled concurrent-load test involving 50 simulated users. Under this condition, SafeScape V2 recorded an average response time of 0.54 seconds and an error rate of 0%. This result indicates that the portal maintained responsive behavior under the specific concurrent-user condition established for the study.

The findings are consistent with Saputri et al. (2024), who examined performance optimization in a Laravel-based web application and emphasized the importance of evaluating response time under varying workloads. Kundan and Makwana (2025) also discussed database-query and API-response optimization techniques in Laravel-based applications, while Erdelt (2024) examined the use of established database performance benchmarks in cloud-native environments. These studies provide supporting context for evaluating application response time, database behavior, and performance under defined workloads. However, the present results describe SafeScape V2 under the specific testing environment and workload used in this study and do not establish equivalent performance under different hosting conditions or larger workloads.

Overall, the results demonstrate that all predefined performance scenarios satisfied their acceptance criteria and that the system remained responsive during the 50-user simulated load test. Nevertheless, these findings should be interpreted within the specified testing conditions. They do not establish unlimited scalability or guarantee identical performance under different hosting environments, network conditions, or larger production workloads.

4.1.3 Security and Maintainability Evidence

Security was evaluated separately through eight predefined internal test cases covering authentication, authorization, input validation, session management, data protection and access control, file-upload validation, and controlled error handling.

All eight security test cases passed, producing a 100.00% internal security-test pass rate. The tests confirmed that invalid login attempts were rejected, administrator-only functions were protected from unauthorized users, protected information could not

be accessed through the tested URL or parameter-manipulation scenarios, unsupported file types were rejected, sessions required reauthentication after logout, and controlled errors did not expose sensitive application or database information.

These findings demonstrate that the security controls included in the predefined internal test instrument operated according to their intended behavior. However, the testing represents controlled internal verification and not a comprehensive penetration test; therefore, the result should not be interpreted as evidence that the deployed platform contains no undiscovered vulnerabilities.

Professional IT-expert evaluation provides additional evidence relevant to the architectural qualities examined under Research Objective 1.

Table 8. IT Expert Software-Quality Results Relevant to Research Objective 1

ISO/IEC 25010:2023 CHARACTERISTIC	MEAN	SD	INTERPRETATION
Performance Efficiency	3.83	0.36	Very Good / Agree
Reliability	4.40	0.34	Excellent / Strongly Agree
Security	4.63	0.29	Excellent / Strongly Agree
Maintainability	3.93	0.34	Very Good / Agree
Flexibility	3.83	0.28	Very Good / Agree

The IT experts rated Security at 4.63 (SD = 0.29) and Reliability at 4.40 (SD = 0.34), both within the Excellent / Strongly Agree interpretation. Performance Efficiency obtained 3.83 (SD = 0.36), Maintainability obtained 3.93 (SD = 0.34), and Flexibility obtained 3.83 (SD = 0.28), all interpreted as Very Good / Agree. The complete ISO/IEC 25010:2023 evaluation is reported in Section 4.5.2.

Table 8 presents the IT experts' evaluation of selected software-quality characteristics relevant to Research Objective 1. Among the five characteristics presented, Security received the highest mean of 4.63 (SD = 0.29), followed by Reliability with 4.40 (SD = 0.34). These results indicate that the IT experts gave favorable assessments of the system's security controls and dependable operation.

Performance Efficiency and Flexibility received the comparatively lowest mean scores, both obtaining 3.83. Maintainability received a mean of 3.93. Although these characteristics were still interpreted as Very Good / Agree, their results identify areas that may benefit from continued technical refinement. In particular, the supplementary comments of the IT experts also identified code organization and performance optimization as areas for improvement.

The results may be considered alongside previous studies on software architecture and quality evaluation. Ariyanto et al. (2024) examined Laravel-based REST API development and discussed the relationship between framework architecture, code organization, and application performance. Dandan et al. (2026) applied a software-quality model based on ISO/IEC 25010 in evaluating a local government information system, while Olipas et al. (2022) examined the quality of a web-based information management system using an established software-quality framework. These studies provide context for evaluating the organization and quality of web-based systems. In the present study, the IT experts' ratings represent their professional assessment of the selected software-quality characteristics and should be distinguished from the system's measured response times and internal security-test results.

The complete software-quality evaluation, including the other selected ISO/IEC 25010:2023 characteristic, Functional Suitability, is presented in Section 4.5.2. Overall, the results indicate favorable professional assessments while identifying Performance Efficiency, Maintainability, and Flexibility as areas for continued improvement.

4.1.4 Alignment with DICT GWHS Requirements

Relevant DICT Government Web Hosting Service requirements were considered as architectural and development-alignment criteria for SafeScape V2. The developed architecture incorporates structured application routing, authentication and authorization controls, validation mechanisms, access control, responsive cross-device presentation, data-management mechanisms, and controlled error handling.

These characteristics support the study's objective of developing a structured and secure public-facing web architecture. However, the study did not conduct a formal DICT GWHS certification or compliance assessment. Consequently, the findings should be interpreted as evidence that relevant GWHS considerations informed system development rather than as a declaration that SafeScape V2 has been formally certified as GWHS-compliant.

This distinction is important because the technical tests conducted in the study evaluate the behavior of the developed system under the researchers' predefined requirements, while formal government hosting compliance would require a separate assessment process.

4.1.5 Overall Findings for Research Objective 1

The results indicate that Research Objective 1 was achieved within the defined scope of the study. The existing Berong SafeScape portal was restructured into an enhanced Laravel 11-based system supported by React, Inertia.js, PostgreSQL, Redis, Laravel Octane, and related web-development technologies. The implemented

architecture provides a structured foundation for the portal's educational, artificial-intelligence, accessibility, and administrative components.

Technical performance testing showed that all 10 predefined performance scenarios met their acceptance criteria, with recorded response times ranging from 0.15 to 1.40 seconds and no recorded errors. Under the specific 50-user simulated concurrent-load test, the system achieved an average response time of 0.54 seconds with a 0% error rate.

All eight controlled security test cases also passed. Complementary IT-expert evaluation showed favorable results for Reliability and Security, while Performance Efficiency, Maintainability, and Flexibility remained within the Very Good / Agree category and were identified as areas where continued technical improvement may be beneficial.

Taken together, the findings provide evidence that the enhanced architecture satisfied the technical performance, controlled scalability, security, and software-quality requirements evaluated within the study. Relevant GWHS requirements were considered during development as alignment criteria; however, the results do not constitute a formal GWHS compliance certification.

Because no controlled quantitative baseline of the previous portal was available, the findings describe the measured behavior and quality of SafeScape V2 rather than a numerical before-and-after improvement over the previous implementation.

4.2 Results of Research Objective 2 – RAG Chatbot Evaluation

Research Objective 2 focused on the development, integration, and evaluation of a Retrieval-Augmented Generation (RAG)-based chatbot powered by the Gemini API and grounded in Bureau of Fire Protection knowledge resources. The chatbot was evaluated in terms of Response Accuracy, Contextual Relevance, Response

Completeness, Knowledge Retrieval Effectiveness, and Hallucination/Error Rate for automated fire-safety information dissemination, as specified in the research objectives.

SafeScape V2 implemented a localized retrieval pipeline using a 500-record fire-safety knowledge base derived from validated BFP safety resources and stored in a CSV file. During interaction, the user's query undergoes text preprocessing before being compared with the dataset through a weighted keyword-matching approach. The system assigns scores based on matches in the questions, answers, categories, and subcategories of the knowledge base. The highest-ranking records are then retrieved and supplied as contextual information to the Gemini API for response generation. This approach allows the chatbot to incorporate relevant localized fire-safety information into its generated responses.

4.2.1 Implementation of the RAG-Based BFP Assistant

The RAG-based conversational component was implemented through the BFP Assistant interface of SafeScape V2. Users may submit fire-safety questions through text or voice input, after which the system retrieves relevant fire-safety information from the localized knowledge base and supplies the retrieved context to the Gemini API for generation of the final response.

Figure 18 presents an actual interaction with the implemented BFP Assistant. In the example, the user asks how to respond when a small fire starts at night, and the chatbot returns a fire-safety response through the conversational interface. However, retrieval quality itself was evaluated separately through the controlled RAG testing rather than inferred solely from the displayed chatbot response.

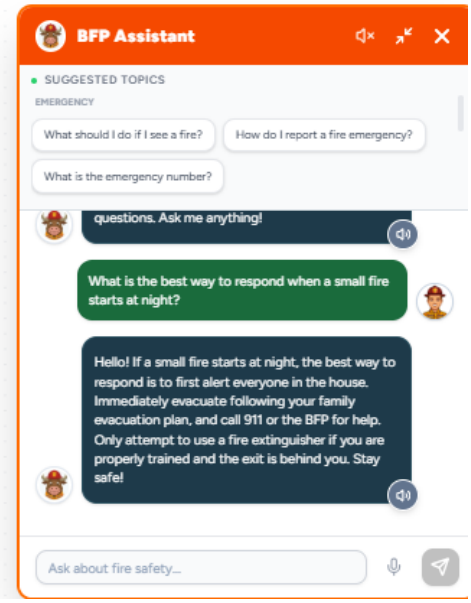


Figure 18. RAG-Based BFP Assistant of SafeScape V2

The implemented process follows the RAG pipeline established in Chapter III. The user's query undergoes text preprocessing and weighted keyword matching against the localized BFP knowledge base. The retrieved question-and-answer records are ranked according to their matching scores, and the highest-ranking entries are selected as contextual information. These retrieved records are combined with the user's original question, system instructions, and available conversation history before being supplied to the Gemini API for response generation. When the Gemini API is unavailable, the system uses its offline retrieval mechanism to provide the highest-ranking matching record from the local dataset.

The retrieval-based approach is consistent with the RAG evaluation framework discussed by Es et al. (2024), who emphasized the evaluation of retrieval and generated-response quality. Krishna et al. (2025) also discussed the importance of evaluating factuality, retrieval, and reasoning in RAG systems, while Knollmeyer et al. (2026) reviewed multiple dimensions for assessing RAG performance. These studies provide

context for the architecture and evaluation of the BFP Assistant. The figure itself illustrates an implemented interaction and should not be treated as proof of chatbot accuracy.

4.2.2 Internal Technical Evaluation of the RAG Chatbot

The RAG chatbot underwent controlled internal evaluation by the researchers/developers using 15 fire-safety test questions. The test set covered fire prevention, fire-extinguisher use, emergency response, evacuation procedures, LPG safety, electrical safety, children's fire safety, smoke safety, home fire safety, and community preparedness. Each response was examined for factual correctness, accuracy, contextual relevance, completeness, relevance of the retrieved knowledge, and the presence of unsupported or erroneous information.

Table 9. Technical Evaluation Results of the RAG Chatbot

EVALUATION MEASURE	RESULT
Total Test Queries	15
Factually Correct Responses	15
Response Accuracy	100.00%
Mean Accuracy Rating	4.60
Mean Contextual Relevance	4.60
Mean Response Completeness	4.33
Queries with Relevant Retrieved References	15
Knowledge Retrieval Effectiveness	100.00%

Table 9 continued…

EVALUATION MEASURE	RESULT
Responses with Identified Hallucination/Error	0
Hallucination/Error Rate	0.00%

Table 9 presents the results of the internal technical evaluation of the Retrieval-Augmented Generation (RAG) chatbot using 15 controlled fire-safety test queries. All 15 responses were classified as factually correct, resulting in a response accuracy of 100.00%. The chatbot also obtained a mean accuracy rating of 4.60 and a mean contextual relevance rating of 4.60. These results indicate that, within the conducted test, the chatbot provided responses that were assessed as factually accurate and relevant to the submitted fire-safety questions.

The findings also show that all 15 test queries were associated with relevant retrieved references, resulting in a knowledge retrieval effectiveness of 100.00%. Furthermore, no hallucination or error was identified among the evaluated responses, resulting in a hallucination/error rate of 0.00%. These results demonstrate that the chatbot was able to retrieve relevant information from its knowledge base and generate responses that were consistent with the expected fire-safety information during the controlled evaluation. This is consistent with Es et al. (2024), who explained that RAG evaluation should consider the retriever's ability to identify relevant context, the generator's ability to use that context faithfully, and the overall quality of the generated response. Similarly, Krishna et al. (2025) emphasized that evaluating RAG systems requires examining factuality, retrieval, and reasoning together rather than relying on a single measure of answer accuracy.

Among the three response-quality ratings, accuracy and contextual relevance obtained the highest mean scores of 4.60, while response completeness received the comparatively lowest mean score of 4.33. Although completeness was the lowest-rated dimension, its result remained relatively favorable compared with the other evaluated measures. This suggests that the chatbot's responses were generally accurate and relevant, but there may still be opportunities to provide more complete explanations or additional details when answering certain fire-safety questions. Knollmeyer et al. (2026) likewise highlighted the importance of evaluating RAG systems through multiple dimensions, including context relevance, faithfulness, answer relevance, and correctness, because each dimension examines a different aspect of system quality.

Overall, the results indicate that the RAG chatbot performed favorably in the 15 controlled test queries, particularly in factual accuracy, contextual relevance, and knowledge retrieval. However, the 100.00% accuracy and 0.00% hallucination/error rate should be interpreted within the scope of the test set and should not be considered proof that the chatbot will produce error-free responses to all possible user questions. These findings support the technical functionality of the RAG chatbot in providing relevant fire-safety information based on the tested knowledge base.

4.2.3 Supporting BFP Domain Validation

The internal technical results were complemented by professional domain validation from the 15 BFP personnel evaluators. Since the complete BFP evaluation is already presented in Section 4.5.3, only the chatbot-related results directly relevant to Research Objective 2 are summarized in this section.

Table 10 presents the results of the BFP domain validation of the RAG chatbot, conducted by 15 BFP personnel evaluators. The chatbot obtained an overall domain-

validation mean of 4.78 (SD = 0.31). Among the three evaluation criteria, Completeness received the highest mean of 4.82 (SD = 0.27), followed by Accuracy with 4.77 (SD = 0.36), while Contextual Relevance received a mean of 4.75 (SD = 0.35). These results indicate that the BFP evaluators gave favorable assessments of the chatbot's ability to provide complete, accurate, and contextually relevant fire-safety information.

Table 10. BFP Domain Validation Results Relevant to the RAG Chatbot

EVALUATION CRITERION	MEAN	SD
Accuracy	4.77	0.36
Contextual Relevance	4.75	0.35
Completeness	4.82	0.27
Overall Chatbot Domain Validation	4.78	0.31

The highest mean for Completeness suggests that the evaluators generally considered the chatbot's responses sufficiently detailed for the fire-safety questions assessed. Accuracy also received a favorable mean of 4.77, indicating positive professional assessments of the information presented. Contextual Relevance obtained the comparatively lowest mean of 4.75; however, the difference between the three criteria is small. Therefore, this result should not be interpreted as a major weakness but as an area that may still benefit from further refinement in matching responses to the specific situations and questions raised by users.

These findings are supported by Hostetter et al. (2024), who examined chatbot responses to fire-engineering questions involving fire prevention, evacuation, building-code compliance, and fire suppression. Their study demonstrated that chatbots can

provide useful responses to fire-safety-related questions, while also identifying areas that require improvement. Similarly, Hsu et al. (2024) evaluated ChatNetZero, a RAG-based chatbot, through source-based factual checking and expert surveys. Their findings emphasized that expert assessment of response quality and relevance is important, and that users may prefer responses that provide sufficient context and detail. In addition, Xie et al. (2025), through their development and evaluation of WildfireGPT, demonstrated the value of expert-led assessment when evaluating RAG-based systems intended for natural-hazard information and decision support.

The topic-level results further support the favorable evaluation of the chatbot. Fire Prevention and Fire Extinguisher Use both obtained an overall mean of 4.80, followed by LPG/Electrical Safety with 4.78 and Evacuation Procedures with 4.73. Although Evacuation Procedures received the lowest topic-level mean, its score remained close to the other evaluated topics. This indicates that the BFP evaluators generally viewed the chatbot's responses favorably across the fire-safety topics included in the validation.

Overall, the results provide professional domain-based support for the chatbot's response quality. Unlike the internal technical evaluation in Table 9, which examined the chatbot's performance through controlled test queries, Table 10 reflects the professional judgment of BFP personnel regarding the suitability of its fire-safety information. These findings provide complementary evidence that the developed RAG chatbot can support the dissemination of fire-safety information within the evaluated scope. However, the results represent the assessments of the 15 BFP evaluators and should not be interpreted as proof that every response will be correct or complete in all possible situations.

4.2.4 Overall Findings for Research Objective 2

The findings indicate that Research Objective 2 was achieved within the defined scope of the study. SafeScape V2 successfully implemented a Gemini API-powered Retrieval-Augmented Generation chatbot grounded through a localized weighted vector retrieval system based on BFP fire-safety knowledge resources.

Across the 15 controlled test queries, the chatbot achieved 100.00% Response Accuracy, 100.00% Knowledge Retrieval Effectiveness, and a 0.00% Hallucination/Error Rate. Mean ratings of 4.60 for Accuracy, 4.60 for Contextual Relevance, and 4.33 for Response Completeness further demonstrated favorable response quality under the controlled evaluation conditions.

Professional domain validation by the BFP personnel provided additional supporting evidence, with the chatbot receiving an overall mean of 4.78 (SD = 0.31) for Accuracy, Contextual Relevance, and Completeness. These findings indicate that the implemented RAG mechanism was capable of retrieving relevant fire-safety knowledge and supporting context-grounded responses for the evaluated questions.

Nevertheless, the findings remain dependent on the scope, quality, and completeness of the localized BFP knowledge base and the limited controlled question set used during technical testing. Accordingly, the results should not be generalized as evidence that the system meets predefined criteria for unrestricted or out-of-domain inquiries.

4.3 Results of Research Objective 3 – Voice-Enabled Accessibility and User Interaction

Research Objective 3 focused on the design, implementation, and evaluation of a voice-enabled accessibility framework incorporating Web Speech API-based Speech-to-Text (STT) and Text-to-Speech (TTS), age-appropriate educational content, and

user-centered UI/UX design. The objective further examined the system in terms of usability, accessibility, ease of interaction, and user engagement.

The evaluation combined controlled technical testing of the voice-enabled functions with supporting community end-user results. The detailed community evaluation is presented in Section 4.5.4; therefore, only the findings directly relevant to Research Objective 3 are summarized in this section.

4.3.1 Voice-Enabled Accessibility Functions

SafeScape V2 integrated Speech-to-Text and Text-to-Speech within the BFP Assistant. The microphone control allows users to provide spoken input, while the speaker control allows chatbot responses to be delivered through audible speech. These functions provide alternative interaction methods for users who may find conventional text input or reading less convenient.

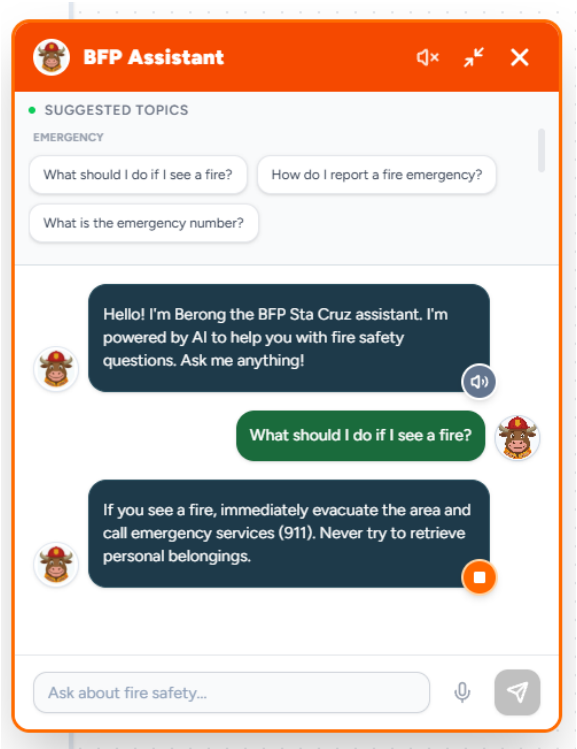


Figure 19. Voice-Enabled BFP Assistant with Speech-to-Text and Text-to-Speech Controls

Table 11. Technical Results of the Voice-Enabled Accessibility Functions

TECHNICAL MEASURE	RESULT
STT Test Utterances	10
Correctly Recognized Utterances	10
STT Recognition Accuracy	100.00%
TTS Test Cases Passed	5
TTS Pass Rate	100.00%
Word Error Rate (WER)	0.00%
TTS Functional Test Cases	5

Table 11 presents the technical results of the voice-enabled accessibility functions, specifically Speech-to-Text (STT) and Text-to-Speech (TTS). The STT component was tested using ten predefined utterances, and all ten were correctly recognized, resulting in a recognition accuracy of 100.00%. Based on the 46 reference words, no substitutions, deletions, or insertions were recorded after normalization, resulting in a Word Error Rate (WER) of 0.00%. Meanwhile, the TTS component passed all five functional test cases, achieving a 100.00% pass rate. These results show that both voice-enabled functions successfully met their predefined technical requirements during the conducted tests.

The results are consistent with Prado et al. (2023), who emphasized the importance of addressing different dimensions of digital accessibility in educational environments. Their systematic review highlighted the need to consider

communication, methodological, and instrumental barriers when developing accessible digital systems. Similarly, El Fakir et al. (2025) discussed how STT and TTS can provide alternative ways for learners to interact with educational content by combining auditory and textual forms of communication. These findings support the use of voice-enabled functions as additional interaction methods in the SafeScape V2 portal.

Furthermore, Feng et al. (2024) reported that automatic speech recognition systems may produce different recognition results across speaker groups, including differences related to age and regional or non-native accents. This highlights the importance of interpreting the 100.00% recognition accuracy obtained in the present study within its controlled testing conditions. Further testing involving different speakers, accents, and background-noise conditions may provide additional evidence of the Speech-to-Text function's performance in varied real-world situations. They do not establish that the system will achieve the same performance across all speakers, accents, or environmental conditions.

Overall, the findings indicate that the STT and TTS functions operated successfully under the predefined technical test scenarios. However, further testing involving more speakers, varied pronunciation patterns, background noise, and different devices may provide a broader assessment of their performance and accessibility.

4.3.2 Educational Content and Responsive User Interface

The accessibility framework was complemented by a visually oriented educational interface designed to present fire-safety lessons using simplified text, multimedia content, visible progress indicators, and straightforward navigation.

Figure 20 shows one of the implemented fire-safety learning modules. The interface combines a prominent lesson title, concise instructional text, multimedia material, visible course progress, and simplified navigation. These elements were designed to support younger learners and users with different levels of digital experience.

The design of digital learning materials may be considered alongside the accessibility and educational-design principles discussed by Prado et al. (2023), the digital game-based training findings of Ren et al. (2023), and the human-centered design considerations reviewed by Hussain et al. (2024). These studies provide context for presenting educational content through understandable layouts and accessible learning materials. The figure illustrates the implemented learning module and does not independently establish an improvement in learners' knowledge or performance.

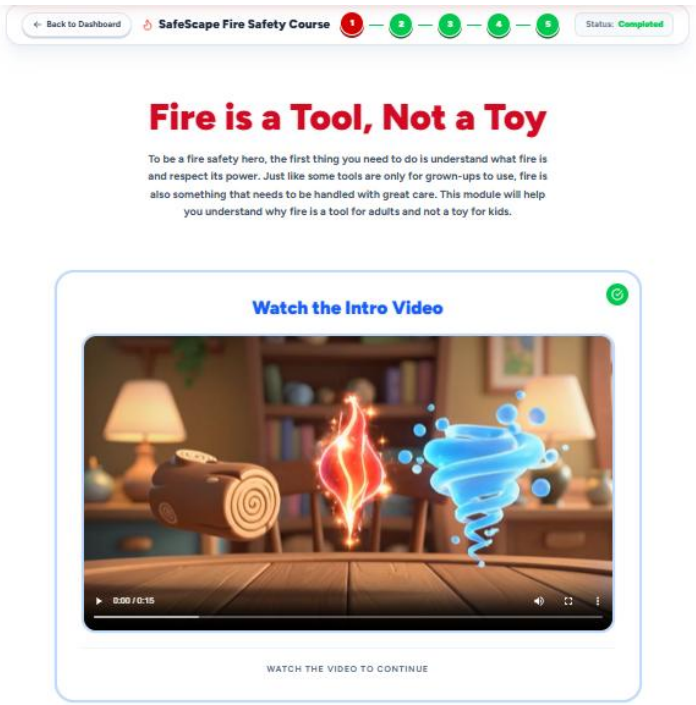


Figure 20. Fire-Safety Learning Module of SafeScape V2

SafeScape V2 was also implemented as a responsive web application to support access from narrower mobile screens. The mobile interface retains the major system

functions while reorganizing content and navigation for a smaller display. The relevance of responsive design is supported by the community-user device profile, in which 48 of the 50 respondents, or 96%, reported smartphone use.

The responsive presentation of the portal is relevant to the accessibility concerns discussed by Jou et al. (2024), Barrios and Moreno (2024), and Bokingkito et al. (2025). Their studies provide context for considering digital accessibility, user access, and the availability of government digital services across different user conditions. The figure demonstrates how the SafeScape V2 interface is presented on a mobile screen; the actual device-use distribution is reported separately in Table 29.



Figure 21. Responsive Mobile Interface of SafeScape V2

Internal accessibility and user-task testing further showed that all 11 predefined user tasks were successfully completed without recorded assistance. These tasks included registration, login, locating fire-safety content, interacting with the chatbot, using STT and TTS, completing learning activities, viewing progress, submitting

feedback, and navigating through a mobile device. These accessibility and user-task results are also included in the consolidated internal testing summary presented in Section 4.5.1.

4.3.3 Supporting Community End-User Results

The community end-user evaluation provides complementary evidence regarding how the implemented accessibility and interaction features were perceived by actual users. Since the complete item-level results are already presented in Section 4.5.4, Table 12 summarizes only the dimensions most directly related to Research Objective 3.

Table 12. Community End-User Results Relevant to Research Objective 3

EVALUATION AREA	MEAN	SD	VERBAL INTERPRETATION
Usability and Ease of Use	4.35	0.60	Strongly Agree
Content Quality and Relevance	4.41	0.67	Strongly Agree
Perceived Usefulness and Learning Effectiveness	4.44	0.79	Strongly Agree
Accessibility and Inclusiveness	4.29	0.71	Strongly Agree
Satisfaction with Application Design and Layout	4.52	0.61	Very Satisfied
Satisfaction with Ease of Use and Navigation	4.42	0.67	Very Satisfied
Satisfaction with Accessibility	4.34	0.77	Very Satisfied

Table 12 shows that Satisfaction with Application Design and Layout received the highest mean of 4.52, followed by Perceived Usefulness and Learning Effectiveness at 4.44. Satisfaction with Accessibility received a mean of 4.34, while Accessibility and Inclusiveness obtained the comparatively lowest mean of 4.29. Despite this difference, all reported evaluation areas remained within their respective favorable verbal interpretations.

These results indicate that community end-users generally provided favorable assessments of the portal's design, usability, and accessibility. Ren et al. (2023) discussed the role of developmentally appropriate design in digital learning interventions, while Hussain et al. (2024) examined human-centered interface considerations in emergency applications. Bokingkito et al. (2025) also discussed the relationship between accessibility and the functional delivery of digital government services. These studies provide context for the design and accessibility considerations reflected in the present evaluation. However, the mean scores alone do not establish that specific interface features caused the favorable ratings or that Internet connectivity was the primary reason for the comparatively lower accessibility score.

4.3.4 Overall Findings for Research Objective 3

The results indicate that Research Objective 3 was achieved within the defined scope of the study. SafeScape V2 successfully implemented Speech-to-Text and Text-to-Speech interaction, responsive interface behavior, visually supported educational content, and user-centered navigation.

Technical testing showed 100.00% STT Recognition Accuracy, a WER of 0.00%, a 100.00% TTS functional pass rate, and successful completion of all 11

predefined accessibility and user-task scenarios. Community end-users likewise provided favorable evaluations for usability, accessibility, content presentation, perceived usefulness, and interface satisfaction.

The technical and user-perception results therefore provide complementary evidence regarding the system's accessibility and ease of interaction. However, the findings should be interpreted within the study's limitations: STT performance was measured under controlled conditions, Internet connectivity remained the weakest accessibility-related indicator, and user engagement was supported through usefulness, satisfaction, interface feedback, and supplementary comments rather than through a separate standardized engagement scale.

4.4 Results of Research Objective 4 – Adaptive Learning and Administrative Dashboard

Research Objective 4 focused on the development of an intelligent administrative dashboard with user-engagement and assessment analytics and an adaptive-learning engine that determines appropriate Easy, Medium, or Hard difficulty levels for module quizzes and the final examination based on learner profile and assessment performance.

The adaptive-learning engine consisted of two Random Forest classification models. The Module Quiz Difficulty Model used Age, Grade Level, and Pre-Assessment Score as predictor variables, while the Final Examination Difficulty Model used Age, Grade Level, Pre-Assessment Score, and the learner's Module 1 to Module 4 quiz scores. Each model was developed using a separate synthetic dataset containing 5,000 learner records. An 80/20 train-test split was applied, resulting in 4,000 training records and 1,000 unseen testing records for each model.

The performance of the two classifiers was evaluated using Accuracy, Precision, Recall, F1-Score, per-class classification results, confusion matrices, and learning curves. In addition, the implemented adaptive-learning and administrative-dashboard components were subjected to internal system verification.

4.4.1 Module Quiz Difficulty Model

The Module Quiz Difficulty Model classified the appropriate difficulty level of module quizzes using the learner's Age, Grade Level, and Pre-Assessment Score. Evaluation was conducted on the 1,000 records reserved as the unseen testing partition.

Table 13. Module Quiz Difficulty Random Forest Classification Results

DIFFICULTY CLASS	PRECISION	RECALL	F1-SCORE	SUPPORT
Easy	0.9594	0.9220	0.9403	359
Medium	0.8462	0.9595	0.8993	321
Hard	0.9588	0.8719	0.9133	320
Macro Average	0.9214	0.9178	0.9176	1,000
Weighted Average	0.9229	0.9180	0.9185	1,000

Table 13 shows that the Easy class obtained the highest F1-score of 0.9403, followed by the Hard class at 0.9133 and the Medium class at 0.8993. The Medium class also recorded the lowest precision of 0.8462, indicating that some records predicted as Medium belonged to another actual difficulty class. The confusion matrix provides further detail regarding these classification errors, particularly the 35 actual Hard records that were predicted as Medium. Abu-Issa et al. (2024) discussed the use

of learner-profile information in supporting instructional adaptation, while Endla et al. (2025) examined adaptive-learning algorithms for personalized education. Chrysafiadi and Virvou (2024) also presented a fuzzy-logic approach to personalized tutoring decisions. These studies provide context for using learner-related information in adaptive educational systems. However, because the present model was trained and tested using synthetic learner-performance records, the reported classification results should be interpreted within the generated dataset and should not be treated as evidence of equivalent accuracy among actual learners.

The Medium class recorded the lowest precision at 0.8462. This result indicates that some records predicted as Medium were actually classified as Easy or Hard in the test data. In particular, the confusion matrix shows that 35 actual Hard records were predicted as Medium. This pattern identifies the distinction between adjacent difficulty classes as an area for further evaluation. Chrysafiadi and Virvou (2024) discussed the use of learner-related information and fuzzy-logic strategies in personalized tutoring, which provides context for adaptive instructional decisions. Nevertheless, the present study does not establish that the observed misclassifications resulted from actual learner behavior because the dataset consisted of synthetically generated records.

Figure 22 presents the classification results of the Module Quiz Difficulty Model for the 1,000 unseen testing records. Of the 359 actual Easy records, 331 were correctly classified as Easy, while 21 were classified as Medium and seven as Hard. Of the 321 actual Medium records, 308 were correctly classified as Medium, while eight were classified as Easy and five as Hard. Of the 320 actual Hard records, 279 were correctly classified as Hard, while six were classified as Easy and 35 as Medium. Overall, 918 records were correctly classified, corresponding to an accuracy of 91.80%.

The confusion matrix therefore shows that most records were assigned to their corresponding classes, while the largest group of classification errors involved actual Hard records predicted as Medium. The use of class-level performance measures and confusion matrices is consistent with the evaluation of classification models discussed by Abu-Issa et al. (2024), Endla et al. (2025), and Villegas-Ch et al. (2025). These results should be interpreted within the synthetic-data conditions of the present study.

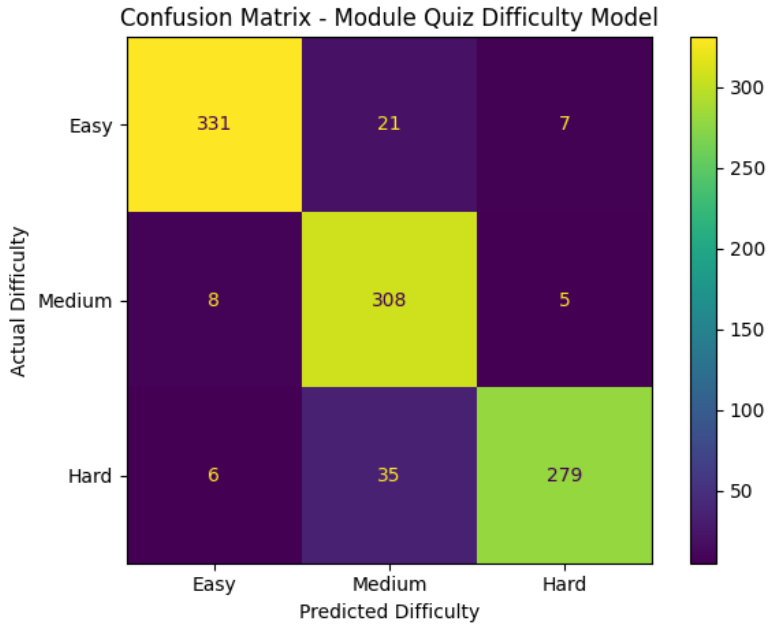


Figure 22. Confusion Matrix of the Module Quiz Difficulty Model

The learning curve of the Module Quiz Difficulty Model shows that the difference between the training and cross-validation scores became smaller as the number of training examples increased. At larger training sizes, both scores stabilized within a relatively narrow range. This pattern does not indicate severe overfitting under the synthetic-data conditions used in the study and suggests that the classifier was able to maintain relatively stable performance when applied to unseen records generated from the same data-generation process.

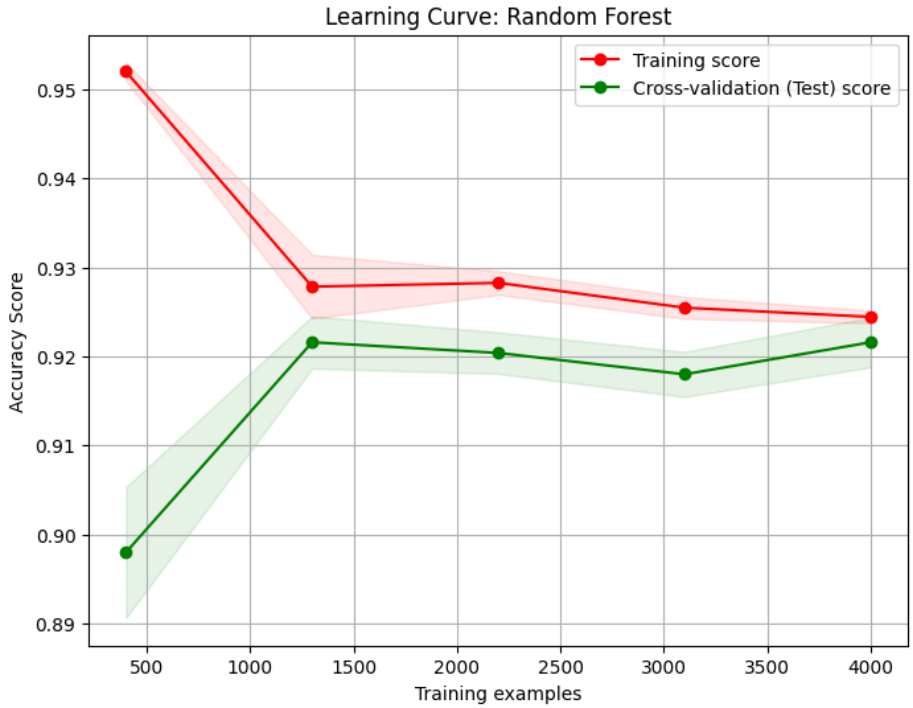


Figure 23. Learning Curve of the Module Quiz Difficulty Model

Figure 23 presents the learning curve of the Module Quiz Difficulty Model. The training and cross-validation scores became closer as the number of training examples increased and eventually stabilized within a relatively narrow range. This pattern suggests that the model maintained relatively consistent performance on records generated from the same synthetic-data process. However, the learning curve does not establish that the model will perform similarly on actual learner records. Abu-Issa et al. (2024) discussed the use of learner-profile information in adaptive instruction, while Endla et al. (2025) and Villegas-Ch et al. (2025) examined the use of learner information and performance data in adaptive educational systems. These studies provide context for the model's use of learner-related variables, but further validation using actual learner data is necessary to assess its performance in a real educational setting.

4.4.2 Final Examination Difficulty Model

Table 14. Final Examination Difficulty Random Forest Classification Results

DIFFICULTY CLASS	PRECISION	RECALL	F1-SCORE	SUPPORT
Easy	0.9877	0.9758	0.9817	330
Medium	0.9707	0.9822	0.9764	337
Hard	0.9940	0.9940	0.9940	333
Macro Average	0.9841	0.9840	0.9840	1,000
Weighted Average	0.9841	0.9840	0.9840	1,000

The Final Examination Difficulty Model determines the appropriate Easy, Medium, or Hard difficulty level for the final examination based on the learner's Age, Grade Level, Pre-Assessment Score, and accumulated quiz scores from Modules 1 to 4. The model was evaluated using a separate 1,000-record unseen testing partition.

Table 14 shows that the Final Examination Difficulty Model achieved an overall accuracy of 98.40%, with a macro F1-score of 98.40%. Among the three classes, Hard obtained the highest F1-score of 0.9940, followed by Easy at 0.9817 and Medium at 0.9764. These results indicate that the model correctly classified most of the records in the unseen testing partition. The model used Age, Grade Level, Pre-Assessment Score, and the recorded quiz scores from Modules 1 to 4 as predictor variables. Villegas-Ch et al. (2025) examined the use of learner interaction and performance information in adaptive intelligent tutoring systems, while Abrar et al. (2025) discussed personalized learning pathways supported by learner-related information. Chrysafiadi and Virvou

(2024) also presented an approach for personalizing instructional decisions using learner information. These studies provide context for the use of performance-related variables in adaptive learning. However, the higher accuracy of the final examination model should not be attributed solely to its additional predictors because the study did not conduct a controlled feature-ablation comparison. Both models were evaluated using separate synthetic datasets.

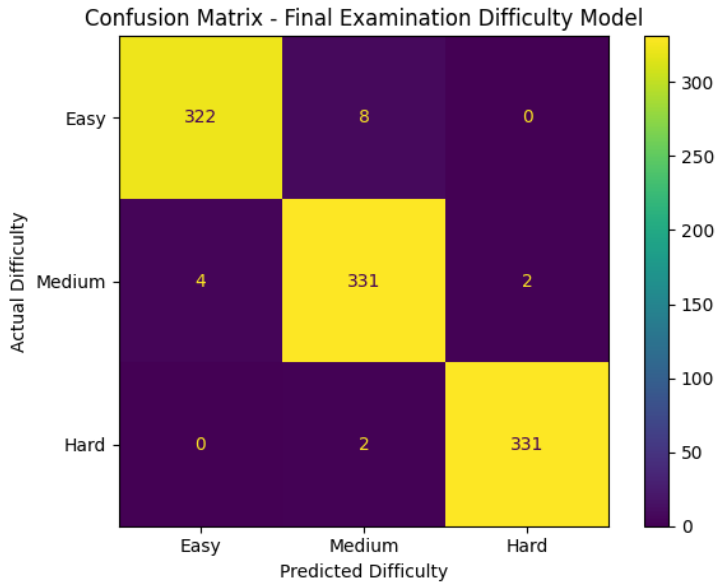


Figure 24. Confusion Matrix of the Final Examination Difficulty Model

Figure 24 presents the classification results of the Final Examination Difficulty Model. The confusion matrix shows that the model correctly classified most of the 1,000 unseen testing records. It recorded no misclassification between the two opposite classes, Easy and Hard. However, eight actual Easy records were classified as Medium, while the remaining errors involved adjacent difficulty classes. Overall, 984 records were correctly classified, corresponding to an accuracy of 98.40%. This result indicates strong classification performance within the synthetic testing dataset. Villegas-Ch et al. (2025), Endla et al. (2025), and Chrysafiadi and Virvou (2024) provide related context on adaptive learning systems that use learner-related information to support

instructional decisions. Nevertheless, the result does not establish that the same classification performance will occur when the model is applied to actual learners.

Figure 25 presents the learning curve of the Final Examination Difficulty Model. The training and cross-validation scores remained close to each other and stabilized above 98% as the training sample increased. This pattern indicates consistent performance on the synthetic records used in the experiment. However, the learning curve only describes model behavior within the data-generation process used in this study. It does not establish equivalent generalization to actual learner records. Abrar et al. (2025), Endla et al. (2025), and Villegas-Ch et al. (2025) discuss the use of learner information and assessment performance in adaptive educational systems, providing context for the model's selected predictors. Further testing with actual learner data is still needed to evaluate its performance in a real educational environment.

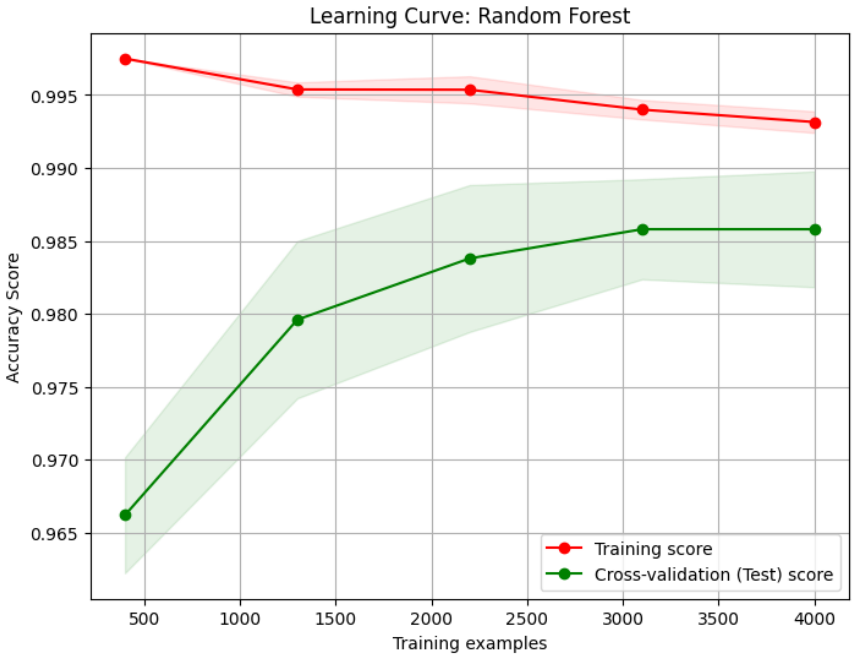


Figure 25. Learning Curve of the Final Examination Difficulty Model

4.4.3 Comparative Performance of the Two Random Forest Models

Table 15 presents the comparative predictive performance of the two Random Forest classifiers. Both models demonstrated high classification performance, although

the Final Examination Difficulty Model consistently obtained higher scores across the reported metrics.

Table 15 compares the predictive performance of the two Random Forest models. The Module Quiz Difficulty Model obtained an accuracy of 91.80% and a macro F1-score of 91.76%, while the Final Examination Difficulty Model achieved an accuracy of 98.40% and a macro F1-score of 98.40%.

Table 15. Comparative Performance of the Adaptive Learning Models

EVALUATION METRIC	MODULE QUIZ DIFFICULTY MODEL	FINAL EXAMINATION DIFFICULTY MODEL
Accuracy	91.80%	98.40%
Macro Precision	92.14%	98.41%
Macro Recall	91.78%	98.40%
Macro F1-Score	91.76%	98.40%
Weighted Precision	92.29%	98.41%
Weighted Recall	91.80%	98.40%
Weighted F1-Score	91.85%	98.40%

The final examination model therefore recorded higher values across the reported evaluation metrics. One difference between the models is their predictor variables: the module quiz model used Age, Grade Level, and Pre-Assessment Score, whereas the final examination model also used the learner's quiz scores from Modules

1 to 4. Villegas-Ch et al. (2025), Abrar et al. (2025), and Abu-Issa et al. (2024) discussed the use of learner profiles and assessment-related information in adaptive educational systems. These studies provide context for the selection of learner-related predictors. However, the difference in accuracy cannot be attributed solely to the additional quiz-score variables because the two models were trained and tested using separate synthetic datasets rather than through a controlled experiment using the same data and feature-ablation procedure.

4.4.4 Adaptive Learning Integration and Administrative Dashboard

The model-level evaluation was complemented by internal verification of the adaptive-learning integration and administrative dashboard. Six controlled adaptive-learning scenarios were executed, consisting of three Module Quiz Difficulty scenarios and three Final Examination Difficulty scenarios. All six scenarios returned their expected Easy, Medium, or Hard classifications.

The 6/6 system verification result should be distinguished from the predictive accuracy of the two Random Forest models. The system scenarios verify that SafeScape V2 successfully receives and applies the model output, whereas the 91.80% and 98.40% accuracy values measure classification performance across the corresponding 1,000-record unseen testing partitions. The adaptive-learning integration result is also included in the consolidated internal testing summary presented in Section 4.5.1.

The administrative dashboard was likewise verified through six internal test areas covering total users, learning participation, assessment results, user feedback, reports, and data filtering. All six dashboard tests passed, with displayed values verified against their corresponding records. The dashboard verification result is also included in the consolidated internal testing summary presented in Section 4.5.1.

BFP personnel provided complementary operational evidence, rating the statement that the administrative dashboard provides the required monitoring functions at 4.93 (SD = 0.26), interpreted as Excellent / Strongly Agree.

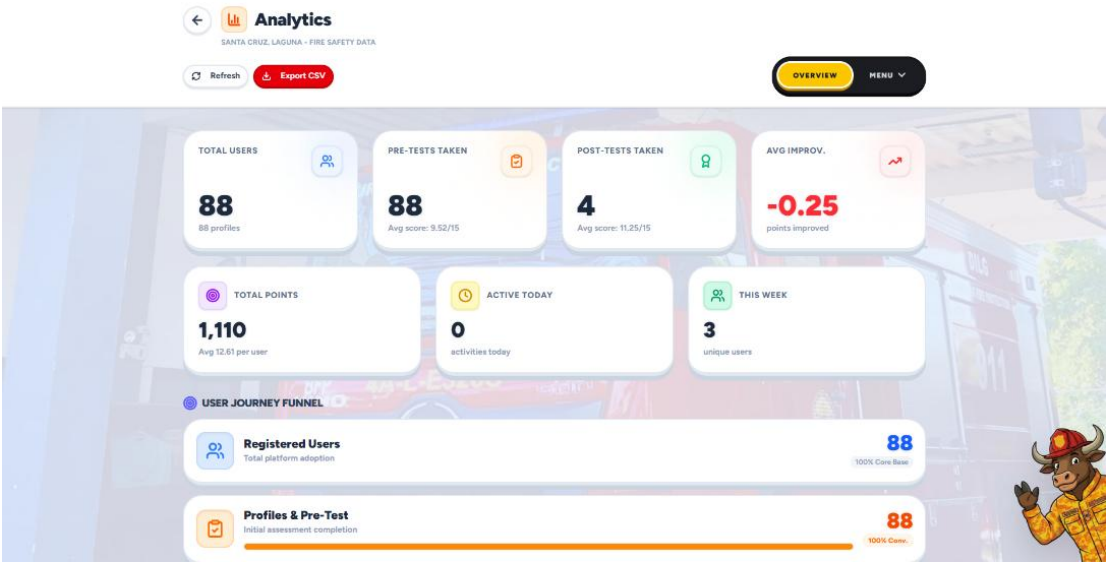


Figure 26. SafeScape V2 Administrative Dashboard

The dashboard's monitoring and information-display functions are consistent with the use of analytics and decision-support tools discussed by Esbieto et al. (2026). Roque et al. (2024) also examined the integration of digital information services in a local government setting, while Dandan et al. (2026) discussed the evaluation of a local government information system using software-quality characteristics. These studies provide context for the role of administrative information systems in organizing and presenting data. The figure illustrates the implemented dashboard, while its internal verification results and BFP personnel evaluation are reported separately in Sections 4.4.4 and 4.5.3.3.

The result is consistent with Esbieto et al. (2026), who reported that analytics dashboards can support data-driven governance by providing administrators with automated reporting and visual information for decision-making.

4.4.5 Overall Findings for Research Objective 4

The results indicate that Research Objective 4 was achieved within the defined scope of the study. The Module Quiz Difficulty Model achieved an accuracy of 91.80% with a macro F1-Score of 91.76%, while the Final Examination Difficulty Model achieved an accuracy of 98.40% with a macro F1-Score of 98.40%. The corresponding confusion matrices showed 918 and 984 correct classifications out of 1,000 unseen testing records, respectively.

The six adaptive-learning system scenarios also produced their expected difficulty outputs, demonstrating successful integration of the classifiers with the web application. For the administrative component, all six internal dashboard verification areas passed, while BFP personnel rated the dashboard's monitoring functionality at 4.93 (SD = 0.26).

Taken together, the findings demonstrate the successful implementation of a Random Forest-based adaptive-learning mechanism for Easy, Medium, or Hard assessment difficulty classification and an administrative dashboard for monitoring user engagement and assessment-related information.

The model-performance results should nevertheless be interpreted with consideration of the synthetic-data limitation. Both classifiers were trained and tested using simulated learner-performance records; therefore, their measured performance may differ when subsequently evaluated using accumulated real learner interaction and assessment data.

4.5 Results of Research Objective 5 – System and Stakeholder Evaluation

Research Objective 5 presents the overall technical verification and stakeholder evaluation of the developed SafeScape V2. The assessment incorporates internal

technical testing, software-quality evaluation, domain and operational validation, and community end-user evaluation.

Internal technical testing was conducted solely by the researchers/developers to verify the actual behavior of the system through functional, integration, AI chatbot, Speech-to-Text, Text-to-Speech, accessibility and user-task, adaptive-learning, administrative-dashboard, performance, and security testing. These technical results were recorded separately from the external stakeholder evaluation.

For the external evaluation, ten (10) Information Technology experts assessed the selected ISO/IEC 25010:2023 software quality characteristics of Functional Suitability, Performance Efficiency, Reliability, Security, Maintainability, and Flexibility. Fifteen (15) Bureau of Fire Protection personnel assessed the factual accuracy, contextual relevance, completeness, and operational appropriateness of the fire-safety information and AI-assisted functions, while fifty (50) community end-users evaluated the usability, accessibility, ease of interaction, perceived usefulness, satisfaction, and overall acceptability of the platform.

4.5.1 Internal System Testing by the Researchers/Developers

The internal technical testing of SafeScape V2 was conducted by the researchers/developers using the Individual Researcher/Developer System Testing and Verification Tool. The testing process employed predefined test cases, expected system behaviors, controlled test accounts, and component-specific technical measures to determine whether the implemented system functions operated according to their intended requirements.

The internal testing covered functional testing, integration testing, Artificial Intelligence (AI) chatbot verification, Speech-to-Text (STT), Text-to-Speech (TTS), accessibility and user-task testing, adaptive-learning verification, administrative-

dashboard testing, performance testing, and security testing. The internal technical testing was conducted separately from the external evaluations performed by the Information Technology experts, Bureau of Fire Protection personnel, and community end-users, in accordance with the testing and evaluation procedures established in Chapter III.

4.5.1.1 Consolidated Internal System Testing Results

Table 16. Summary of Internal System Testing Results

TESTING AREA	TEST CASES	PASSED	FAILED	PASS RATE
Functional Testing	24	24	0	100.00%
Integration Testing	5	5	0	100.00%
Speech-to-Text Testing	10	10	0	100.00%
Text-to-Speech Testing	5	5	0	100.00%
Accessibility and User-Task Testing	11	11	0	100.00%
Adaptive Learning System Testing	6	6	0	100.00%
Administrative Dashboard Testing	6	6	0	100.00%
Performance Testing	10	10	0	100.00%
Security Testing	8	8	0	100.00%

Table 16 continued…

TESTING AREA	TEST CASES	PASSED	FAILED	PASS RATE
Overall	85	85	0	100.00%

Note. The 15 AI chatbot test queries were analyzed separately using response-quality and retrieval measures and were therefore not included in the binary Pass/Fail total.

Table 16 shows that all 85 predefined binary test cases passed, resulting in an overall pass rate of 100.00% under the controlled testing conditions. The tests covered functional testing, integration testing, Speech-to-Text, Text-to-Speech, accessibility and user tasks, adaptive-learning integration, administrative-dashboard verification, performance testing, and security testing. These results indicate that the tested functions operated according to their expected behavior during internal verification. Ramadhani et al. (2025) discussed the use of structured web application architecture, while Ariyanto et al. (2024) examined the organization and performance of Laravel-based REST API implementations. Pratama and Raharja (2023) also examined server-side framework performance and resource use. These studies provide context for the system architecture and development approach. However, the 100.00% pass rate should be attributed to the observed test results and not assumed to have been caused solely by the use of the Laravel MVC architecture.

The integration tests also passed all five predefined scenarios, indicating that the tested components communicated and operated according to their expected behavior during the verification process. These tests covered the integration of the

Gemini API chatbot and the Random Forest classification models with the main web portal. Sermet and Demir (2021) discussed the integration of intelligent web components, while Ramadhani et al. (2025) and Ariyanto et al. (2024) examined structured web application development and component organization. These studies provide supporting context for the system's modular implementation. Nevertheless, the results represent controlled internal testing and do not guarantee that the same behavior will be maintained under every production environment or future system update.

4.5.1.2 Summary of Component-Specific Technical Results

In addition to the consolidated Pass/Fail testing, several system components were evaluated using technical measures specific to their corresponding research objectives. Since these results are presented in detail under Sections 4.1 to 4.4, only their principal findings are summarized in Table 17 to avoid repetition.

The RAG chatbot was evaluated through 15 controlled fire-safety questions and achieved 100.00% Response Accuracy, 100.00% Knowledge Retrieval Effectiveness, and a 0.00% Hallucination/Error Rate within the defined test set. These values are discussed together with Contextual Relevance and Response Completeness under Research Objective 2.

Table 17. Summary of Component-Specific Technical Results

COMPONENT	PRINCIPAL INTERNAL RESULT	DETAILED SECTION
RAG Chatbot	15 controlled queries; 100.00% Response Accuracy; 100.00% Knowledge Retrieval Effectiveness; 0.00% Hallucination/Error Rate	Section 4.2

Table 17 continued…

COMPONENT	PRINCIPAL INTERNAL RESULT	DETAILED SECTION
Speech-to-Text	10/10 utterances correctly recognized; 100.00% Recognition Accuracy; 0.00% WER	Section 4.3
Text-to-Speech	5/5 functional test cases passed	Section 4.3
Accessibility and User Tasks	11/11 tasks successfully completed; no recorded assistance required	Section 4.3
Adaptive Learning Integration	6/6 controlled system-level scenarios produced the expected difficulty classification	Section 4.4
Administrative Dashboard	6/6 dashboard verification areas passed, and corresponding displayed data were verified	Section 4.4
Performance and Scalability	10/10 scenarios met acceptance criteria; response times ranged from 0.15–1.40 s; 50-user test averaged 0.54 s with 0% error rate	Section 4.1
Security	8/8 controlled security test cases passed	Section 4.1

Table 17 summarizes the principal internal technical results of the system components. The RAG chatbot achieved 100.00% response accuracy and knowledge retrieval effectiveness, with a 0.00% hallucination/error rate across the 15 controlled test queries. The Speech-to-Text component correctly recognized all ten predefined utterances, resulting in a 0.00% Word Error Rate. The performance tests also recorded an average response time of 0.54 seconds during the 50-user simulated load test, with no recorded errors. Ndum et al. (2026) discussed source provenance and verification in safety-critical RAG systems, while Saputri et al. (2024) examined performance optimization in Laravel-based applications. Feng et al. (2024) discussed differences in automatic speech-recognition performance across speaker groups. These studies provide context for the different technical measures used in the present evaluation. However, the reported results describe the specific test queries, speech utterances, devices, and workload used in this study. They do not establish that the chatbot will never generate an unsupported response, that performance will remain unchanged under all workloads, or that speech recognition will be identical across all speakers and environments.

4.5.2 ISO/IEC 25010:2023 Software Quality Evaluation by Ten IT Experts

The software quality evaluation of SafeScape V2 involved ten (10) Information Technology experts selected based on their relevant academic or professional background in information technology and related fields. The IT experts evaluated the developed portal using selected software product quality characteristics based on ISO/IEC 25010:2023, namely Functional Suitability, Performance Efficiency, Reliability, Security, Maintainability, and Flexibility. The ten IT experts participated exclusively in the ISO/IEC 25010:2023 software quality evaluation, while internal technical testing was conducted separately by the researchers/developers.

The quantitative responses of the IT experts were summarized using frequency, percentage, mean, and standard deviation. Mean scores ranging from 4.21 to 5.00 were interpreted as Excellent / Strongly Agree, 3.41 to 4.20 as Very Good / Agree, 2.61 to 3.40 as Good / Moderately Agree, 1.81 to 2.60 as Fair / Disagree, and 1.00 to 1.80 as Poor / Strongly Disagree.

4.5.2.1 Profile of the IT Expert Respondents

Table 18. Profile of the IT Expert Respondents According to Years of IT/Software Experience (N = 10)

YEARS OF IT/SOFTWARE EXPERIENCE	FREQUENCY	PERCENTAGE
1–3 years	9	90.00%
4–6 years	1	10.00%
7 years and above	0	0.00%
Total	10	100.00%

Table 18 shows that nine of the ten IT expert respondents, or 90.00%, reported one to three years of IT or software experience, while one respondent, or 10.00%, reported four to six years. No respondent reported seven or more years of experience. These results describe the professional experience of the selected evaluation panel. The distribution alone does not establish that early-career professionals are more qualified than senior professionals to evaluate the system. The suitability of the respondents should instead be considered in relation to the selection criteria and the software-quality characteristics assessed in this study. Ariyanto et al. (2024) discussed the technical considerations involved in Laravel-based application development, while Olipas et al.

(2022) and Dandan et al. (2026) presented examples of software-quality evaluation in web-based and local information systems. These studies provide context for the technical nature of the evaluation but do not explain the experience distribution of the present respondents.

Table 19. Profile of the IT Expert Respondents According to Primary Area of Expertise

AREA OF EXPERTISE	FREQUENCY	PERCENTAGE
Software Development	9	90.00%
Database Administration	1	10.00%
System Architecture	1	10.00%
Cybersecurity	1	10.00%
Artificial Intelligence / Machine Learning	0	0.00%
Other – Integration Development	1	10.00%

Note. Respondents could indicate more than one area of expertise; therefore, frequencies and percentages may exceed the total number of respondents and 100%, respectively.

Table 19 shows that Software Development was the most frequently selected area of expertise, reported by nine respondents or 90.00%. Database Administration, System Architecture, Cybersecurity, and Other – Integration Development were each selected by one respondent or 10.00%, while Artificial Intelligence / Machine Learning was not selected by any respondent. Since respondents were allowed to select more than one area, the frequencies and percentages are not expected to total exactly ten

respondents and 100.00%. The results indicate that the evaluation panel primarily had a software-development background. This profile is relevant to the assessment of the portal's software-quality characteristics, including functional suitability, performance efficiency, reliability, security, maintainability, and flexibility. Ariyanto et al. (2024), Olipas et al. (2022), and Dandan et al. (2026) provide related examples of web-application development and software-quality evaluation. However, the absence of respondents who selected AI/ML as their primary area of expertise should be noted as a characteristic of the panel and should not be interpreted as proof that AI-related technical validation was unnecessary.

4.5.2.2 Overall ISO/IEC 25010:2023 Software Quality Evaluation

Table 20. Summary of the ISO/IEC 25010:2023 Software Quality Evaluation of SafeScape V2

ISO/IEC 25010:2023 CHARACTERISTICS	MEAN	SD	VERBAL INTERPRETATION
Functional Suitability	4.73	0.31	Excellent / Strongly Agree
Performance Efficiency	3.83	0.36	Very Good / Agree
Reliability	4.40	0.34	Excellent / Strongly Agree
Security	4.63	0.29	Excellent / Strongly Agree
Maintainability	3.93	0.34	Very Good / Agree
Flexibility	3.83	0.28	Very Good / Agree
Overall ISO Evaluation	4.23	0.15	Excellent / Strongly Agree

Legend: 4.21–5.00 = Excellent / Strongly Agree; 3.41–4.20 = Very Good / Agree; 2.61–3.40 = Good / Moderately Agree; 1.81–2.60 = Fair / Disagree; 1.00–1.80 = Poor / Strongly Disagree.

Note. The overall ISO mean was computed from the 18 items corresponding to the six selected ISO/IEC 25010:2023 characteristics. Standard deviations for the characteristics and overall result were computed from the corresponding respondent-level composite scores.

Table 20 shows that SafeScape V2 obtained an overall ISO/IEC 25010:2023 mean of 4.23 (SD = 0.15), verbally interpreted as Excellent / Strongly Agree. The result indicates a generally favorable professional assessment of the software quality of the developed portal based on the six selected characteristics.

Among the evaluated characteristics**,** Functional Suitability obtained the highest mean of 4.73 (SD = 0.31), followed by Security with 4.63 (SD = 0.29) and Reliability with 4.40 (SD = 0.34). These results indicate that the IT experts particularly viewed the system favorably with respect to its implemented functionality, protection mechanisms, and dependable system behavior.

The use of ISO/IEC 25010:2023 as a structured software-quality evaluation framework is supported by Dandan et al. (2026), whose localized barangay management-system evaluation demonstrated the applicability of the standard in measuring explicit software-quality characteristics within a Philippine municipal context. The present evaluation similarly applies selected measurable software-quality characteristics to assess SafeScape V2 from a professional IT perspective.

Performance Efficiency and Flexibility both obtained the comparatively lowest characteristic mean of 3.83, while Maintainability obtained 3.93. Although all three

were interpreted as Very Good / Agree, the lower ratings indicate areas in which the system may still benefit from continued optimization and refinement.

These findings are consistent with the architectural literature reviewed in Chapter II. Ariyanto et al. (2024) noted that Laravel's MVC architecture provides advantages in code organization and maintainability, while also recognizing the need to evaluate the framework's actual processing overhead. Similarly, Saputri et al. (2024) demonstrated the importance of caching, workload optimization, and load testing in improving Laravel application response time and scalability under increased user demand.

The overall findings therefore suggest that while the IT experts evaluated SafeScape V2 positively in terms of software quality, the results also identify performance efficiency, maintainability, and flexibility as relevant priorities for continued technical refinement.

4.5.2.3 Overall Evaluation of the Developed System

In addition to the six selected ISO/IEC 25010:2023 characteristics, the IT expert instrument contained two items addressing the respondents' overall assessment of the developed system.

Table 21. Overall Evaluation of SafeScape V2 by the IT Experts

EVALUATION	MEAN	SD	VERBAL INTERPRETATION
Overall Evaluation Q19	4.60	0.52	Excellent / Strongly Agree
Overall Evaluation Q20	4.50	0.53	Excellent / Strongly Agree

Table 21 continued…

EVALUATION	MEAN	SD	VERBAL INTERPRETATION
Overall	4.55	0.28	Excellent / Strongly Agree

Legend: 4.21–5.00 = Excellent / Strongly Agree; 3.41–4.20 = Very Good / Agree; 2.61–3.40 = Good / Moderately Agree; 1.81–2.60 = Fair / Disagree; 1.00–1.80 = Poor / Strongly Disagree.

Table 21 presents the IT experts' overall assessment of SafeScape V2. Overall Evaluation Q19 received a mean of 4.60 (SD = 0.52), while Overall Evaluation Q20 obtained a mean of 4.50 (SD = 0.53). The combined overall mean was 4.55 (SD = 0.28), interpreted as Excellent / Strongly Agree. These results indicate that the ten IT experts gave favorable overall assessments of the developed system. Olipas et al. (2022) examined the quality of a web-based information management system, while Ariyanto et al. (2024) discussed the technical organization of Laravel-based applications. Dandan et al. (2026) also demonstrated the application of a software-quality model in evaluating a local information system.

These studies provide context for evaluating web-based systems through technical and quality-related criteria. Nevertheless, the combined mean of 4.55 represents the perceptions of the ten IT experts and should be interpreted as an overall stakeholder assessment rather than formal certification or a substitute for separate technical, security, and operational validation.

4.5.2.4 Supplementary IT Expert Comments and Recommendations

The IT expert evaluation instrument also included five open-ended questions concerning the major technical strengths of SafeScape V2, identified limitations or defects, recommendations concerning performance, reliability and security, recommendations concerning maintainability and future expansion, and recommendations for the AI chatbot, adaptive-learning component, and related system integrations.

Consistent with the research methodology, the responses were summarized as supplementary evaluator feedback to identify commonly mentioned observations and recommendations. They were not treated as a separate qualitative or thematic research analysis. Chapter III specifies that open-ended evaluator comments are supplementary to the primary quantitative and technical findings.

Table 22. Summary of Supplementary IT Expert Feedback (N = 10)

OPEN-ENDED QUESTION	DESCRIPTIVE CATEGORY	FREQUENCY	PERCENTAGE
Major Technical Strengths (Q21)	UI / UX and Usability	5	50.00%
	Interactive Learning Features	3	30.00%
	AI/Adaptive Learning Integration	1	10.00%

Table 22 continued…

OPEN-ENDED QUESTION	DESCRIPTIVE CATEGORY	FREQUENCY	PERCENTAGE
	Admin/Role Management	1	10.00%
Technical Limitations (Q22)	Code Organization / Maintainability	3	30.00%
	Performance Optimization	2	20.00%
	No Critical Issue	2	20.00%
	Cross-Browser STT Compatibility	1	10.00%
	Security / Stress Validation Needed	1	10.00%
	Geographic Expansion	1	10.00%
Performance, Reliability, and Security Recommendations (Q23)	Performance/Scalability	4	40.00%

Table 22 continued…

OPEN-ENDED QUESTION	DESCRIPTIVE CATEGORY	FREQUENCY	PERCENTAGE
	No Additional Recommendation	4	40.00%
	Security / Error Handling	2	20.00%
Maintainability and Future Expansion Recommendations (Q24)	Code Maintainability / Organization	7	70.00%
	Coding Standards / Dependency	1	10.00%
	No Additional Recommendation	1	10.00%
	Geographic Scalability	1	10.00%
AI, Adaptive Learning, and Integration Recommendations (Q25)	No Additional Improvement / Positive Feedback	7	70.00%

Table 22 continued…

OPEN-ENDED QUESTION	DESCRIPTIVE CATERGORY	FREQUENCY	PERCENTAGE
	Chatbot Grounding Citations	2	20.00%
	Chatbot FAQ Expansion	1	10.00%

Table 22 summarizes the supplementary comments of the ten IT experts. UI/UX and Usability was the most frequently mentioned technical strength, with five respondents or 50.00% identifying this category. Interactive Learning Features followed with three responses or 30.00%, while AI/Adaptive Learning Integration and Admin/Role Management each received one response or 10.00%. These results indicate that UI/UX and usability were the most mentioned strengths among the respondents. Ren et al. (2023) examined digital game-based training and learner engagement, while Hussain et al. (2024) discussed human-centered design in evacuation applications. Barrios and Moreno (2024) also examined accessibility and digital inclusion in e-government initiatives. These studies provide context for the importance of usable and accessible digital interfaces, although the present frequencies reflect only the feedback provided by the ten IT experts.

The results in Table 22 show that Code Organization and Maintainability was identified as a technical limitation by three out of ten IT experts, equivalent to 30.00% of the panel. Separately, seven out of ten IT experts, or 70.00%, mentioned Code Maintainability and Organization as an area for future improvement under the maintainability and future expansion recommendations. These results indicate that code

organization was both an observed area for refinement and a recurring recommendation for future development. The two percentages represent responses to different open-ended questions and should therefore be interpreted separately. Ramadhani et al. (2025) discussed the importance of organized application architecture in web-based system development, while Ariyanto et al. (2024) examined the relationship between Laravel architecture, code organization, and application performance. Kundan and Makwana (2025) also discussed optimization and maintenance considerations in Laravel-based applications. These studies provide context for the experts' recommendations concerning code organization and maintainability.

4.5.3 BFP Personnel Evaluation

The domain and operational evaluation of SafeScape V2 involved 15 Bureau of Fire Protection (BFP) personnel. The BFP evaluators were selected based on their professional knowledge and experience in fire safety and public-safety operations. Their role was to validate the factual accuracy, contextual relevance, completeness, and appropriateness of the fire-safety information and AI-assisted functions, as well as the relevance of the educational and administrative features of the developed system. This evaluator role is consistent with the domain and operational validation established in the research methodology.

The BFP evaluation consisted of three components: evaluator profile, AI chatbot accuracy and relevance assessment, and system operational evaluation. The questionnaire also contained open-ended questions that allowed the BFP personnel to provide supplementary comments and recommendations regarding information dissemination, chatbot responses, administrative functions, continued system operation, and possible local expansion.

4.5.3.1 Profile of the BFP Personnel Evaluators

Table 23. Profile of the BFP Personnel Evaluators According to Rank/Position (N = 15)

RANK/POSITION	FREQUENCY	PERCENTAGE
FO1	6	40.00%
FO2	2	13.33%
SFO3	2	13.33%
SFO2	1	6.67%
SFO1	1	6.67%
FO1/FSI	1	6.67%
FO2/Fire Truck Driver/Operator	1	6.67%
FO2/PIO Clerk	1	6.67%
Total	15	100.00%

Table 23 shows the rank and position distribution of the 15 BFP personnel who participated in the evaluation. Fire Officer 1 (FO1) respondents formed the largest group, with six respondents or 40.00%. The remaining respondents represented different ranks and specialized positions, including FO2, SFO3, SFO2, SFO1, and personnel assigned to fire-safety inspection, fire-truck operation, and public information duties. This distribution indicates that the evaluation included personnel from different ranks and work assignments. Longa and Perena (2024) discussed the

operational responsibilities and emergency-response capabilities of BFP personnel, while Sabordo (2026) examined communication and coordination concerns within fire-protection operations. Carsola and Bagtaso (2026) also discussed personnel and operational challenges in local fire-protection settings. These studies provide context for the professional setting of the evaluation. However, the rank distribution in the present study should be interpreted only as the profile of the selected respondents; it does not establish that FO1 personnel are inherently more qualified to evaluate the portal than higher-ranking or specialized personnel.

Table 24. Profile of the BFP Personnel Evaluators According to Years in Service

YEARS IN SERVICE	FREQUENCY	PERCENTAGE
1–3 years	1	6.67%
4–6 years	5	33.33%
7–10 years	5	33.33%
10+ years	4	26.67%
Total	15	100.00%

Table 24 shows that respondents with four to six years of service and those with seven to ten years of service each accounted for five respondents or 33.33%. Four respondents or 26.67% had more than ten years of service, while one respondent or 6.67% had one to three years of service. The distribution indicates that most members of the evaluation panel had at least four years of service. Longa and Perena (2024) examined the emergency-response capabilities and operational experiences of BFP personnel, while Sabordo (2026) discussed communication and coordination

challenges in fire-protection operations. Araojo et al. (2024) also examined the role of institutional information and legacy systems in public-safety decision-making. These studies provide context for the operational knowledge relevant to the evaluation. However, the present results do not establish that a particular length of service automatically makes a respondent more qualified to evaluate an AI chatbot or administrative dashboard.

4.5.3.2 AI Chatbot Accuracy and Relevance Evaluation

Table 25. BFP Evaluation of the AI Chatbot by Fire-Safety Topic

FIRE - SAFETY TOPIC	ACCURACY MEAN	CONTEXTUAL RELEVANCE MEAN	COMPLETENESS MEAN	OVERALL MEAN	SD
Fire Prevention	4.73	4.87	4.80	4.80	0.30
Fire Extinguisher Use	4.87	4.73	4.80	4.80	0.37
Evacuation Procedures	4.67	4.73	4.80	4.73	0.42
LPG/Electrical Safety	4.80	4.67	4.87	4.78	0.35
Overall	4.77	4.75	4.82	4.78	0.31

Note. The BFP chatbot instrument used a five-point rating scale, with higher scores representing stronger domain validation of the chatbot responses.

Table 25 shows that the AI chatbot obtained an overall domain-validation mean of 4.78 (SD = 0.31) across the four evaluated fire-safety topics. Among the three evaluation criteria, Completeness obtained the highest overall mean of 4.82 (SD = 0.27), followed by Accuracy at 4.77 (SD = 0.36) and Contextual Relevance at 4.75 (SD = 0.35).

At the topic level, Fire Prevention and Fire Extinguisher Use both obtained the highest overall mean of 4.80, followed by LPG/Electrical Safety at 4.78. Evacuation Procedures obtained the comparatively lowest overall mean of 4.73. Nevertheless, all topic-level means remained close to the maximum rating of five, indicating a highly favorable domain assessment of the responses generated by the chatbot.

The results support the use of domain-grounded retrieval for safety-related information. Chae et al. (2025) evaluated retrieval and question answering using a corpus derived from 117 fire-safety codes and emphasized that retrieval quality and generated-answer quality should be assessed separately when RAG is applied within a regulated information domain. The present BFP evaluation similarly assesses multiple dimensions of generated responses rather than relying on a single overall judgment.

The favorable BFP ratings are also consistent with research emphasizing the importance of grounding generated responses in verifiable domain sources. Ndum et al. (2026) demonstrated that source provenance and response-verification mechanisms in a safety-critical RAG environment produced high compliance rates and substantially reduced hallucination compared with general-purpose ungrounded models. These

findings reinforce the relevance of grounding SafeScape V2's chatbot in BFP fire-safety resources and subjecting its responses to professional domain validation.

The findings may also be considered alongside Xie et al. (2025), who examined the use of retrieval-augmented generation in natural-hazard information and decision-support applications. Their work provides additional context for evaluating AI-generated information in public-safety-related domains. In the present study, the favorable ratings across the four fire-safety topics indicate that the BFP evaluators considered the chatbot responses appropriate within the topics and questions included in the evaluation. These ratings should not be generalized to every possible fire-safety inquiry or emergency.

4.5.3.3 System Operational Evaluation

The BFP personnel also evaluated the operational relevance of SafeScape V2 based on six indicators covering fire-safety information dissemination, chatbot appropriateness, educational modules, administrative monitoring, information accessibility, and community fire-safety engagement.

Table 26. BFP Personnel Evaluation in Terms of System Operational Relevance

INDICATOR	MEAN	SD	VERBAL INTERPRETATION
The system provides the functions necessary for fire-safety information dissemination.	4.87	0.35	Excellent / Strongly Agree
The chatbot provides appropriate fire-safety information.	4.87	0.35	Excellent / Strongly Agree

Table 26 continued…

INDICATOR	MEAN	SD	VERBAL INTERPRETATION
The educational modules operate according to their intended purpose.	4.80	0.41	Excellent / Strongly Agree
The system improves access to fire-safety information.	4.93	0.26	Excellent / Strongly Agree
The administrative dashboard provides the required monitoring functions.	4.93	0.26	Excellent / Strongly Agree
The system provides an effective platform for community fire-safety engagement.	4.87	0.35	Excellent / Strongly Agree
Overall	4.88	0.28	Excellent / Strongly Agree

Legend: 4.21–5.00 = Excellent / Strongly Agree; 3.41–4.20 = Very Good / Agree; 2.61–3.40 = Good / Moderately Agree; 1.81–2.60 = Fair / Disagree; 1.00–1.80 = Poor / Strongly Disagree.

Table 26 presents the BFP personnel's evaluation of SafeScape V2 in terms of its operational functions. The system's ability to improve access to fire-safety information and the administrative dashboard's monitoring functions both obtained the highest mean of 4.93 (SD = 0.26). The overall mean of 4.88 (SD = 0.28) indicates a favorable evaluation of the system's operational relevance. Esbieto et al. (2026) discussed the use of analytics and decision-support tools in local government planning,

while Roque et al. (2024) examined the integration of a conversational information system in a local government setting. Amil (2024) also discussed the role of community participation in disaster risk reduction. These studies provide context for the system's information, monitoring, and community-engagement functions. However, the ratings represent the BFP personnel's assessments and do not establish that the portal has independently reduced emergency-response time or replaced existing fire-safety activities.

The educational modules received the comparatively lowest mean of 4.80 (SD = 0.41), although the result remained within the Excellent / Strongly Agree interpretation. This indicates that the evaluators still gave the educational modules a favorable assessment, while the slightly lower score identifies an area that may be considered for future refinement. The result alone does not establish that the evaluators rated the modules lower because they require physical fire-safety drills.

4.5.3.4 Supplementary BFP Comments and Recommendations

Table 27. Summary of Supplementary BFP Personnel Feedback

OPEN-ENDED QUESTION	DESCRIPTIVE CATEGORY	FREQUENCY	PERCENTAGE
Information Dissemination and Community Engagement (N=15)	Effective Information/Knowledge Support	6	40.00%
	Accessibility and Convenience	3	20.00%
	Interactive Learning/Engagement	2	13.33%
	Continued Engagement Improvement	2	13.33%

Table 27 continued…

OPEN-ENDED QUESTION	DESCRIPTIVE CATEGORY	FREQUENCY	PERCENTAGE
	No Comment	2	13.33%
Fire-Safety Content and Chatbot Appropriateness (N=15)	Appropriate/Accurate/Relevant	11	73.33%
	Grounded in Fire-Safety References/RA 9514	2	13.33%
	Regular Review/Updating Recommended	1	6.67%
	No Comment	1	6.67%
Additional Administrative/Monitoring Features (N=15)	Existing Features Sufficient	10	66.67%
	Monitoring/Feedback/Activity Tracking	3	20.00%
	Add FSIS Guidelines/References	1	6.67%
	Funding/Resource Support	1	6.67%
Continued Operation and Local Expansion (N=15)	No Additional Improvement	7	46.67%
	Additional Educational/Interactive Content	3	20.00%
	Funding/Expansion Support	2	13.33%
	Technical/UI/AI Improvements	2	13.33%
	Promotion/School Coordination	1	6.67%

Table 27 summarizes the supplementary feedback provided by the 15 BFP personnel. For fire-safety content and chatbot appropriateness, 11 respondents or 73.33% described the responses as appropriate, accurate, or relevant. For administrative and monitoring features, ten respondents or 66.67% considered the existing functions sufficient. Other respondents suggested additional activity tracking, educational content, and improvements to system features. These results indicate that most of the BFP respondents who provided feedback viewed the chatbot and administrative functions favorably, while some identified areas for further development. Chae et al. (2025) discussed the evaluation of retrieval and question-answering in a fire-safety information domain, while Ndum et al. (2026) examined source provenance and verification in safety-critical RAG systems. Xie et al. (2025) also explored RAG-supported information and decision support for natural hazards. These studies provide context for the importance of grounded information and expert validation. However, the comments in Table 27 represent the views of the present BFP respondents and do not independently prove that the chatbot's architecture caused their favorable assessments.

4.5.4 Community End-User Evaluation

The community end-user evaluation involved 50 respondents from Sta. Cruz, Laguna who represented the intended users of the developed SafeScape V2 platform. The evaluation was conducted to determine the respondents' perceptions of the system in terms of overall acceptability, usability and ease of use, content quality and relevance, perceived usefulness and learning effectiveness, accessibility and inclusiveness, and satisfaction and future use.

The quantitative responses were summarized using frequency, percentage, mean, and standard deviation. A five-point Likert scale was used, wherein mean scores

ranging from 4.21 to 5.00 were interpreted as Strongly Agree**,** 3.41 to 4.20 as Agree, 2.61 to 3.40 as Neutral, 1.81 to 2.60 as Disagree, and 1.00 to 1.80 as Strongly Disagree.

4.5.4.1 Profile of the Community End-User Respondents

Table 28. Demographic Profile of the Community End-User Respondents (N = 50)

VARIABLE	CATEGORY	FREQUENCY	PERCENTAGE
Age	Below 18	5	10%
	18–24	26	52%
	25–34	2	4%
	35–44	3	6%
	45–54	12	24%
	55 and above	2	4%
Gender	Male	30	60%
	Female	20	40%
	Prefer not to say	0	0%
Role	Student	30	60%
	Teacher/Instructor	2	4%
	Fire Safety Personnel	0	0%
	Community Resident	0	0%
	Office Staff/Employee	6	12%
	Other	12	24%
Prior Fire Safety Training	None	16	32%
	Once	11	22%

Table 28 continued…

VARIABLE	CATEGORY	FREQUENCY	PERCENTAGE
	2–3 times	15	30%
	More than 3 times	8	16%

Table 28 presents the demographic profile of the 50 community end-user respondents. The largest age group was 18–24 years old, comprising 26 respondents or 52.00%, while students accounted for 30 respondents or 60.00%. In terms of prior fire-safety training, 16 respondents or 32.00% reported having no previous training. These results describe the characteristics of the respondents who participated in the evaluation. They should not be interpreted as representing the entire population of Sta. Cruz, Laguna or as proof that the portal naturally attracts younger users. Barrios and Moreno (2024) discussed the relationship between digital government services and digital inclusion, while Jou et al. (2024) examined factors affecting accessibility of digital government applications. Jakaria (2025) also discussed digital inclusion and access challenges in the Philippine context. These studies provide context for considering age, digital access, and participation when developing public-facing digital services. However, the present sample does not establish why fewer older adults participated or whether the respondents without prior fire-safety training were unable to access traditional BFP seminars.

4.5.4.2 Application Use and Device Profile

Table 29. Application Use and Device Profile of the Community End-Users

VARIABLE	CATEGORY	FREQUENCY	PERCENTAGE
Frequency of Application Use	Only once	39	78%

Table 29 continued…

VARIABLE	CATEGORY	FREQUENCY	PERCENTAGE
	2–3 times	7	14%
	Weekly	2	4%
	More than once a week	2	4%
Digital Device Used	Smartphone	48	96%
	Tablet	3	6%
	Laptop	8	16%
	Desktop Computer	3	6%

Note. Respondents were allowed to select more than one device; therefore, the percentages for device use do not total 100%.

Table 29 shows that 39 respondents or 78.00% had used the application only once at the time of evaluation, while seven respondents or 14.00% had used it two to three times. Four respondents reported using the application weekly or more frequently. These results indicate that most participants evaluated the platform after limited exposure. Therefore, the succeeding satisfaction and acceptability findings should primarily be interpreted as reflecting the respondents' initial or short-term experience rather than sustained long-term use.

Smartphones were the most frequently selected devices, reported by 48 respondents or 96.00%. Laptops were selected by eight respondents or 16.00%, while tablets and desktop computers were each selected by three respondents or 6.00%. Since respondents could select more than one device, the percentages do not total 100.00%.

The high frequency of smartphone use supports the importance of maintaining a responsive interface for different screen sizes. Jou et al. (2024) examined accessibility factors in digital government applications, while Barrios and Moreno (2024) discussed digital inclusion in e-government services. Bokingkito et al. (2025) also examined the availability and accessibility of Philippine e-government services. These studies provide context for considering device access and digital accessibility when developing public-facing web applications.

4.5.4.3 Overall Community End-User Evaluation

Table 30. Summary of the Community End-User Evaluation of SafeScape V2

EVALUATION AREA	MEAN	SD	VERBAL INTERPRETATION
Overall Acceptability	4.32	0.90	Strongly Agree
Usability and Ease of Use	4.35	0.60	Strongly Agree
Content Quality and Relevance	4.41	0.67	Strongly Agree
Perceived Usefulness and Learning Effectiveness	4.44	0.79	Strongly Agree
Accessibility and Inclusiveness	4.29	0.71	Strongly Agree

Table 30 continued…

EVALUATION AREA	MEAN	SD	VERBAL INTERPRETATION
Satisfaction and Future Use	4.40	0.68	Very Satisfied
Overall Composite	4.37	0.67	Highly Favorable

Note. Agreement-based dimensions were interpreted using the following scale: 4.21–5.00 = Strongly Agree; 3.41–4.20 = Agree; 2.61–3.40 = Neutral; 1.81–2.60 = Disagree; and 1.00–1.80 = Strongly Disagree. The Satisfaction and Future Use dimension was interpreted using the corresponding satisfaction anchors: 4.21–5.00 = Very Satisfied; 3.41–4.20 = Satisfied; 2.61–3.40 = Neutral; 1.81–2.60 = Dissatisfied; and 1.00–1.80 = Very Dissatisfied. The term "Highly Favorable" is used only as a descriptive summary of the overall composite because the composite combines agreement-based and satisfaction-based dimensions.

Table 30 presents the summary of the community end-user evaluation of SafeScape V2. The overall composite mean was 4.37 (SD = 0.67), indicating a favorable evaluation among the 50 respondents. Perceived Usefulness and Learning Effectiveness received the highest mean of 4.44 (SD = 0.79), followed by Content Quality and Relevance at 4.41 (SD = 0.67) and Satisfaction and Future Use at 4.40 (SD = 0.68). Accessibility and Inclusiveness received the comparatively lowest mean of 4.29 (SD = 0.71), although it remained within the Strongly Agree interpretation. Ferrer (2025) discussed disaster preparedness and resilience in the Philippine context, while Parallag et al. (2023) examined a web-based emergency coordination application. Amil (2024) also discussed the role of community participation in disaster risk reduction.

These studies provide context for the use of digital platforms in supporting public information and preparedness activities. However, the composite mean represents the respondents' perceptions and should not be interpreted as objective proof that the system directly improved disaster preparedness or that the comparatively lower accessibility score was caused by Internet connectivity.

4.5.4.4 Overall Acceptability

Table 31. Community End-User Evaluation in Terms of Overall Acceptability

INDICATOR	MEAN	SD	VERBAL INTERPRETATION
General acceptability of the platform	4.50	0.89	Strongly Agree
Overall	4.32	0.90	Strongly Agree
Willingness to recommend the platform	4.40	0.99	Strongly Agree
Appropriateness of the platform	4.34	1.00	Strongly Agree
Willingness to use the platform again	4.36	1.03	Strongly Agree
Preference for using the application	3.98	1.08	Agree
Extent to which the platform met expectations	4.36	0.96	Strongly Agree

Legend: 4.21–5.00 = Strongly Agree; 3.41–4.20 = Agree; 2.61–3.40 = Neutral; 1.81–2.60 = Disagree; 1.00–1.80 = Strongly Disagree.

Table 31 shows that General Acceptability of the Platform received the highest mean of 4.50 (SD = 0.89), followed by Willingness to Recommend the Platform at 4.40 (SD = 0.99). The overall mean was 4.32 (SD = 0.90), interpreted as Strongly Agree. Preference for Using the Application received the comparatively lowest mean of 3.98 (SD = 1.08), interpreted as Agree.

The results indicate that respondents generally accepted the platform and expressed willingness to recommend and use it again. However, the comparatively lower rating for the preference item suggests that respondents were less likely to agree with using the application instead of purely face-to-face fire-safety lectures. Parallag et al. (2023) examined a web application for emergency coordination, while Ferrer (2025) discussed disaster preparedness and resilience in the Philippines. Amil (2024) also examined community participation in disaster risk reduction. These studies provide context for the use of digital tools in public-safety information dissemination. The findings of the present study nevertheless represent the respondents' reported preferences and do not establish that digital learning should replace face-to-face fire-safety instruction.

4.5.4.5 Usability and Ease of Use

Table 32. Community End-User Evaluation in Terms of Usability and Ease of Use

INDICATOR	MEAN	SD	VERBAL INTERPRETATION
Ease of understanding and navigating the interface	4.34	0.69	Strongly Agree
Clarity of buttons, menus, and icons	4.44	0.61	Strongly Agree
Ease of learning how to use the application	4.34	0.75	Strongly Agree
Ability to complete lessons and activities without confusion	4.26	0.66	Strongly Agree
System performance on the device used	4.30	0.79	Strongly Agree
Rare occurrence of technical errors	4.40	0.86	Strongly Agree
Overall	4.35	0.60	Strongly Agree

Legend: 4.21–5.00 = Strongly Agree; 3.41–4.20 = Agree; 2.61–3.40 = Neutral; 1.81–2.60 = Disagree; 1.00–1.80 = Strongly Disagree.

Table 32 shows that Clarity of Buttons, Menus, and Icons received the highest mean of 4.44 (SD = 0.61), followed by Rare Occurrence of Technical Errors at 4.40 (SD = 0.86). The overall mean was 4.35 (SD = 0.60), interpreted as Strongly Agree. Ability to Complete Lessons and Activities Without Confusion received the comparatively lowest mean of 4.26 (SD = 0.66), although it also remained within the Strongly Agree interpretation. These results indicate that respondents generally found the interface understandable and easy to use, while the completion of learning activities may still benefit from further refinement. Hussain et al. (2024) discussed human-centered design in evacuation applications, while Ren et al. (2023) examined digital game-based training and learning engagement. Barrios and Moreno (2024) also discussed accessibility and digital inclusion in e-government services. These studies provide context for the importance of clear navigation and user-centered interface design. However, the present results do not establish that Tailwind CSS alone caused the favorable usability ratings.

4.5.4.6 Content Quality and Relevance

Table 33. Community End-User Evaluation in Terms of Content Quality and Relevance

INDICATOR	MEAN	SD	VERBAL INTERPRETATION
Accuracy and trustworthiness of fire-safety content	4.48	0.71	Strongly Agree
Relevance of examples and scenarios to the user's environment	4.48	0.74	Strongly Agree
Clarity of actions before, during, and after a fire	4.44	0.76	Strongly Agree

Table 33 continued…

INDICATOR	MEAN	SD	VERBAL INTERPRETATION
Helpfulness of visual aids	4.34	0.77	Strongly Agree
Reinforcement of understanding through quizzes and activities	4.36	0.83	Strongly Agree
Appropriateness of lesson length	4.36	0.75	Strongly Agree
Overall	4.41	0.67	Strongly Agree

Legend: 4.21–5.00 = Strongly Agree; 3.41–4.20 = Agree; 2.61–3.40 = Neutral; 1.81-2.60 = Disagree; 1.00–1.80 = Strongly Disagree.

Table 33 shows that Accuracy and Trustworthiness of Fire-Safety Content and Relevance of Examples and Scenarios to the User's Environment tied for the highest mean of 4.48. Clarity of Actions Before, During, and After a Fire received a mean of 4.44, while Helpfulness of Visual Aids obtained the comparatively lowest mean of 4.34. The overall mean was 4.41 (SD = 0.67), interpreted as Strongly Agree. These results indicate that respondents generally viewed the fire-safety content as accurate, relevant, and understandable. Saflor and Kudhal (2026) discussed the importance of message clarity and community trust in digital warning systems, while Faiz et al. (2025) examined contextual learning in fire-hazard awareness campaigns. Ren et al. (2023) also examined digital learning interventions. These studies provide context for the use of relevant content and learning materials in educational platforms. However, the ratings reflect respondents' perceptions of the presented content and do not independently measure actual knowledge retention or changes in fire-safety behavior.

4.5.4.7 Perceived Usefulness and Learning Effectiveness

Table 34. Community End-User Evaluation in Terms of Perceived Usefulness and Learning Effectiveness

INDICATOR	MEAN	SD	VERBAL INTERPRETATION
Perceived increase in fire-prevention knowledge	4.46	0.91	Strongly Agree
Improved understanding of fire-emergency response	4.40	0.88	Strongly Agree
Confidence in fire-safety skills and preparedness	4.36	0.88	Strongly Agree
Increased awareness of fire hazards	4.44	0.88	Strongly Agree
Perceived contribution to reducing fire-related risks	4.40	0.90	Strongly Agree
Overall usefulness for fire-safety education	4.56	0.70	Strongly Agree
Overall	4.44	0.79	Strongly Agree

Legend: 4.21–5.00 = Strongly Agree; 3.41–4.20 = Agree; 2.61–3.40 = Neutral; 1.81–2.60 = Disagree; 1.00–1.80 = Strongly Disagree.

Table 34 shows that Overall Usefulness for Fire-Safety Education received the highest mean of 4.56 (SD = 0.70), while Confidence in Fire-Safety Skills and Preparedness received the comparatively lowest mean of 4.36 (SD = 0.88). The overall

mean was 4.44 (SD = 0.79), interpreted as Strongly Agree. These results indicate that respondents generally perceived the platform as useful for supporting fire-safety education. Ferrer (2025) discussed disaster preparedness and resilience in the Philippine context, while Abrar et al. (2025) examined AI-supported personalized learning pathways. Amil (2024) also discussed community participation in disaster risk reduction. These studies provide context for the use of digital learning tools in public-safety education. However, the present results measure perceived usefulness and self-reported learning-related outcomes; they do not establish that the portal independently improved actual emergency-response skills or replaced practical fire-safety training.

4.5.4.8 Accessibility and Inclusiveness

Table 35. Community End-User Evaluation in Terms of Accessibility and Inclusiveness

INDICATOR	MEAN	SD	VERBAL INTERPRETATION
Readability of text size, colors, and layout	4.30	0.81	Strongly Agree
Understandability for users with different digital skill levels	4.36	0.72	Strongly Agree
Consideration of users with diverse needs	4.32	0.74	Strongly Agree
Accessibility under limited Internet connectivity or data	4.08	0.94	Agree

Table 35 continued…

INDICATOR	MEAN	SD	VERBAL INTERPRETATION
Clarity of instructions and prompts	4.38	0.85	Strongly Agree
Overall	4.29	0.71	Strongly Agree

Legend: 4.21–5.00 = Strongly Agree; 3.41–4.20 = Agree; 2.61–3.40 = Neutral; 1.81–2.60 = Disagree; 1.00–1.80 = Strongly Disagree.

Table 35 shows that Clarity of Instructions and Prompts received the highest mean of 4.38 (SD = 0.85), followed by Understandability for Users with Different Digital Skill Levels at 4.36 (SD = 0.72). Accessibility under Limited Internet Connectivity or Data received the lowest mean of 4.08 (SD = 0.94), interpreted as Agree. The overall mean was 4.29 (SD = 0.71), interpreted as Strongly Agree. These results indicate that respondents generally viewed the platform as accessible and understandable, although access under limited Internet connectivity or data received a comparatively lower assessment. Prado et al. (2023) discussed accessibility considerations in digital education, while El Fakir et al. (2025) examined the use of Speech-to-Text and Text-to-Speech in educational settings. Bokingkito et al. (2025) also examined accessibility and digital service delivery in Philippine e-government websites. These studies provide context for considering accessibility, alternative interaction methods, and infrastructure limitations in digital platforms. The lower score in the present study identifies an area for further improvement but does not, by itself, establish the specific technical cause of the respondents' ratings.

4.5.4.9 Satisfaction and Future Use

Table 36. Community End-User Evaluation in Terms of Satisfaction and Future Use

INDICATOR	MEAN	SD	VERBAL INTERPRETATION
Overall satisfaction with the application	4.46	0.76	Very Satisfied
Satisfaction with application design and layout	4.52	0.61	Very Satisfied
Satisfaction with fire-safety content and activities	4.28	0.97	Very Satisfied
Satisfaction with ease of use and navigation	4.42	0.67	Very Satisfied
Satisfaction with accessibility	4.34	0.77	Very Satisfied
Overall	4.40	0.68	Very Satisfied

Legend: 4.21–5.00 = Very Satisfied; 3.41–4.20 = Satisfied; 2.61–3.40 = Neutral; 1.81–2.60 = Dissatisfied; 1.00–1.80 = Very Dissatisfied.

Table 36 shows that Satisfaction with Application Design and Layout received the highest mean of 4.52 (SD = 0.61), followed by Overall Satisfaction with the Application at 4.46 (SD = 0.76). Satisfaction with Fire-Safety Content and Activities received the comparatively lowest mean of 4.28 (SD = 0.97), although all indicators remained within the Very Satisfied interpretation. The overall mean was 4.40 (SD = 0.68). Ren et al. (2023) examined digital game-based training and learner engagement, while Ackerman et al. (2025) discussed human-centered design for underserved

populations. Faiz et al. (2025) also examined contextual learning in fire-hazard awareness campaigns. These studies provide context for considering interface design, accessibility, and interactive learning materials. However, the present ratings represent the respondents' satisfaction at the time of evaluation. The fact that most respondents had used the application only once does not establish the reason for the lower rating on fire-safety content and activities or predict whether users will continue using the system over time.

4.5.4.10 Supplementary End-User Comments and Suggestions

Table 37. Summary of Supplementary Community End-User Feedback

OPEN-ENDED QUESTION	DESCRIPTIVE CATEGORY	FREQUENCY	PERCENTAGE
Features Liked Most (n=28)	Interactive Learning Features	7	25.00%
	UI/UX and Visual Design	7	25.00%
	Usability and Accessibility	7	25.00%
	Overall Features/Functionality	4	14.29%
	Fire-Safety Content/Awareness	2	7.14%
	AI BFP Assistant	1	3.57%

Table 37 continued…

OPEN-ENDED QUESTION	DESCRIPTIVE CATEGORY	FREQUENCY	PERCENTAGE
Areas for Improvement (n=16)	UI/Visual Design	6	37.50%
	No Major Improvement/Satisfied	5	31.25%
	Typography/Readability	2	12.50%
	More Interactive Activities	2	12.50%
	Additional Content	1	6.25%
Influence on Fire-Safety Preparedness (n=23)	Increased Knowledge/Understanding	10	43.48%
	Improved Preparedness/Response	7	30.43%
	General Educational Benefit	2	8.70%

Table 37 continued…

OPEN-ENDED QUESTION	DESCRIPTIVE CATEGORY	FREQUENCY	PERCENTAGE
	Increased Confidence	2	8.70%
	Improved Retention Through Interactivity	1	4.35%
	No Perceived New Learning	1	4.35%
Other Comments/Suggestions (n=6)	Interactivity/Gamification	2	33.33%
	Positive Feedback/No Concern	2	33.33%
	UX/Simulation Improvement	1	16.67%
	Text Readability/Visibility	1	16.67%

Note. Percentages were calculated using only respondents who provided substantive responses to each open-ended question.

Table 37 summarizes the supplementary comments and suggestions provided by the community end-users. The number of responses differed across the open-ended questions; therefore, the percentages were calculated using only the respondents who provided substantive answers to each specific question.

For the features-liked question, 28 respondents provided answers. Interactive Learning Features, UI/UX and Visual Design, and Usability and Accessibility each received seven responses, representing 25.00% per category. These results indicate that the three categories were equally and most frequently mentioned as positive features of the platform. Ackerman et al. (2025) discussed the value of human-centered design for underserved populations, while Ren et al. (2023) examined digital game-based training and learning engagement. Prado et al. (2023) also discussed accessibility in digital education. These studies provide context for the importance of usability, accessibility, and interactive learning design. However, the responses in Table 37 represent the feedback of the present respondents and should not be interpreted as proof that the same preferences apply to all community users.

For the areas-for-improvement question, 16 respondents provided substantive answers. UI/Visual Design received the highest frequency, with six responses or 37.50%. No Major Improvement/Satisfied received five responses or 31.25%, while Typography/Readability and More Interactive Activities each received two responses or 12.50%. For the influence-on-fire-safety-preparedness question, 23 respondents provided answers. Ten respondents, equivalent to 43.48% of those who answered this question, mentioned Increased Knowledge/Understanding, while seven respondents or 30.43% reported Improved Preparedness/Response. These findings indicate that some respondents perceived benefits in their understanding and preparedness after using the portal.

For the other comments and suggestions question, six respondents provided substantive answers. Interactivity/Gamification and Positive Feedback/No Concern each received two responses, representing 33.33% per category. UX/Simulation Improvement and Text Readability/Visibility each received one response, or 16.67%. These responses provide additional suggestions for improving the platform's presentation and interactive features.

Overall, the supplementary feedback indicates that respondents appreciated the platform's interactive learning features, visual design, usability, and accessibility, while also identifying opportunities to improve its visual presentation, readability, and interactive activities. These findings should be interpreted according to the number of respondents who answered each open-ended question rather than as percentages of all 50 community end-users. Prado et al. (2023) discussed accessibility considerations in digital education, while Faiz et al. (2025) examined contextual learning in fire-hazard awareness campaigns. Ackerman et al. (2025) also discussed human-centered design for underserved populations. These studies provide context for the importance of accessibility and user-centered improvement, but they do not establish that the suggestions made by the present respondents will necessarily produce the same outcomes in other communities.

CHAPTER V

SUMMARY, CONCLUSIONS, AND RECOMMENDATIONS

This chapter presents the summary of findings, conclusions, and recommendations derived from the development, technical testing, and stakeholder evaluation of SafeScape V2: An AI-Driven Responsive Web Portal for Fire Safety Information Dissemination. The findings are summarized according to the five research objectives of the study. The conclusions are based on the results presented in Chapter IV, while the recommendations identify areas for continued improvement, validation, and possible future expansion of the developed system.

5.1 Summary of Findings

The study developed and evaluated SafeScape V2 as an enhanced fire-safety information and educational platform for BFP Santa Cruz, Laguna. The findings are organized according to the five research objectives.

1. Laravel Architecture and System Performance

SafeScape V2 was implemented using a Laravel-based architecture with React, Inertia.js, PostgreSQL, Redis, and Laravel Octane. All ten performance scenarios met their acceptance criteria, with response times ranging from 0.15 to 1.40 seconds. The 50-user simulated load test recorded an average response time of 0.54 seconds with a 0% error rate, while all eight controlled security tests passed.

2. RAG-Based BFP Assistant

The chatbot achieved 100.00% Response Accuracy, 100.00% Knowledge Retrieval Effectiveness, and a 0.00% Hallucination/Error Rate across 15 controlled fire-safety questions. BFP personnel also provided a favorable chatbot domain-validation mean of 4.78 (SD = 0.31).

3. Accessibility and Voice-Enabled Functions

The STT component achieved 100.00% Recognition Accuracy and 0.00% WER for the predefined utterances, while all five TTS scenarios and all eleven accessibility/user-task scenarios were successfully completed. Community end-users also provided favorable perceptions regarding usability, accessibility, ease of interaction, and overall acceptability.

4. Adaptive Learning and Administrative Dashboard

The Module Quiz Difficulty Model achieved 91.80% accuracy, while the Final Examination Difficulty Model achieved 98.40% accuracy. All six adaptive-learning integration scenarios and all six dashboard verification areas passed. However, both classification models were developed and evaluated using synthetic learner-performance data; therefore, these accuracy results reflect performance under the simulated data conditions of the study and should not be generalized as equivalent performance on actual learner data.

5. Overall System Evaluation

All 85 predefined internal binary test cases passed under the controlled testing conditions established in the study. The ten IT experts gave the system an overall ISO/IEC 25010:2023 mean of 4.23 (SD = 0.15), the BFP personnel gave an operational evaluation mean of 4.88 (SD = 0.28), and the 50 community end-users gave an overall composite mean of 4.37 (SD = 0.67), indicating generally favorable technical, professional, and user evaluations.

5.2 Conclusions

Based on the findings of the study, the researchers conclude that SafeScape V2 was successfully developed and evaluated as an enhanced fire-safety information and educational platform for BFP Santa Cruz, Laguna. The conclusions are presented according to the five research objectives of the study.

1. The Laravel-based redevelopment provided a suitable technical foundation for SafeScape V2 and satisfied the defined performance, controlled scalability, and security requirements. Further optimization of performance efficiency, maintainability, and flexibility may still improve the system.

2. The RAG-based BFP Assistant could provide relevant and grounded fire-safety responses within the evaluated scope. Its continued reliability depends on the quality and maintenance of the underlying fire-safety knowledge base.

3. The STT, TTS, responsive interface, and accessibility-oriented functions supported alternative methods of interacting with the system. The results indicate favorable accessibility and usability under the specific browser, device, speech, and testing conditions used in the study.

4. The two Random Forest models successfully supported Easy, Medium, and Hard assessment-difficulty classification, while the administrative dashboard fulfilled its monitoring and analytics functions. However, the models require future validation using actual learner data because their current performance was measured using synthetic datasets.

5. The combined internal testing, IT-expert evaluation, BFP validation, and community end-user evaluation indicate that SafeScape V2 satisfactorily met the technical, domain-specific, accessibility, and user-acceptability requirements examined in the study.

5.3 Recommendations

Based on the findings, conclusions, identified limitations, and feedback obtained from the evaluators, the following recommendations are proposed for the continued improvement and future development of SafeScape V2:

1. Validate the adaptive-learning models using actual learner data

Future researchers and system developers are encouraged to retrain and evaluate the Module Quiz Difficulty Model and Final Examination Difficulty Model using accumulated real learner assessment and interaction records. Although the models demonstrated high classification performance, their current results are based on synthetic learner-performance datasets and may differ when applied to actual users.

2. Continue updating and validating the RAG chatbot knowledge base

The fire-safety knowledge resources used by the BFP Assistant should be periodically reviewed and updated in coordination with BFP personnel to maintain the factual accuracy and operational relevance of its responses. Future versions may also provide clearer references or citations to the corresponding BFP guidelines or fire-safety sources when appropriate, reflecting the supplementary recommendations provided by the IT experts.

3. Improve system performance, code organization, maintainability, and scalability

Future development should continue optimizing application response time, database operations, caching, and code structure. The IT-expert evaluation identified performance optimization and code organization as recurring areas for refinement, with maintainability and organization receiving particular attention in the supplementary recommendations.

4. Conduct broader compatibility, security, and load testing

Future testing should involve additional browsers, devices, network environments, and larger concurrent-user workloads. Further vulnerability assessment, stress testing, and cross-browser verification of the Speech-to-Text functionality are also recommended to provide stronger evidence of system reliability and security

beyond the controlled conditions used in the present study. IT-expert feedback specifically identified cross-browser STT compatibility and additional security/stress validation as areas for consideration.

5. Continue improving accessibility and interface usability

Future versions should further refine text readability, interface presentation, interactive educational activities, and operation under limited Internet connectivity. Community end-users rated accessibility under limited Internet connectivity or data comparatively lower than the other accessibility indicators, indicating an area that may benefit from additional optimization.

6. Conduct longer-term user evaluation

Future studies should evaluate SafeScape V2 after users have interacted with the system over a longer period. In the present evaluation, 39 of the 50 community respondents, or 78%, had used the application only once; therefore, future evaluations may examine sustained usability, satisfaction, engagement, and learning-related outcomes after repeated system use.

7. Consider gradual expansion after further validation

Continued operation within Sta. Cruz, Laguna should be accompanied by monitoring, content refinement, and stakeholder feedback. Possible expansion to additional schools, communities, or other municipalities may be considered after sufficient technical validation, stakeholder coordination, resources, and localized fire-safety content are available. BFP feedback also identified continued operation and possible local expansion as areas for future consideration.

8. Conduct formal standards assessment when required for future government deployment

Since the present study considered relevant DICT Government Web Hosting Service requirements only as architectural alignment criteria, a future production or government-hosted implementation may undergo a separate formal GWHS compliance or certification assessment when required by the appropriate government authority.

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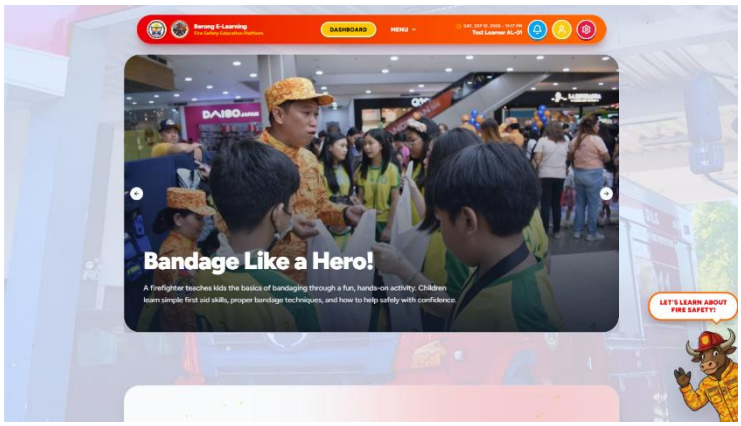
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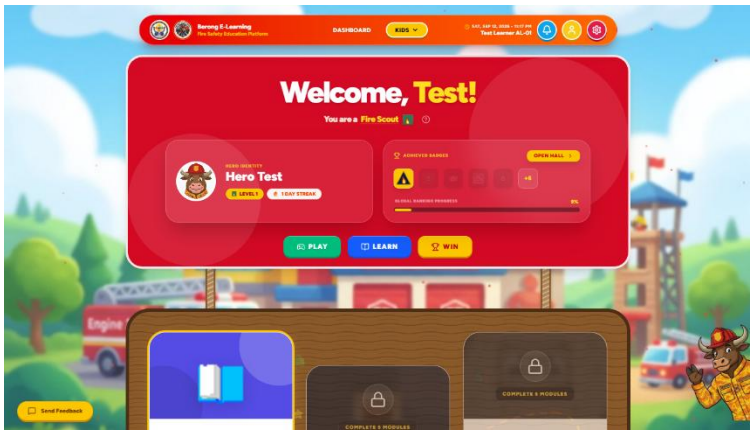
APPENDIX A

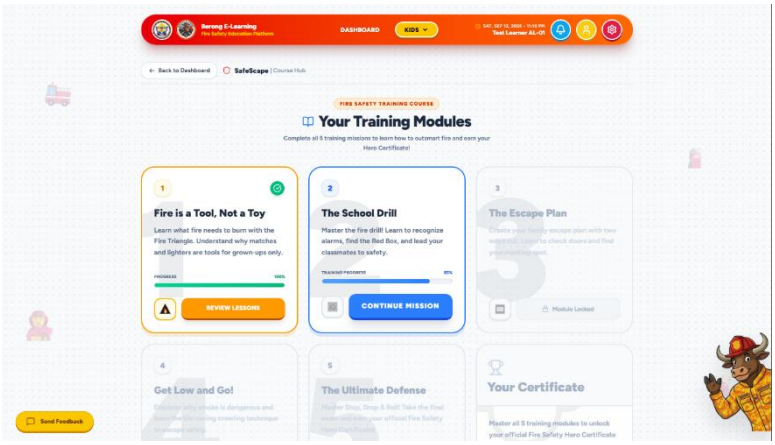
(TECHNICAL BACKGROUND)

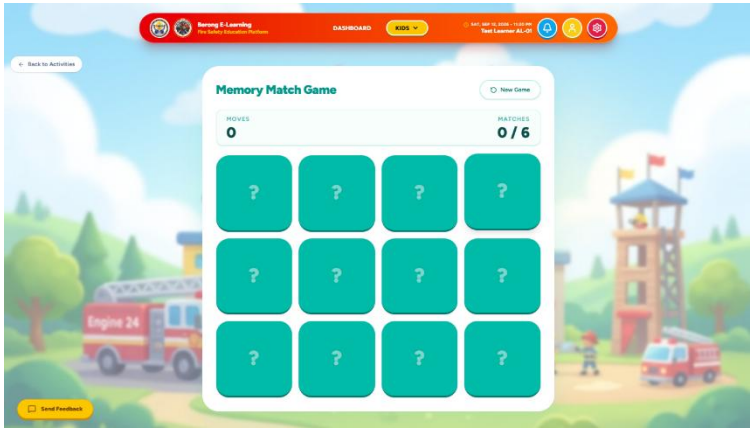
Planning & Requirement Analysis Phase

SAFESCAPE V2 Web Application Storyboard (Kid / Learner)

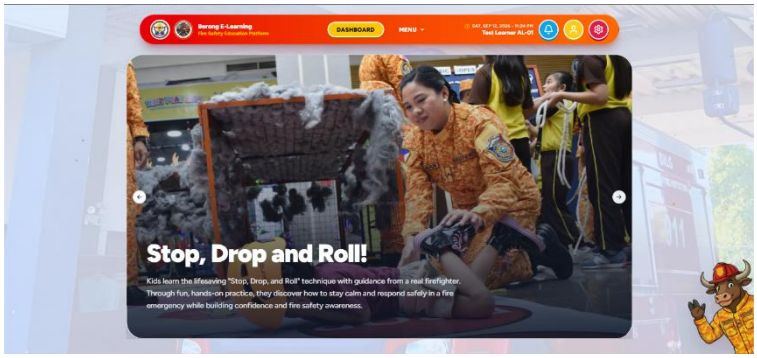


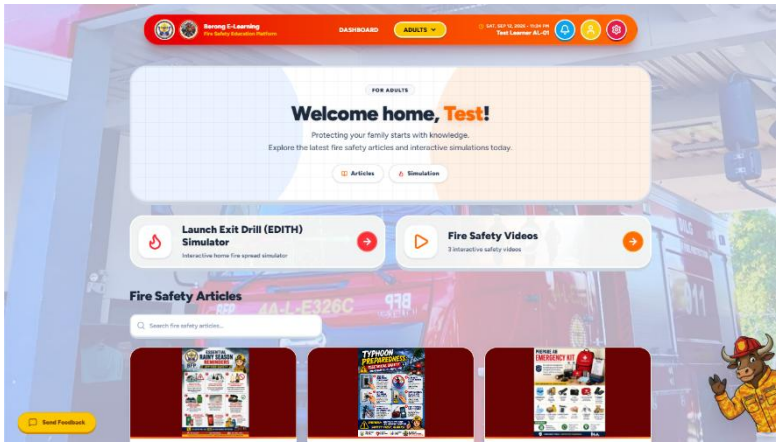


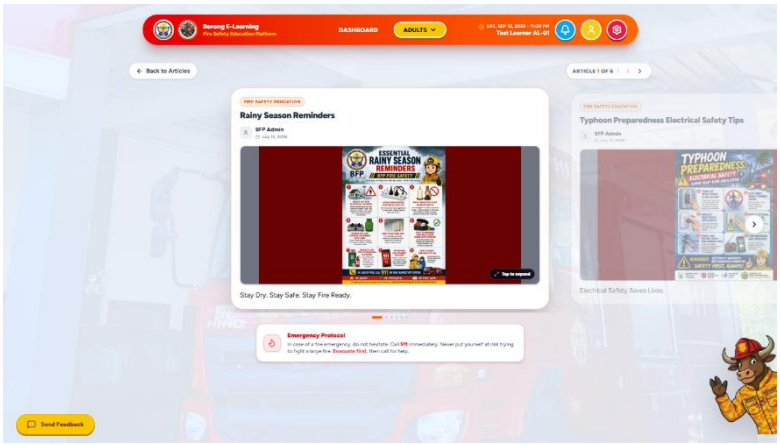


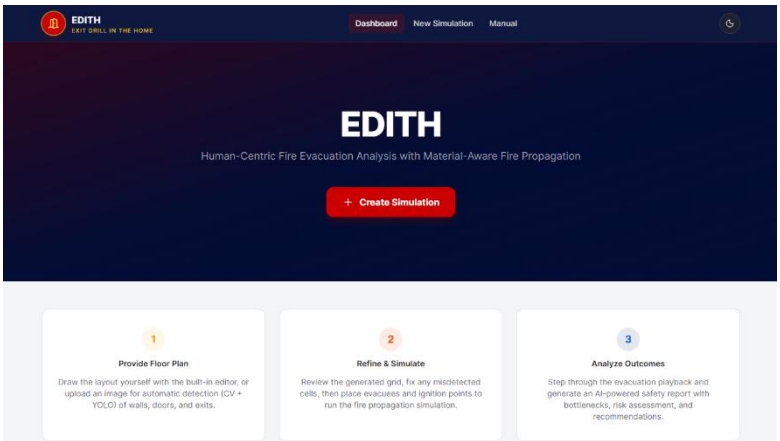


SAFESCAPE V2 Web Application Storyboard (Adult)

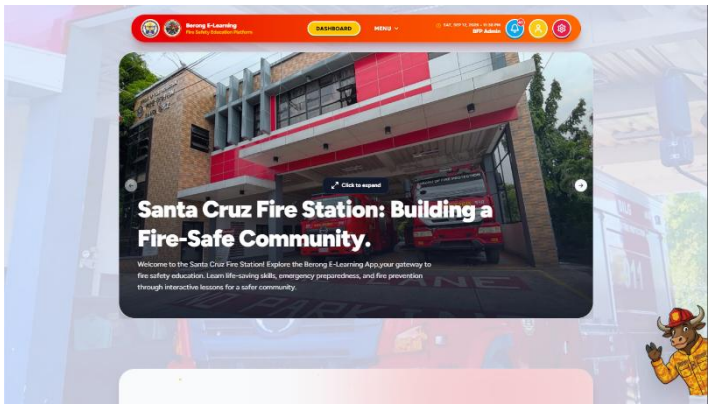


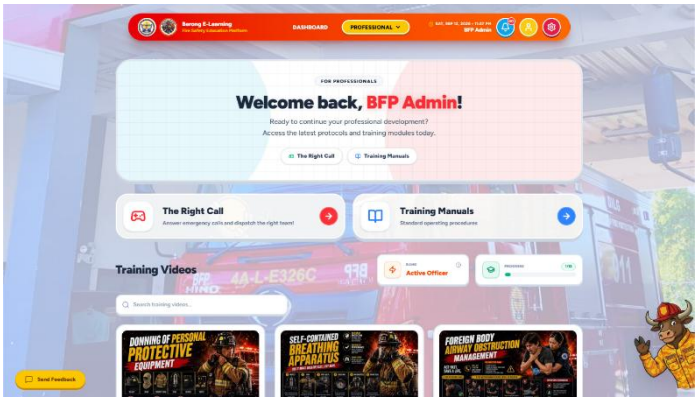


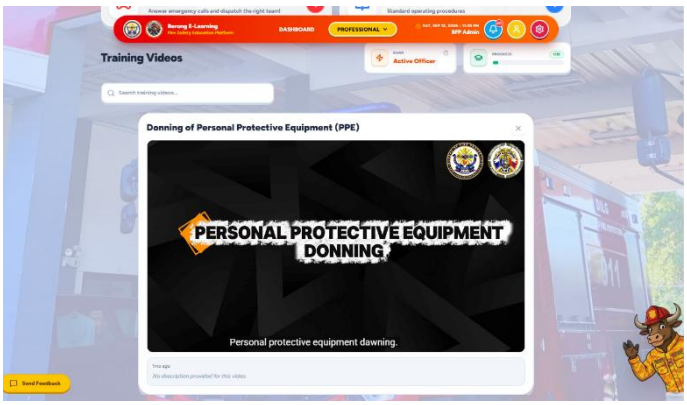


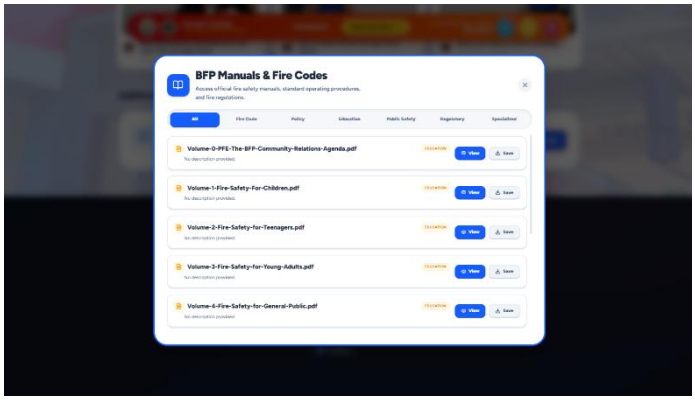


SAFESCAPE V2 Web Application Storyboard (BFP Personnel)

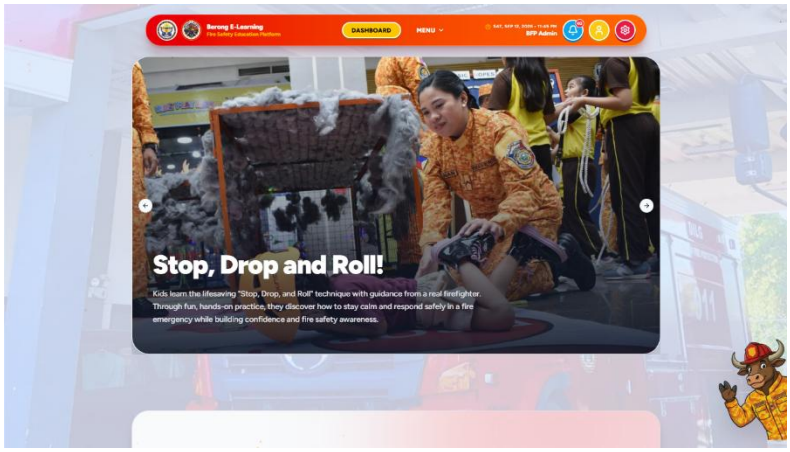


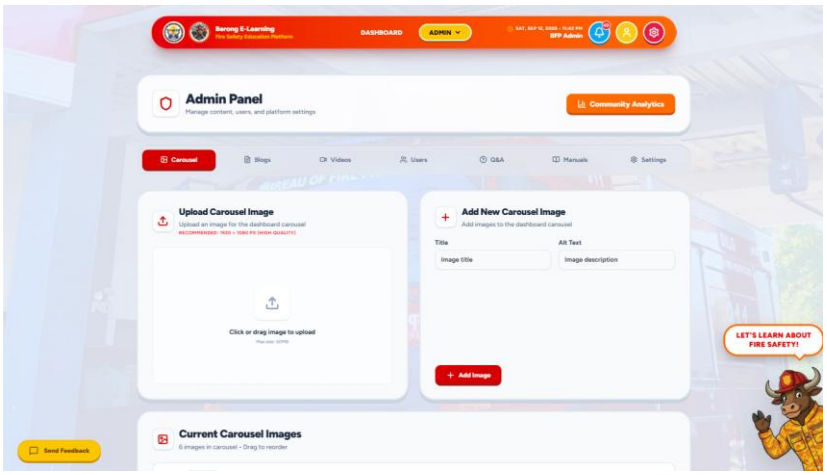


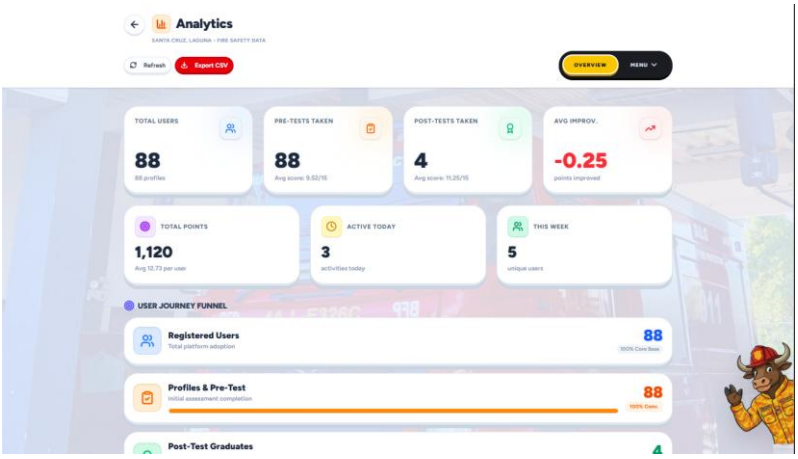




SAFESCAPE V2 Web Application Storyboard (Administrator)

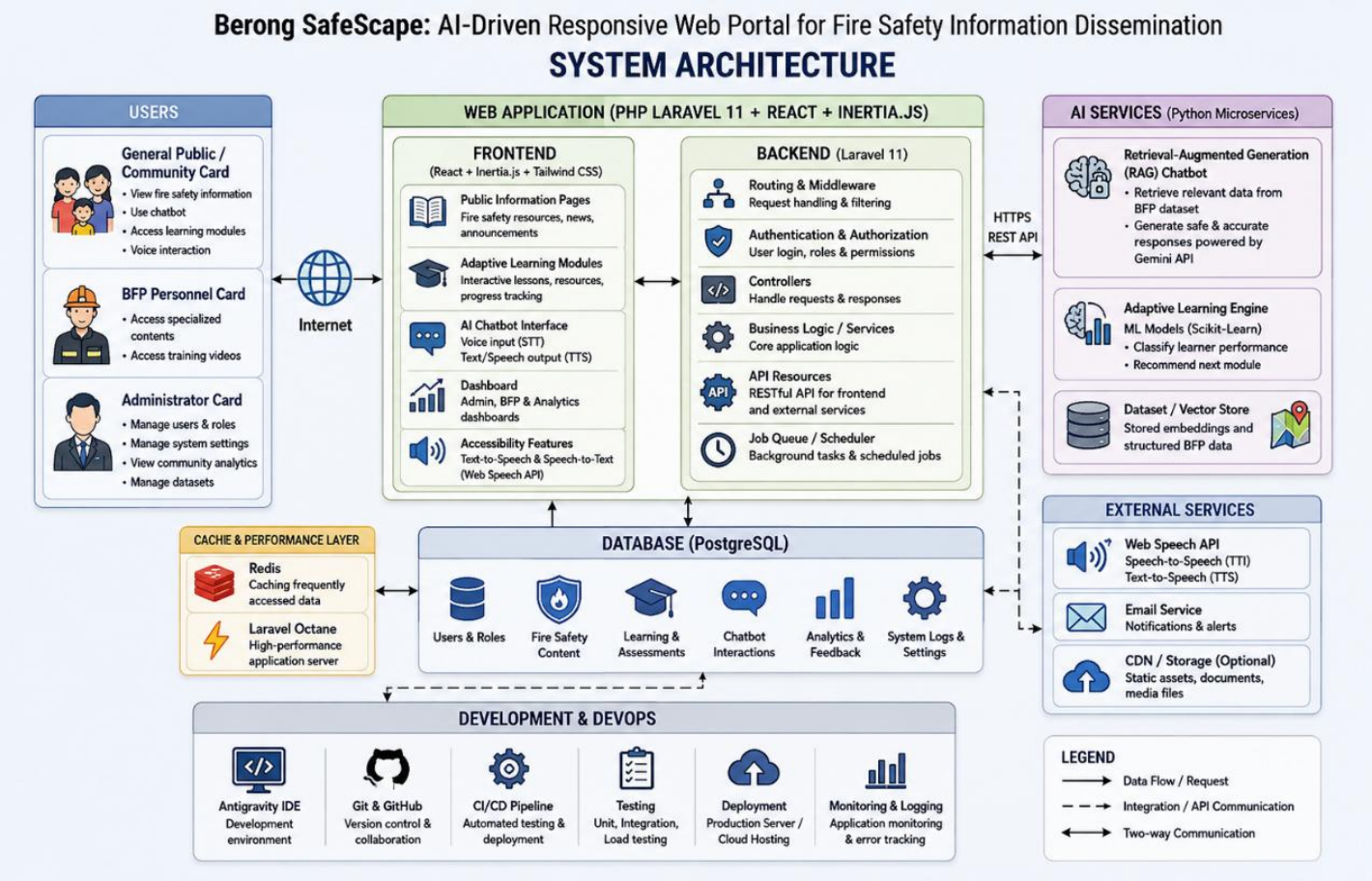






Specification and Design Phase

System Architecture



Project Schedule

Gantt Chart

SafeScape V2

Phase	Main Activities	April 2026	May 2026	June 2026	July 2026	August 2026	September 2026
Requirements & Planning	BFP consultation; review existing portal; requirements gathering; literature review						
System & UI Design	Storyboard; system architecture; database and interface planning						
Core System Development	Laravel 11 migration; React/Inertia frontend; PostgreSQL; authentication and content pages						
AI & Accessibility Integration	RAG chatbot + Gemini API; STT/TTS; Random Forest adaptive learning						
Integration, Testing & Revision	System integration; functional, performance and security testing; bug fixing						
Stakeholder Evaluation	BFP validation; IT expert ISO/IEC 25010:2023 evaluation; community end-user evaluation						
Analysis & Finalization	Statistical analysis, interpretation of results, and manuscript finalization						

Hardware and Software

Resources

Hardware Resources

- 512Gb SSD
- 16GB RAM
- Ryzen 5 8645HS
- NVIDIA RTX 3050 6GB

Software Resources

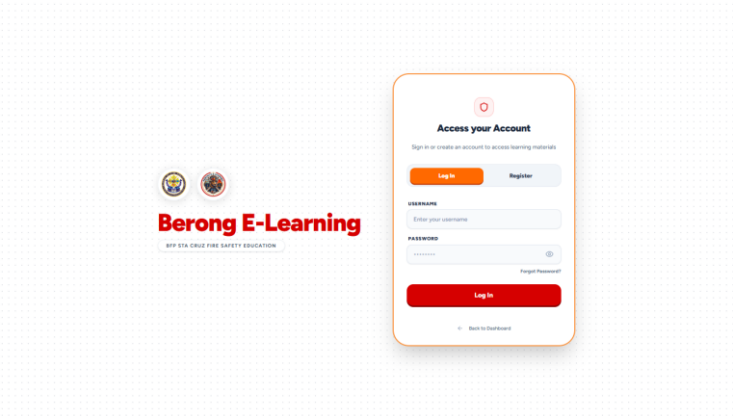
- Antigravity IDE
- PHP 8.2+
- Laravel 11
- React
- TypeScript / JavaScript
- Inertia.js
- Vite
- Tailwind CSS
- PostgreSQL
- Redis
- Laravel Octane
- Python
- Scikit-learn
- Google Gemini API
- Web Speech API
- Git and GitHub
- Google Chrome / Microsoft Edge

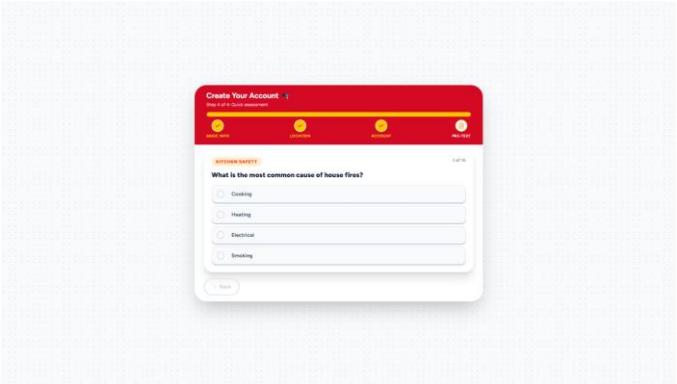
Development and Testing Tools

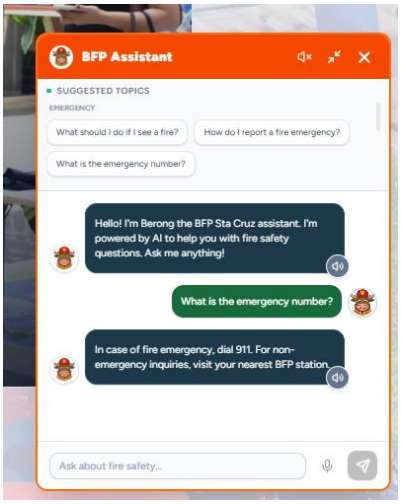
- PHPUnit
- Laravel Pint
- Faker
- Mockery
- Npm / Node.js

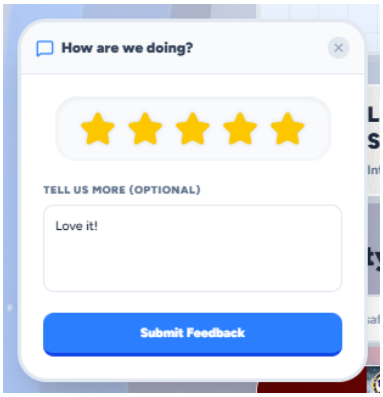
Input/Output/Reports Screen Shots

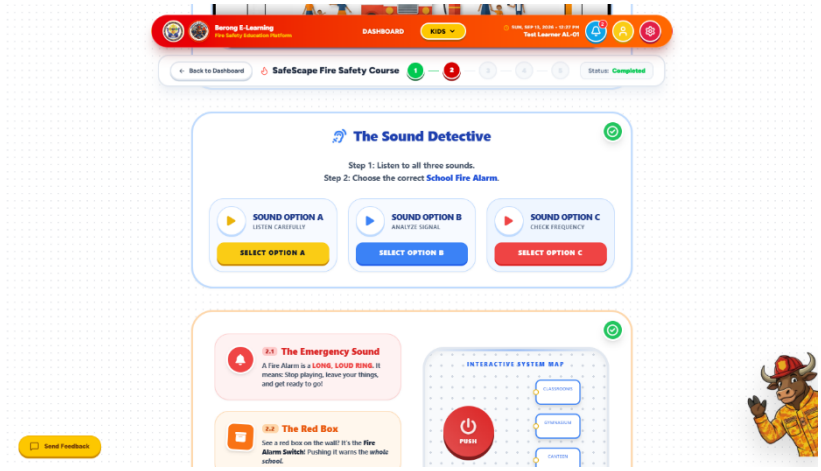
Input and Output Screens

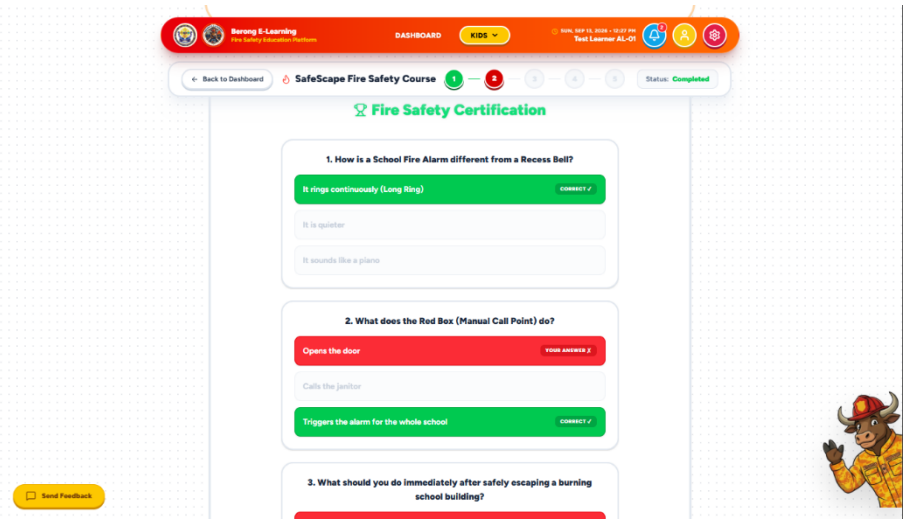


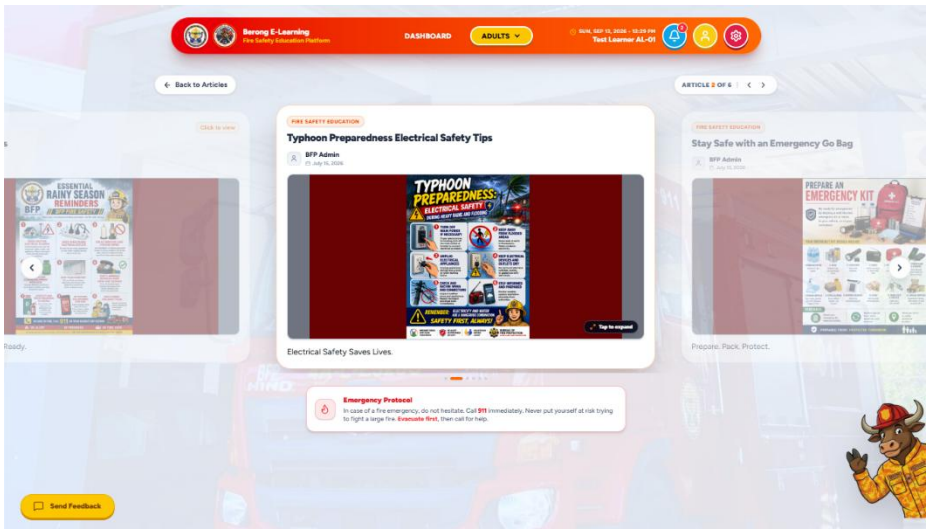




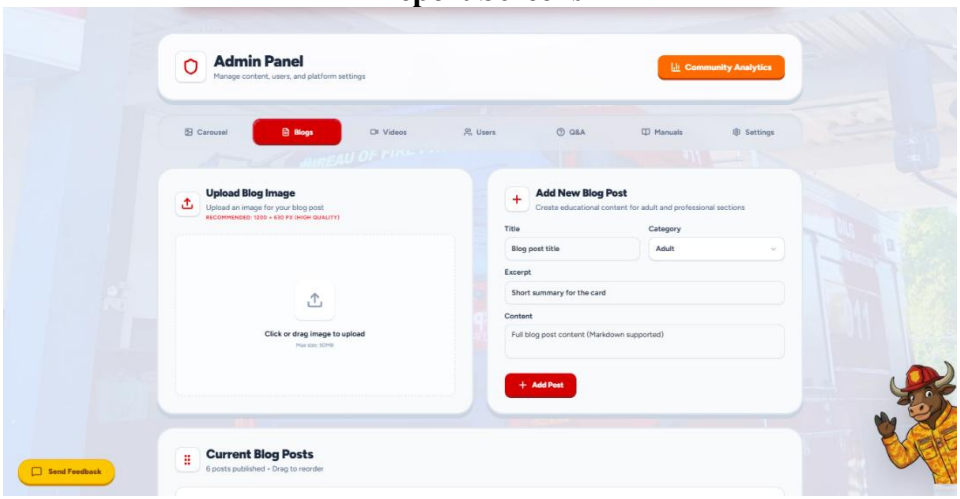


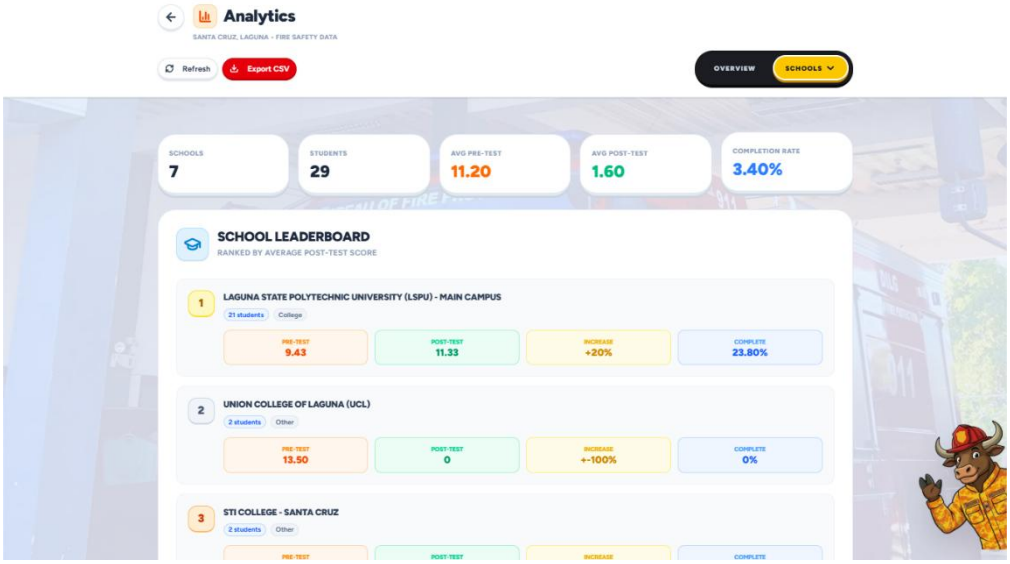

Berong E-Learning
DASHBOARD
KIDS
Back to Dashboard
SafeScape Fire Safety Course
Status: Completed
The Sound Detective
Step 1: Listen to all three sounds.
Step 2: Choose the correct School Fire Alarm.
SOUND OPTION A
LISTEN CAREFULLY
SELECT OPTION A
SOUND OPTION B
ANALYZE SIGNAL
SELECT OPTION B
SOUND OPTION C
CHECK FREQUENCY
SELECT OPTION C
2.1 The Emergency Sound
A Fire Alarm is a LONG, LOUD RING. It means: Stop playing, leave your things, and get ready to go!
2.2 The Red Box
See a red box on the wall? It's the Fire Alarm Switch! Pushing it warns the whole school.
INTERACTIVE SYSTEM MAP
PUSH
Send Feedback

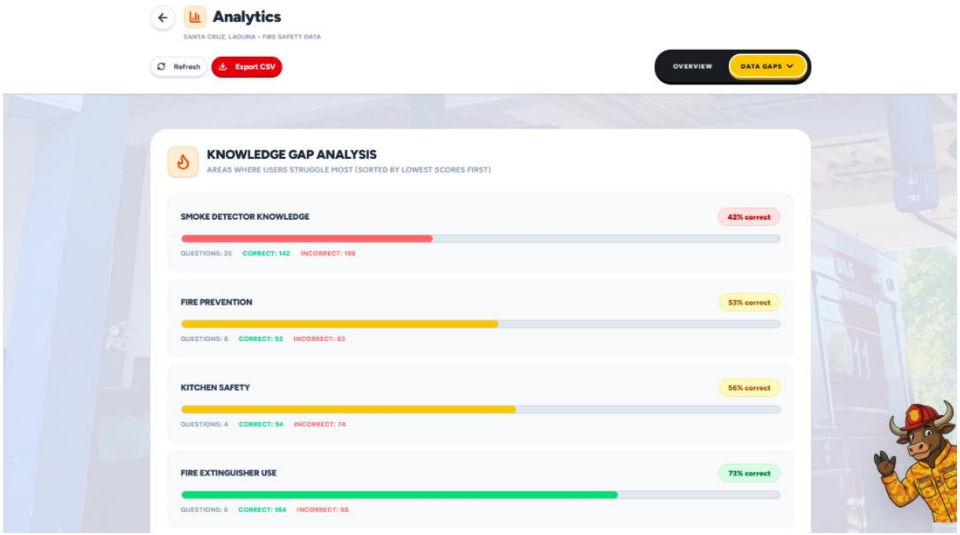

Berong E-Learning
DASHBOARD
KIDS
Back to Dashboard
SafeScape Fire Safety Course
Status: Completed
Fire Safety Certification
1. How is a School Fire Alarm different from a Recess Bell?
It rings continuously (Long Ring)
CORRECT ✓
It is quieter
It sounds like a piano
2. What does the Red Box (Manual Call Point) do?
Opens the door
YOUR ANSWER ✗
Calls the janitor
Triggers the alarm for the whole school
CORRECT ✓
3. What should you do immediately after safely escaping a burning school building?
Send Feedback


Berong E-Learning
DASHBOARD
ADULTS
Back to Articles
ARTICLE 2 OF 6
Typhoon Preparedness Electrical Safety Tips
BFP Admin
TYPHOON PREPAREDNESS
ELECTRICAL SAFETY
Tap to expand
Electrical Safety Saves Lives.
Emergency Protocol
Stay Safe with an Emergency Go Bag
PREPARE AN EMERGENCY KIT
Send Feedback

Report Screens







Testing & Evaluation Instruments

ENHANCED BERONG SAFESCAPE WEB PORTAL
INDIVIDUAL RESEARCHER/DEVELOPER SYSTEM TESTING AND VERIFICATION TOOL

I. TESTING RECORD

Information	Response
System Version / Build:	
Date of Testing:	
Testing Environment / Device:	
Test Data / Dataset Used:	

II. PURPOSE OF THE INSTRUMENT

This Individual Researcher/Developer System Testing and Verification Tool is designed to systematically assess the functionality, performance, reliability, security, accessibility, artificial intelligence capabilities, adaptive learning functions, and overall technical operation of the enhanced Berong SafeScape Web Portal.

Sections III-XIV of this instrument shall be conducted only by the Researcher/Developer during internal testing and pre-deployment verification. No external evaluator or IT expert shall conduct these internal system tests. The Researcher/Developer shall use designated test accounts, controlled test records, and non-production, anonymized, or aggregate data whenever possible to avoid exposing the personal information of existing users. These internal tests are separate from the ISO/IEC 25010:2023 evaluation instruments beginning in Section XV, which remain unchanged.

III. SYSTEM FUNCTIONAL TESTING

Purpose: This test verifies whether the major functions of the Enhanced Berong SafeScape Web Portal operate according to their defined functional requirements**.**

Instructions for the Researcher/Developer: Execute each functional test case using the specified test input or condition. Use designated test accounts and test records only. Record the actual result and compare it with the expected result. Mark the test as Pass when the actual result satisfies the expected result and Fail when the expected behavior is not achieved. Document any identified defect, unexpected behavior, or deviation in the Remarks column.

Test Case ID	Module/Function	Test Scenario	Test Input	Expected Result	Actual Result	Status	Remarks
FT-01	Registration	Register a new user	Valid test registration information	Account is successfully created			
FT-02	Registration	Submit incomplete registration	Required field omitted	System displays appropriate validation message			

Test Case ID	Module/Function	Test Scenario	Test Input	Expected Result	Actual Result	Status	Remarks
FT-03	Login	Login using valid credentials	Valid test username/password	User successfully accesses account			
FT-04	Login	Login using invalid credentials	Incorrect password	Access is denied and appropriate message displayed			
FT-05	User Profile	Update profile	Valid test profile information	Information is successfully updated			
FT-06	Fire Safety Content	View fire safety information	Select information category	Correct content is displayed			
FT-07	Search	Search for fire safety information	Valid keyword	Relevant results are returned			
FT-08	Educational Module	Open learning module	Select module	Correct module loads successfully			
FT-09	Educational Module	Complete learning activity	Valid responses	Activity is processed and result recorded			
FT-10	Assessment	Submit assessment	Completed assessment	Score is correctly calculated and recorded			
FT-11	Chatbot	Submit fire safety question	Valid question	Chatbot provides appropriate response			
FT-12	STT	Submit voice input	Valid spoken command/question	Speech is converted to text			
FT-13	TTS	Activate text-to-speech	Select readable content	Text is converted to speech			
FT-14	Feedback	Submit feedback	Valid feedback	Feedback is recorded successfully			

Test Case ID	Module/Function	Test Scenario	Test Input	Expected Result	Actual Result	Status	Remarks
FT-15	Dashboard	View user statistics	Authorized researcher/developer admin test account	Current statistics are displayed			
FT-16	Dashboard	Generate report	Select report parameters	Correct report is generated			
FT-17	Adaptive Learning	Complete learning activity	User completes module	Learning profile/progress is updated			
FT-18	User Management	manage user	Authorized researcher/developer admin test account	User record is successfully managed			
FT-19	Access Control	Access restricted page	Unauthorized user	Access is denied			
FT-20	Notification	Publish announcement	Authorized administrator	Announcement is displayed to intended users			
FT-21	Database	Save transaction	Valid test transaction	Data is correctly stored			
FT-22	Database	Retrieve transaction	Existing test record	Correct record is retrieved			
FT-23	Logout	Logout from account	Select logout	Session is terminated			
FT-24	Error Handling	Submit invalid data	Invalid input	System handles error appropriately without crashing			

IV. SYSTEM INTEGRATION TESTING

Purpose: This test verifies that the major software components of the Enhanced Berong SafeScape Web Portal correctly exchange data and operate together as an integrated system.

Test ID	Components	Integration Scenario	Expected Result	Actual Result	Status
INT-01	Laravel – PostgreSQL	Save test user or learning data through the portal	Submitted test data is correctly stored in PostgreSQL and associated with the correct test account		
INT-02	PostgreSQL – Laravel	Retrieve an existing test user, learning, or dashboard record	Correct test database information is retrieved and displayed by the application		
INT-03	Laravel – Gemini API/RAG	Submit a fire safety question through the chatbot	Request is successfully transmitted to the RAG/Gemini service and a response is returned to the portal		
INT-04	Laravel – Random Forest Service	Submit learner-performance data requiring difficulty prediction	Learner data is processed by the classification service and the predicted difficulty is correctly returned to the portal		
INT-05	External AI Service Failure	Simulate an unavailable, failed, or timed-out external AI request	System handles the failure safely, displays an appropriate fallback/error message, and does not crash		

V. AI CHATBOT TESTING TOOL

B. Gemini API RAG Chatbot Researcher Verification

The chatbot shall be tested using a validated set of fire safety questions derived from BFP-approved or otherwise authoritative fire safety knowledge resources.

AI Response Rating Scale:

5 – Excellent / Completely Accurate
4 – Very Good / Mostly Accurate
3 – Good / Partially Accurate
2 – Fair / Mostly Inaccurate
1 – Poor / Incorrect or Misleading

Response Accuracy Test

Test ID	Category	Test Question	Factually Correct? Y/N	Accuracy (1-5)	Contextual Relevance (1-5)	Completeness (1-5)	Hallucination/ Error? Y/N	Relevant Retrieved Reference? Y/N	Researcher Remarks
AI-01	Prevention	What are common causes of residential fires?							
AI-02	Prevention	How can electrical fires be prevented?							
AI-03	Extinguisher	How should a fire extinguisher be used?							
AI-04	Emergency Response	What should a person do when a fire occurs?							
AI-05	Evacuation	What should you do during a building fire?							
AI-06	LPG Safety	How should LPG cylinders be handled safely?							
AI-07	Electrical Safety	What should be done when an electrical appliance catches fire?							

Test ID	Category	Test Question	Factually Correct? Y/N	Accuracy (1-5)	Contextual Relevance (1-5)	Completeness (1-5)	Hallucination/ Error? Y/N	Relevant Retrieved Reference? Y/N	Researcher Remarks
AI-08	Children	What should children do when they discover a fire?							
AI-09	Smoke	Why should a person stay low when escaping smoke?							
AI-10	Emergency	What information should be provided when reporting a fire emergency?							
AI-11	Prevention	How can fires be prevented during cooking?							
AI-12	Evacuation	Should someone return to a burning building to retrieve belongings?							
AI-13	General Safety	What are important fire safety practices at home?							
AI-14	Community	What can communities do to improve fire preparedness?							

Test ID	Category	Test Question	Factually Correct? Y/N	Accuracy (1-5)	Contextual Relevance (1-5)	Completeness (1-5)	Hallucination/ Error? Y/N	Relevant Retrieved Reference? Y/N	Researcher Remarks
AI-15	Emergency	When should a person evacuate during a fire?							

AI Chatbot Evaluation Summary

Response Accuracy (%)

Response Accuracy = (Number of Factually Correct Responses ÷ Total Number of Test Questions) × 100

Knowledge Retrieval Effectiveness (%)

Knowledge Retrieval Effectiveness = (Number of Queries with Relevant Retrieved References ÷ Total Number of Test Questions) × 100

Hallucination/Error Rate (%)

Hallucination/Error Rate = (Number of Responses with Hallucination/Error ÷ Total Number of Test Questions) × 100

VI. SPEECH-TO-TEXT TESTING

C. Speech Recognition Test The following test shall be conducted using predefined spoken commands or fire safety questions.

Test ID	Spoken Utterance	Expected Transcription	Actual Transcription	Status (Pass/Fail)	Error Details / Remarks
STT-01	Fire safety at home	Fire safety at home			
STT-02	What causes electrical fires?	What causes electrical fires?			
STT-03	How do I use a fire extinguisher?	How do I use a fire extinguisher?			
STT-04	What should I do during a fire?	What should I do during a fire?			
STT-05	How can I prevent a fire?	How can I prevent a fire?			
STT-06	Where is the nearest fire station?	Where is the nearest fire station?			
STT-07	Fire evacuation procedures	Fire evacuation procedures			
STT-08	LPG safety practices	LPG safety practices			

Test ID	Spoken Utterance	Expected Transcription	Actual Transcription	Status (Pass/Fail)	Error Details / Remarks
STT-09	Emergency fire response	Emergency fire response			
STT-10	Fire prevention tips	Fire prevention tips			

Speech-to-Text Evaluation Summary

The completed Speech-to-Text test cases shall be summarized using Recognition Accuracy and Word Error Rate.

Recognition Accuracy (%)

Recognition Accuracy = (Number of Correctly Transcribed Utterances ÷ Total Number of Utterances) × 100

Word Error Rate (WER)

WER = (Substitutions + Deletions + Insertions) ÷ Total Number of Words in the Reference Transcription

Where:

Substitutions (S) = words incorrectly replaced by another word

Deletions (D) = reference words omitted from the system transcription

Insertions (I) = additional words appearing in the transcription

N = total number of words in the reference transcription

The resulting WER may be multiplied by 100 to express the result as a percentage.

VII. TEXT-TO-SPEECH TESTING

D. TTS Functional Test

Test ID	Test Condition	Expected Result	Status (Pass/Fail)	Remarks (Explain if Failed)
TTS-01	Read short text	Text is converted to clear, understandable speech.		
TTS-02	Read long paragraph	Complete content is spoken without cutting off.		
TTS-03	Read fire safety terminology	Technical terms (e.g., "extinguisher", "combustible") are pronounced reasonably well.		
TTS-04	Stop speech	Playback stops completely and resets to the beginning.		
TTS-05	Test Text-to-Speech using the supported desktop and mobile browser/device environment	Text-to-Speech activates and produces understandable speech on the tested supported browser/device environment without causing the portal to malfunction		

VIII. ACCESSIBILITY AND USER-TASK FUNCTIONAL TESTING

E. Researcher-Simulated User Task Testing

1.

Task ID	User Task	Status (Pass/Fail)	Issue Observed? (Yes/No)	Remarks / Errors Encountered
UT-01	Register an account			
UT-02	Log in			
UT-03	Find a fire safety article			
UT-04	Ask the chatbot a question			
UT-05	Use Speech-to-Text			
UT-06	Listen to content using Text-to-Speech			
UT-07	Open a learning module			
UT-08	Complete a learning activity			
UT-09	View learning progress			
UT-10	Submit feedback			
UT-11	Navigate using a mobile device			

IX. ADAPTIVE LEARNING ENGINE TESTING

F. Adaptive Learning Test Scenarios (Random Forest Evaluation)

Question Difficulty Reference Key: The system dynamically serves different question variants based on the Random Forest model's classification. To verify the assigned difficulty, the Researcher/Developer shall observe the specific phrasing of the first few questions presented:

- EASY Classification Indicator: The system asks direct, simple questions (e.g., "What are the three things fire needs to burn (The Fire Triangle)?" or "What should you do if you find matches or a lighter lying around?").
- MEDIUM Classification Indicator: The system asks slightly more analytical questions (e.g., "Which specific elements must combine to form the Fire Triangle?" or "If you see a lighter on a table, why tell an adult?").
- HARD Classification Indicator: The system asks exclusion or reasoning questions (e.g., "Which of the following is NOT part of the Fire Triangle?" or "Why must you never try to pick up a lighter?").

Test ID	Testing Phase	Learner Condition (Inputs)	Expected Difficulty Classification	Observed Difficulty Classification	Correct (Y/N)	Remarks
AL-01	Module Quiz	**Account:** test_al01 (Age: 9, Grade: 3, Pre-Test: 4/15)	Easy			
AL-02	Module Quiz	**Account:** test_al02 (Age: 14, Grade: 8, Pre-Test: 8/15)	Medium			
AL-03	Module Quiz	**Account:** test_al03 (Age: 18, Grade: 12, Pre-Test: 13/15)	Hard			
AL-04	Final Exam	**Account:** test_al06 (Pre-Test: 14/15, Module Avg: 5/5)	Hard			
AL-05	Final Exam	**Account:** test_al07 (Pre-Test: 4/15, Module Avg: 5/5)	Medium			
AL-06	Final Exam	**Account:** test_al08 (Pre-Test: 14/15, Module Avg: 1/5)	Easy			

X. ADMINISTRATIVE DASHBOARD TESTING

G. Dashboard Validation

Test ID	Dashboard Function	Expected Result	Status (Pass/Fail)	Data Verified? (Y/N)	Remarks (Explain if Failed)
AD-01	Total users	Displays correct aggregate total user count without exposing personal user data			
AD-02	Learning participation	Displays correct participation			
AD-03	Assessment results	Displays correct results			
AD-04	User feedback	Displays recorded feedback			
AD-05	Reports	Generates correct report			
AD-06	Data filtering	Displays correct filtered results			

XI. PERFORMANCE TESTING

H. Performance Test Record

Performance Testing Basis

The performance testing of the Enhanced Berong SafeScape Web Portal is guided by the Performance Efficiency characteristic of ISO/IEC 25010:2023, particularly the subcharacteristics of time behaviour and capacity. Response time is measured to determine how efficiently the system performs its functions under specified conditions, while concurrent-user testing is conducted to observe system behavior under a defined simulated load. The numerical response-time and load thresholds used in this study are researcher-defined acceptance criteria for the developed system and are not presented as numerical limits prescribed by ISO/IEC 25010:2023.

Test ID	Function	Test Load (Users/Reqs)	Researcher-Defined Acceptance Criterion	Actual Response Time	Error Rate (%)	Status (Pass/Fail)	Remarks
PT-01	Homepage	1 User / 10 Reqs	≤ 3 seconds average				
PT-02	Login	1 User / 5 Reqs	≤ 2 seconds average				

Test ID	Function	Test Load (Users/Reqs)	Researcher-Defined Acceptance Criterion	Actual Response Time	Error Rate (%)	Status (Pass/Fail)	Remarks
PT-03	Search	1 User / 10 Reqs	≤ 2 seconds average				
PT-04	Learning module	1 User / 5 Reqs	≤ 3 seconds average				
PT-05	Database retrieval	1 User / 10 Reqs	≤ 1 second average				
PT-06	Chatbot	1 User / 10 Reqs	≤ 7 seconds average				
PT-07	Dashboard	1 User / 5 Reqs	≤ 3 seconds average				
PT-08	Report generation	1 User / 3 Reqs	≤ 5 seconds average				
PT-09	Concurrent load (Stress)	50 Users (Simulated)	≤ 5 seconds average				
PT-10	Mobile access	1 User (3G/4G sim)	≤ 4 seconds average				

XII. SECURITY TESTING

I. Security Test Cases

Test ID	Security Area	Test Scenario	Expected Result	Status (Pass/Fail)	Remarks (Explain if Failed)
SEC-01	Authentication	Enter invalid login credentials	Access is denied and an appropriate generic error message is displayed.		

Test ID	Security Area	Test Scenario	Expected Result	Status (Pass/Fail)	Remarks (Explain if Failed)
SEC-02	Authorization	Regular user attempts to access an administrator-only URL/function	Access is denied and the user is redirected to an appropriate page or login page.		
SEC-03	Authorization	Administrator accesses an authorized administrative function	Access is successfully granted.		
SEC-04	Input Validation	Submit potentially malicious or invalid input through applicable system fields	Input is rejected, sanitized, or safely handled without executing unintended code or causing the system to crash.		
SEC-05	Session	Log out, then attempt to access a protected page/function again	The authenticated session is no longer valid and protected functions require reauthentication.		
SEC-06	Data Protection / Access Control	Using two designated test accounts, attempt to access one test account's protected information from the other through URL or parameter manipulation	Unauthorized access to protected test-account data is prevented.		
SEC-07	File Upload	Attempt to upload an unsupported or invalid file type	The file is rejected and an appropriate validation message is displayed.		
SEC-08	Error Handling	Submit an invalid parameter or otherwise trigger a controlled application error	A safe error response is displayed, and sensitive application/database details are not exposed.		

XIII. INDIVIDUAL RESEARCHER/DEVELOPER OVERALL TEST RESULT AND RECOMMENDATION

Overall Result	Check One
Acceptable / Passed	☐
Conditionally Acceptable – Revision Required	☐
Not Acceptable – Major Revision Required	☐

General Comments / Recommendations:

_____
_____
_____
_____
_____
_____
_____
_____

XIV. INDIVIDUAL RESEARCHER/DEVELOPER TEST COMPLETION

The Researcher/Developer confirms that the testing activities recorded in Sections III-XIII were conducted using the specified Enhanced Berong SafeScape Web Portal test procedures and controlled test accounts/data, and that the reported results reflect the observed system behavior during internal testing.

Role	Confirmation / Signature	Date
Researcher/Developer		

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ISO/IEC 25010:2023 SOFTWARE QUALITY EVALUATION

Fire E-Learning App User Acceptability Survey

Purpose: This survey aims to assess users' acceptability, usability, and satisfaction with the fire e-learning app, including its effectiveness in improving fire safety knowledge and preparedness. Your responses will guide improvements to the app.

Participation: Your participation is voluntary. All responses are confidential and will be analyzed in aggregate form only.

Instructions: Please answer all questions honestly based on your experience using the fire e-learning app. For Likert-scale questions, tick the option that best represents your opinion.

Section A. Respondent Profile

1. Age
◯ Below 18
◯ 18–24
◯ 25–34
◯ 35–44
◯ 45–54
◯ 55 and above

2. Gender
◯ Female
◯ Male
◯ Prefer not to say

3. Role / Affiliation
◯ Student
◯ Teacher / Instructor
◯ Fire safety personnel
◯ Community resident
◯ Office staff / employee
◯ Other (please specify): _____

4. Prior fire safety training (before using the app)
◯ None
◯ Once
◯ 2–3 times
◯ More than 3 times

5. Digital device used to access the app (select all that apply)
☐ Smartphone
☐ Tablet
☐ Laptop
☐ Desktop computer

6. Frequency of using the fire e-learning app
◯ Only once
◯ 2–3 times
◯ Weekly
◯ More than once a week

Section B. Overall Acceptability

Please rate your agreement for items 7–12.

Scale:
1 – Strongly Disagree | 2 – Disagree | 3 – Neutral | 4 – Agree | 5 – Strongly Agree

7. I find the fire e-learning app acceptable as a tool for learning about fire safety.
◯ 1 ◯ 2 ◯ 3 ◯4 ◯ 5

8. I would recommend this fire e-learning app to others.
◯ 1 ◯ 2 ◯ 3 ◯4 ◯ 5

9. Using this app is an appropriate way to deliver fire safety training for people like me.
◯ 1 ◯ 2 ◯ 3 ◯4 ◯ 5

10. I am willing to use this fire e-learning app again in the future.
◯ 1 ◯ 2 ◯ 3 ◯4 ◯ 5

11. I prefer using this app compared to purely face-to-face lectures about fire safety.
◯ 1 ◯ 2 ◯ 3 ◯4 ◯ 5

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12. Overall, the fire e-learning app meets my expectations.
○ 1 ○ 2 ○ 3 ○4 ○ 5

Section C. Usability and Ease of Use

Scale:
1 – Strongly Disagree | 2 – Disagree | 3 – Neutral | 4 – Agree | 5 – Strongly Agree

13. The app interface is easy to understand and navigate.
○ 1 ○ 2 ○ 3 ○4 ○ 5
14. The buttons, menus, and icons are clearly labeled and understandable.
○ 1 ○ 2 ○ 3 ○4 ○ 5
15. It was easy for me to learn how to use the app without needing much help.
○ 1 ○ 2 ○ 3 ○4 ○ 5
16. I can complete lessons and activities without getting lost or confused.
○ 1 ○ 2 ○ 3 ○4 ○ 5
17. The app works well on the device I am using (e.g., loads quickly, responds properly).
○ 1 ○ 2 ○ 3 ○4 ○ 5
18. I rarely experience technical errors (e.g., crashing, freezing, broken links) while using the app.
○ 1 ○ 2 ○ 3 ○4 ○ 5

Section D. Content Quality and Relevance

Scale:
1 – Strongly Disagree | 2 – Disagree | 3 – Neutral | 4 – Agree | 5 – Strongly Agree

19. The fire safety content is accurate and trustworthy.
○ 1 ○ 2 ○ 3 ○4 ○ 5
20. The examples and scenarios used in the app are relevant to my environment (home, school, office, community).
○ 1 ○ 2 ○ 3 ○4 ○ 5
21. The lessons clearly explain what to do before, during, and after a fire.
○ 1 ○ 2 ○ 3 ○4 ○ 5
22. Visual aids (images, diagrams, videos) help me understand fire safety procedures better.
○ 1 ○ 2 ○ 3 ○4 ○ 5
23. Quizzes and activities help reinforce my understanding of fire safety.
○ 1 ○ 2 ○ 3 ○4 ○ 5
24. The length of each lesson is appropriate (not too long, not too short).
○ 1 ○ 2 ○ 3 ○4 ○ 5

Section E. Perceived Usefulness and Learning Effectiveness

Scale:
1 – Strongly Disagree | 2 – Disagree | 3 – Neutral | 4 – Agree | 5 – Strongly Agree

25. The app increases my knowledge about fire prevention.
○ 1 ○ 2 ○ 3 ○4 ○ 5
26. The app improves my understanding of how to respond during a fire emergency.
○ 1 ○ 2 ○ 3 ○4 ○ 5
27. The app makes me feel more confident about my fire safety skills and preparedness.
○ 1 ○ 2 ○ 3 ○4 ○ 5
28. I am more aware of fire hazards in my surroundings after using the app.
○ 1 ○ 2 ○ 3 ○4 ○ 5
29. The app can help reduce fire-related risks in my home, school, or workplace if people use it.
○ 1 ○ 2 ○ 3 ○4 ○ 5
30. Overall, the fire e-learning app is useful for fire safety education.
○ 1 ○ 2 ○ 3 ○4 ○ 5

Section F. Accessibility and Inclusiveness

Scale:
1 – Strongly Disagree | 2 – Disagree | 3 – Neutral | 4 – Agree | 5 – Strongly Agree

31. The text size, colors, and layout are comfortable for reading.
○ 1 ○ 2 ○ 3 ○4 ○ 5
32. The app is understandable for users with different levels of digital skills.
○ 1 ○ 2 ○ 3 ○4 ○ 5
33. The app's design considers users with diverse needs (e.g., older adults, persons with disabilities).
○ 1 ○ 2 ○ 3 ○4 ○ 5

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34. I can access the app even with limited internet connectivity or data.
○ 1 ○ 2 ○ 3 ○4 ○ 5
35. Instructions and prompts are clear and easy to follow.
○ 1 ○ 2 ○ 3 ○4 ○ 5

Section G. Satisfaction and Future Use

Scale:
1 – Very Dissatisfied | 2 – Dissatisfied | 3 – Neutral | 4 – Satisfied | 5 – Very Satisfied

36. Overall satisfaction with the fire e-learning app.
○ 1 ○ 2 ○ 3 ○4 ○ 5
37. Satisfaction with app design and layout.
○ 1 ○ 2 ○ 3 ○4 ○ 5
38. Satisfaction with fire safety content and activities.
○ 1 ○ 2 ○ 3 ○4 ○ 5
39. Satisfaction with ease of use and navigation.
○ 1 ○ 2 ○ 3 ○4 ○ 5
40. Satisfaction with accessibility (device compatibility, connectivity needs, inclusiveness).
○ 1 ○ 2 ○ 3 ○4 ○ 5

Section H. Open-Ended Feedback

41. What did you like most about the fire e-learning app?

_____
_____

42. What aspects of the app do you think need improvement (e.g., content, design, language, activities)?

_____
_____

43. In your own words, how has this app influenced your fire safety behavior or preparedness?

_____
_____

44. Any other comments or suggestions?

_____
_____

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BERONG SAFESCAPE: TECHNICAL SOFTWARE QUALITY EVALUATION

Purpose: This evaluation form aims to assess selected technical software quality characteristics of the Enhanced Berong SafeScape Web Portal based on the ISO/IEC 25010:2023 software product quality model. The selected characteristics used in this study are Functional Suitability, Performance Efficiency, Reliability, Security, Maintainability, and Flexibility.

Participation: Participation is voluntary. All responses shall be treated confidentially and summarized for research purposes.

Instructions: Please evaluate the system based on your technical assessment and interaction with the developed portal. For each statement, select the rating that best represents your professional evaluation.

SECTION A: EVALUATOR PROFILE

1. Years of IT/Software Experience:

○ 1–3 years
○ 4–6 years
○ 7+ years

2. Primary Area of Expertise:

○ Software Development
○ Database Administration
○ System Architecture
○ Cybersecurity
○ Artificial Intelligence / Machine Learning
○ Other: _____

SECTION B: ISO/IEC 25010:2023 SOFTWARE QUALITY EVALUATION

Rating Scale:

5 = Excellent / Strongly Agree
4 = Very Good / Agree
3 = Good / Moderately Agree
2 = Fair / Disagree
1 = Poor / Strongly Disagree

1. Functional Suitability

Indicator	5	4	3	2	1
1. The system provides the functions required to support its intended fire safety information and educational activities.					
2. The system produces correct and appropriate results for its intended functions.					
3. The implemented functions are appropriate for the tasks performed by the intended users and administrators.					

2. Performance Efficiency

Indicator	5	4	3	2	1
4. The system responds within an acceptable period during normal user interaction.					
5. The chatbot and other API-dependent functions respond within an acceptable period under normal operating conditions.					
6. The system remains sufficiently responsive when multiple requests or users access the portal concurrently.					

3. Reliability

Indicator	5	4	3	2	1
7. The system operates consistently during normal use.					
8. User information, assessment results, and learning progress are reliably stored and retrieved.					
9. The system appropriately handles errors or service failures without causing unnecessary loss of data or application failure.					

4. Security

Indicator	5	4	3	2	1
10. Authentication and authorization mechanisms appropriately restrict access to protected system functions.					
11. The system provides appropriate protection against unauthorized access to user information and restricted records.					
12. User input, session handling, and application errors are managed in a manner that reduces the exposure of sensitive system information.					

5. Maintainability

Indicator	5	4	3	2	1
13. The system architecture and components are organized in a manner that supports maintenance and future modification.					
14. Individual system components can be modified or updated without unnecessarily affecting unrelated functions.					
15. The organization of the system supports the identification, testing, and correction of software defects.					

6. Flexibility

Indicator	5	4	3	2	1
16. The system can accommodate changes in operational or user requirements without requiring a complete redesign.					
17. The system architecture allows additional modules, functions, or services to be integrated when future requirements arise.					
18. The portal can operate appropriately across the supported devices and usage environments defined by the study.					

3. Overall Evaluation

Indicator	5	4	3	2	1
19. The system is technically suitable for use by its intended stakeholders, subject to the identified deployment requirements and limitations.					
20. Overall, I am satisfied with the technical quality of the Enhanced Berong SafeScape Web Portal.					

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Province of Laguna

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SECTION C: OPEN-ENDED FEEDBACK

21. What are the major technical strengths of the Enhanced Berong SafeScape Web Portal?

_____
_____

22. What technical limitations or defects did you identify?

_____
_____

23. What improvements are recommended regarding system performance, reliability, and security?

_____
_____

24. What improvements are recommended regarding maintainability and future system expansion?

_____
_____

25. What improvements are recommended for the AI chatbot, adaptive learning, or related system integrations?

_____
_____

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XVII. BFP PERSONNEL EVALUATION FORM

BERONG SAFESCAPE: DOMAIN & OPERATIONAL VALIDATION

Purpose: This survey aims to assess the operational relevance, factual accuracy of the fire safety content, and the administrative usefulness of the enhanced Berong SafeScape Web Portal.

Instructions: Please evaluate the system based on your professional experience in fire safety and emergency response.

SECTION A: EVALUATOR PROFILE

1. Rank / Position: _____

2. Years in Service:

◯ 1–3 years
◯ 4–6 years
◯ 7–10 years
◯ 10+ years

SECTION B: AI CHATBOT ACCURACY & RELEVANCE

Please test the AI chatbot by asking fire safety questions and rate the responses below.

Rating Scale:

5 = Completely accurate
4 = Mostly accurate
3 = Partially accurate
2 = Mostly inaccurate
1 = Incorrect / Misleading

Fire Safety Topic Tested	Accuracy (1-5)	Contextual Relevance (1-5)	Completeness (1-5)	Remarks / Corrections Needed
Fire Prevention				
Fire Extinguisher Use				
Evacuation Procedures				
LPG / Electrical Safety				

SECTION C: SYSTEM OPERATIONAL EVALUATION

Rating Scale:

5 = Excellent / Strongly Agree
4 = Very Good / Agree
3 = Good / Moderately Agree
2 = Fair / Disagree
1 = Poor / Strongly Disagree

Indicator	5	4	3	2	1
1. The system provides the functions necessary for fire safety information dissemination.					
2. The chatbot provides appropriate fire safety information.					
3. The educational modules operate according to their intended purpose.					
4. The administrative dashboard provides the required monitoring functions.					
5. The system improves access to fire safety information.					
6. The system provides an effective platform for community fire safety engagement.					

SECTION D: OPEN-ENDED FEEDBACK

7. How well does the system support fire safety information dissemination and community engagement?

_____

8. Are the fire safety contents and chatbot responses appropriate for BFP-related information needs?

_____

_____

9. What additional administrative or monitoring features should be considered?

_____

_____

10. What improvements are recommended before operational deployment?

_____

_____

Photo Documentation of System Evaluation


















2026.09.02 14:32

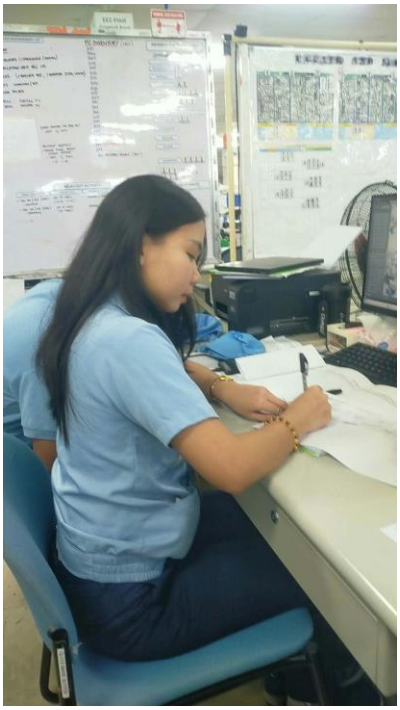

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2026.09.09 16:00


2026.09.02 14:32





Implementation Plan

Development Environment and Configuration

The development of SafeScape V2 was conducted using a structured development environment to support the integration of web technologies, artificial intelligence components, accessibility functions, and database services. Antigravity IDE was used as the primary development environment for managing and maintaining the PHP, Laravel, React, JavaScript, Python, and configuration files of the system.

- **Version Control:** Git and GitHub were utilized for source-code management, version tracking, collaboration, and management of changes throughout the iterative development of the system.

- **Package Management:** Node.js and npm were used to manage the frontend packages and dependencies required by React, Inertia.js, Vite, and Tailwind CSS. PHP dependencies were managed through Composer, while Python libraries required for machine-learning development were managed within the Python development environment.

System Architecture

SafeScape V2 implements a web architecture centered on Laravel 11, React, Inertia.js, PostgreSQL, and integrated artificial intelligence services.

- **Frontend:** The user-facing interface was developed using React with Inertia.js and Tailwind CSS. The frontend provides interfaces for community users, learners, BFP personnel, and administrators and supports responsive interaction across supported devices.

- **Backend:** Laravel 11 serves as the primary backend framework responsible for routing, authentication, authorization, business logic, session management, database operations, and communication with integrated external services.

- **Database:** PostgreSQL is used as the primary relational database for storing user accounts, fire-safety content, learning and assessment records, feedback, chatbot interactions, analytics, and administrative information. Redis and Laravel Octane are integrated to support caching and improved application performance.

Core Artificial Intelligence Implementation

- **Retrieval-Augmented Generation Chatbot:** The BFP Assistant implements a Retrieval-Augmented Generation architecture powered by the Google Gemini API. User queries are converted into embeddings and compared with the

localized fire-safety knowledge base through similarity retrieval. Relevant contextual information is then supplied to the Gemini API to generate a grounded fire-safety response.

- **Adaptive Learning Engine:** Two Random Forest classification models were implemented using Python and Scikit-learn. The Module Quiz Difficulty Model processes the learner's Age, Grade Level, and Pre-Assessment Score to determine an appropriate Easy, Medium, or Hard difficulty level. The Final Examination Difficulty Model additionally considers Module 1 to Module 4 quiz scores to determine the appropriate final-examination difficulty.

Accessibility Implementation

SafeScape V2 integrates the browser-based Web Speech API to provide voice-enabled interaction. Speech-to-Text allows supported users to provide spoken input, while Text-to-Speech enables selected information and chatbot responses to be presented through synthesized speech. These functions complement the system's responsive and user-centered interface.

User Interface Implementation

The SafeScape V2 interface was implemented using React, Inertia.js, and Tailwind CSS to provide responsive and role-appropriate interfaces.

The system provides specialized interfaces for community users and learners, including fire-safety modules, educational videos, games, assessments, fire-safety articles, the BFP Assistant, and applicable evacuation-simulation functions. BFP personnel are provided access to specialized modules and training videos, while authorized administrators access the administrative dashboard, user management, content management, assessment analytics, engagement information, and feedback-monitoring functions.

Deployment Strategy

SafeScape V2 was configured for deployment and use within the intended Sta. Cruz, Laguna setting. Before operational use, the application configuration, database connectivity, environment variables, integrated APIs, authentication controls, and AI components are verified to ensure that the required services function correctly.

Database migration and application configuration are performed according to the selected hosting environment. Sensitive credentials, including API keys, are stored through server-side environment configuration rather than being exposed through the frontend. The system may be further refined and expanded as additional operational requirements and actual learner-interaction data become available.

User's Manual

BERONG E-LEARNING PLATFORM

OFFICIAL USER MANUAL

A Joint Project of the Laguna State Polytechnic University - Santa Cruz Campus (LSPU-SCC) College of Computer Studies (CCS) and the Bureau of Fire Protection (BFP) - Santa Cruz, Laguna

Developers:

Justin Angelo A. Luzande

Kean Gabriel E. Salvahan

Prince Genel R. Umali

Project Adviser:

Mia V. Villarica, DIT

System Expert:

Mark P. Bernardino, MSCS

I. GETTING STARTED & ACCOUNT REGISTRATION

The Berong E-Learning platform uses an intelligent routing system to ensure every user gets the right educational experience based on their age.

Step 1: Access the Platform: Navigate to bfpscberong.app on your web browser.

Step 2: Create an Account: Click on Sign In, then select Register.

Step 3: Enter Details: Fill in the required fields. Important: You must enter your exact birth date. The system uses this data to automatically assign public users to either the Kids or Adults dashboard based on age.

Step 4: Login & Role Routing: Log in using your registered credentials. Public users will immediately be redirected to their age-appropriate dashboard. Note for BFP Personnel: Professional accounts cannot be created via public registration and must be provisioned directly by a System Administrator to ensure secure portal access.

II. THE KIDS DASHBOARD (Ages 17 and below)

Designed to make fire safety fun and engaging through gamification and interactive challenges. By transforming essential life-saving skills into an immersive adventure, this dashboard ensures that younger users remain captivated while mastering critical emergency protocols.

Key Features & Navigation:

SafeScape Fire Safety Course: Click the Start button on the SafeScape card to begin the 5-module learning journey.

- Note on Adaptive Learning: The module quizzes automatically adjust in difficulty based on the child's pre-assessment score, age, and grade level.
- Final Exam: After completing all 5 modules, the system generates a custom final exam based on the pre-assessment score, age, grade level, and user's previous quiz scores.

Badge Hall & Achievements: Track your learning journey through the Achieved Badges panel, which displays your Global Ranking Progress bar.

- Click the yellow Open Hall > button to view your full collection.
- Click on locked badges to see the specific tasks required to earn them (e.g.,
- completing Module 1 to earn the Fire Triangle badge, or scoring 100% on a
- quiz).
- Collect all 13 badges to achieve the ultimate rank of Legendary Fire Hero!

EDITH Simulation: Click Play on the EDITH card to launch the 3D Interactive Home Fire Escape Plan. Use the on-screen controls to navigate the house and practice safe evacuation routes.

Task Master & Mini Games: Click on these sections to play memory games and quizzes to earn extra points and unlock achievements.

The Right Call (Dispatch Simulator): This interactive tool allows kids to simulate answering emergency calls, assessing situations, and dispatching the correct response teams under time pressure.

Fire Safety Videos: Access a library of exciting, kid-friendly educational videos.

III. THE ADULT'S DASHBOARD (Household & Community)

Focused on practical household safety, hazard prevention, and emergency preparedness. This section equips users with the knowledge and tools necessary to protect their families, offering real-world guides and interactive simulations tailored for community resilience.

Key Features & Navigation:

Exit Drill (EDITH) Simulator: Click on the Exit Drill (EDITH) button to access the interactive home fire spread simulator. This tool allows users to map out and visualize safe evacuation routes for their own households.

Fire Safety Articles: Scroll down to browse the library of real-world safety guides. Click on any card (e.g., "Prepare an Emergency Kit" or "Typhoon Preparedness") to read the full, detailed article.

Safety Videos: Click on the Safety Videos section to watch instructional content on fire prevention and response.

IV. THE BFP PROFESSIONAL PORTAL (Active Officers)

A secure workspace dedicated to the continuous training and operational readiness of BFP personnel. It serves as a centralized hub for active officers to access standard operating procedures, advanced training modules, and dispatch simulation tools to keep their skills sharp.

Key Features & Navigation:

The Right Call (Dispatch Simulator): Click the arrow on The Right Call card. This interactive tool allows officers to practice answering emergency calls, assessing situations, and dispatching the correct response teams under time pressure.

Training Manuals & Fire Codes: Click on the Training Manuals card to open the digital library.

- Use the category tabs (All, Fire Code, Policy, Education, etc.) to filter documents.
- Click View to read manuals directly in the browser or click Save to download the PDF for offline access.

• **Training Videos:** Scroll down to access professional-grade modules. Click on any video to track your training progress.

V. SYSTEM SUPPORT & AI ASSISTANCE

If users need help navigating the platform or have specific fire safety questions, the system provides built-in support tools. This ensures that every user has immediate 24/7 access to technical assistance and reliable, intelligent guidance exactly when they need it.

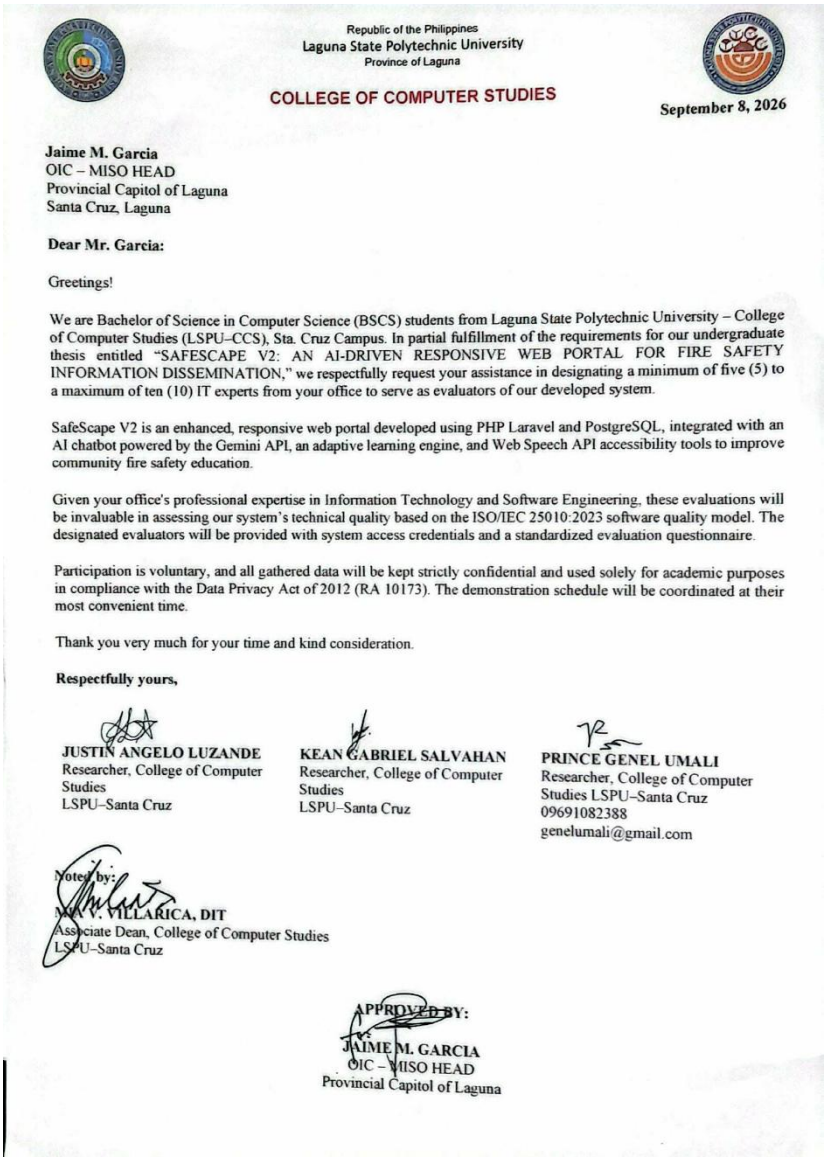
Berong AI Chatbot: Click the floating Chatbot icon on the bottom right. This intelligent virtual assistant uses Retrieval-Augmented Generation (RAG) to provide context-grounded responses to fire-safety inquiries and explain relevant emergency protocols.

System Feedback: We want to hear from you! If you have suggestions for improvement, want to rate your experience, or need to report an issue, click the "Send Feedback" button located at the bottom left of the screen. Your response, along with your 5-star rating, is sent directly to the BFP Administrator Portal, to monitor community satisfaction and track technical reports in real time.

APPENDIX B

(COMMUNICATION LETTERS AND FORMS)

Request Letter



Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

COLLEGE OF COMPUTER STUDIES

September 8, 2026

Jaime M. Garcia
OIC – MISO HEAD
Provincial Capitol of Laguna
Santa Cruz, Laguna

Dear Mr. Garcia:

Greetings!

We are Bachelor of Science in Computer Science (BSCS) students from Laguna State Polytechnic University – College of Computer Studies (LSPU–CCS), Sta. Cruz Campus. In partial fulfillment of the requirements for our undergraduate thesis entitled "SAFESCAPE V2: AN AI-DRIVEN RESPONSIVE WEB PORTAL FOR FIRE SAFETY INFORMATION DISSEMINATION," we respectfully request your assistance in designating a minimum of five (5) to a maximum of ten (10) IT experts from your office to serve as evaluators of our developed system.

SafeScape V2 is an enhanced, responsive web portal developed using PHP Laravel and PostgreSQL, integrated with an AI chatbot powered by the Gemini API, an adaptive learning engine, and Web Speech API accessibility tools to improve community fire safety education.

Given your office's professional expertise in Information Technology and Software Engineering, these evaluations will be invaluable in assessing our system's technical quality based on the ISO/IEC 25010:2023 software quality model. The designated evaluators will be provided with system access credentials and a standardized evaluation questionnaire.

Participation is voluntary, and all gathered data will be kept strictly confidential and used solely for academic purposes in compliance with the Data Privacy Act of 2012 (RA 10173). The demonstration schedule will be coordinated at their most convenient time.

Thank you very much for your time and kind consideration.

Respectfully yours,

JUSTIN ANGELO LUZANDE
Researcher, College of Computer Studies
LSPU–Santa Cruz

KEAN GABRIEL SALVAHAN
Researcher, College of Computer Studies
LSPU–Santa Cruz

PRINCE GENEL UMALI
Researcher, College of Computer Studies LSPU–Santa Cruz
09691082388
genelumali@gmail.com

Noted by:

MARY V. VILLARICA, DIT
Associate Dean, College of Computer Studies
LSPU–Santa Cruz

APPROVED BY:

JAIME M. GARCIA
OIC – MISO HEAD
Provincial Capitol of Laguna

ISO Forms

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

THESIS ADVISER NOMINATION

Student/s Name and Signature : Justin Angelo A. Luzande
: Kean Gabriel E. Salvahan
: Prince Genel R. Umali
:

Degree Sought : Bachelor of Science in Computer Science
Major Field (if applicable) : Intelligent System

I/we, hereby nominate as Thesis Adviser

Name of Faculty : Mia V. Villarica, DIT
College : College of Computer Studies
Field of Specialization :

I, hereby indicate my willingness to serve as the Thesis Adviser.

Signature of Faculty :

Date Signed

Endorsed By:

Mark P. Bernardino, MSCS
Research Coordinator

Recommending Approval:

Mia V. Villarica, DIT
Dean/Associate Dean

APPROVED/DISAPPROVED:

Rosario G. Catapang, PhD
R&D Director/Chairperson

LSPU-RDO-SF-011 Rev. 0 8 August 2018

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

THESIS PANEL MEMBER NOMINATION

Student/s Name and Signature : Justin Angelo A. Luzande
: Kean Gabriel E. Salvahan
: Prince Gend R. Umali
:

Degree Sought : Bachelor of Science in Computer Science
Major Field (if applicable) : Intelligent System

I/we, hereby nominate as Panel Member (System Expert)

Name of Faculty : Mark P. Bernardino, MSCS
College : College of Computer Studies
Field of Specialization : computer science

I, hereby indicate my willingness to serve as a Panel Member.

Signature of Faculty :
Date Signed

Endorsed By:
Mark P. Bernardino, MSCS
Research Coordinator

Recommending Approval:
Mia V. Villarica, DIT
Dean/Associate Dean

APPROVED/DISAPPROVED:
Rosario G. Catipon, PhD
R&D Director/Chairperson

LSPU-RDO-SF-012 Rev. 0 8 August 2018

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

THESIS PANEL MEMBER NOMINATION

Student/s Name and Signature : Justin Angelo A. Luzande
: Kean Gabriel E. Salvahan
: Prince Genel R. Umali
:

Degree Sought : Bachelor of Science in Computer Science
Major Field (if applicable) : Intelligent System

I/we, hereby nominate as Panel Member (Statistician)

Name of Faculty : Edward S. Flores, MSIT
College : College of Computer Studies
Field of Specialization :

I, hereby indicate my willingness to serve as a Panel Member.

Signature of Faculty : _____ Date Signed

Endorsed By:
Mark P. Bernardino, MEGE
Research Coordinator

Recommending Approval:
Ma. V. Villarica, DIT
Dean/Associate Dean

APPROVED/DISAPPROVED:
Rosario G. Catapang, PhD
R&D Director/Chairperson

LSPU-RDO-SF-012 Rev. 0 8 August 2018

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

THESIS PANEL MEMBER NOMINATION

Student/s Name and Signature : Justin Angelo A. Luzande
: Kean Gabriel E. Salvahan
: Prince Genel R. Umali
:

Degree Sought : Bachelor of Science in Computer Science
Major Field (if applicable) : Intelligent System

I/we, hereby nominate as Panel Member (Language Critic)

Name of Faculty : Ms. Cezanne D. Dimaculangan
College : College of Arts and Sciences
Field of Specialization :

I, hereby indicate my willingness to serve as a Panel Member.

Signature of Faculty : _____ Date Signed

Endorsed By:

Mark P. Bernardino, MSCS
Research Coordinator

Recommending Approval:

Ma V. Villarica, DIT
Dean/Associate Dean

APPROVED/DISAPPROVED:

Rosario G. Catapang, PhD
R&D Director/Chairperson

APPENDIX C

(CURRICULUM VITAE OF SYSTEM

TESTERS)

JOHN KERVIN D. EVANGELISTA

AI Enthusiast | Data Scientist | Machine Learning

Nagcarlan, Laguna | 09999631798 | johnkervinevangelista@gmail.com | toneejake.tech

PROFILE

Intelligent systems specialist with a background in building machine learning, computer vision applications, and data management. Experienced in developing full-stack AI solutions and leveraging cloud platforms to deploy and scale data-driven projects that solve real-world problems.

EDUCATION

Bachelor of Science in Computer Science - Major in Intelligent Systems **August 2022 - June 2026**

- Laguna State Polytechnic University Sta. Cruz, Laguna

SKILLS

- **Programming Languages:** Python, JavaScript, SQL, PySpark
- **AI/ML:** PyTorch, Scikit-learn, NLP, RAG, Gen AI, AI Agents, RL, Computer Vision
- **Data Analysis & Visualization:** Pandas, Matplotlib, Seaborn, Tableau, Knime, Power BI, Data profiling
- **Web & Database:** Django, React, Node.js, Next.js, Tailwind CSS, Prisma, PostgreSQL, MongoDB, DBML
- **Cloud & MLops Tools:** Google Cloud, Docker, Hugging Face, Git, GitHub

EXPERIENCE

Solvento | Data Management Intern **February 2026 - April 2026**

Engineered end-to-end ELT pipelines in Databricks using PySpark across a Medallion Architecture, complemented by data modeling, database migration, and SQL-driven data quality validation. Delivered final analytical outputs through interactive Power BI dashboards for business decision-making.

PROJECTS

SafeScape: AI-Driven Fire Safety & Evacuation Simulation

- An application made using PyTorch and reinforcement learning to automate floor plan "digital twin" generation and simulate optimal evacuation policies within a Next.js/Prisma full-stack environment.

Telecommunications Data Pipeline & BI Dashboard

- Architected a full-stack data solution from raw data ingestion and PySpark transformations in Databricks to data quality validation and final Power BI dashboard delivery across a 9-week enterprise simulation.

CERTIFICATIONS AND TRAINING

- **Asia Open Ran Academy:** Artificial Intelligence/Machine Learning **December 2024**
- **Databricks:** Generative AI Fundamentals | AI Agents Fundamentals **October 2025**
- **DataCamp:** Associate Data Scientist in Python | Data Literacy **November 2025**
- **Nvidia:** Getting Started with Deep Learning **November 2025**
- **Udemy:** Professional Certificate of IBM SPSS Statistics Expert **November 2025**
- **Google:** Gemini for Data Scientists **November 2025**
- **Cisco:** Introduction to Data Science and Data Analytics **November 2025**
- **IBM:** AI & Data | IBM Z Skills | Modernization | Security **November 2025**
- **ITTP (TOPCIT):** TOPCIT Level 3: Competent **December 2025**
- **Hack2Skill:** Gen AI Academy 2.0 **December 2025**
- **Certiport:** Cloud Computing **April 2026**

ACHIEVEMENTS

- **Iskolar ng Laguna** **2023**
- **Robotics - Battle Bots Champion** **2025**
- **Data Engineering Pilipinas x Datacamp Scholar** **2026**



ROBIN PAOLO BONA

Bachelor of Science in Information Technology

To utilize experience and qualification in different and challenging position that professionally growth and enhance my future opportunities.

EDUCATION

2018

UNION COLLEGE OF LAGUNA

- Bachelor of Science in Information Technology

CERTIFICATE

2022

NATIONAL CERTIFICATION II

- Computer System Servicing NC II

SKILL

- Troubleshoot 100%
- Technical Support 90%
- Microsoft Office 100%
- Creative 90%
- Team Work 95%

EXPERIENCE

2019 - PRESENT

Municipal Government Santa Cruz, Laguna

Administrative Aide

- Information Technology

2018 - 2019

DEP-ED LAGUNA

Administrative Aide

- Information Technology

2018

SM Calamba

Stock Clerk

- Warehouse Stock Clerk

CONTACT

09650499411

Hushbona@gmail.com

229 Jamaica Subdivision, Santo Angel Central, Santa Cruz, Laguna

REFERENCE

BRYAN BOUVIER BONZA
Municipal Government Santa Cruz, Laguna
HRMO V
09177059405

MELVIN O. BONZA
Municipal Government Santa Cruz, Laguna
ADMINISTRATOR
09967079115

John Marvin A. Tabo

3192 Biya Street
Santo Angel Norte,
09754068759
itsotavs0099@gmail.com

Birthdate

May 29, 2001

EDUCATION

Santo Angel Norte Elementary School

June 2006 - April 2013

Academic Achiever (Grade 6)

Pedro Guevarra Memorial National High School (P.Gueverra Street, Sta Cruz Laguna. Junior High School)

June 2013- April 2017

Academic Achiever

Saint Therese Martin of Lisieux School and Business High School. - Sto Angel Sur, Santa Cruz, Laguna.(Information and Communication Technology)

June 2013- April 2017

Supreme Student Government President

With Honor

Laguna State Polytechnic University - Santa Cruz Campus (Bachelor of Science in Information Technology)

June 2019- July 2023

Dean`s Lister Second Semester AY 2022-2023

CSE PROFESSIONAL PASSER

SKILLS

Computer Literate

Knowledge in Microsoft Office

Hardware Configuration

Installation and Configuration

Computer Troubleshooting

AWARDS

Academic Achiever - Pedro Guevara Memorial National High School

With Honor - Saint Therese Martin Of Lisieux School and Business High School (2017)

With Honor - Saint Therese Martin Of Lisieux School and Business High School (2018)

Dean`s Lister - Laguna State Polytechnic University (2022)



Allen Don Aldrich M. Tobias

Brgy. Sto. Angel Sur Sta. Cruz, Laguna
+63 951 523 7381
allendontobias@gmail.com

Personal Information:

Age: 22
Birthday: Feb 24, 2003
Gender: Male
Citizenship: Filipino
Civil Status: Single
Religion: Catholic
Height: 5'7
Weight: 81 kg
Father: Aldrin M. Tobias
Mother: Genielyn M. Tobias

My Goal:

To gain additional knowledge and challenging experience that could aid in my future development. My main goals are to improve my skills and foster good camaraderie. We all needed one another in order to grow and succeed in life.

Skills and Expertise:

- Proficient in Mircrosoft Office (Word, Excel, and PowerPoint)
- Knowledgeable in using digital camera
- Proficient in using Canva and Capcut
- Programming Languages (HTML, CSS, JS, PHP)

Achievements:

- Student Organization of College of Computer Studies: Auditor (2021-2022)
- Student Organization of College of Computer Studies: Vice President (2022-2023)
- Information Tech Unity Hub: 4th Year Representative (2024-2025)

Educational Attainment:

Primary: Sta. Cruz Central Elementary School (2008-2015)
Secondary:
- Junior High School: Pedro Guevara Memorial National High School (2015-2019)
- Senior High School: Accountancy, Business and Management in Laguna State Polytechnic University (2019-2021)

Tertiary: Bachelor of Science in Information Technology, Major in Web and Mobile Development in Laguna State Polytechnic University (2021-2025)

Experiences:

- Intern at Laguna Provincial Information Office as Creative Team Leader (2/25/25 - 6/21/25)
- Special Program for Employment of Students in Laguna Tourism Culture Arts and Trade Office (7/4/22 - 7/29/22)

References:

- Nelson S. Bautista
  ngcp.nsbautista@gmail.com
- Reymart Joseph P. Pielago, MIT
  rjpielago@lspu.edu.ph

I do hereby certify that the information above I made is true and correct to the best of my knowledge and ability.

Allen Don Aldrich M. Tobias

Pleto, John Ray M.
384 Brgy Calumpang
Liliw Laguna
09498129611
j.pleto321@gmail.com



OBJECTIVE

To acquire valuable knowledge and experience in working in a company and to learn how to be a System developer to learned how to act along with other co-workers. And to acquire some knowledge too in my higher officers and how to develop the confidence from my co-workers

PROJECTS IN ACADEMICS

- 2019 - 2022 "THE DESIGN AND DEVELOPMENT OF AX FITNESS GYM APPOINTMENT AND SCHEDULING SYSTEM" THESIS WE CREATED IN LAGUNA STATE POLYTECHNIC UNIVERSITY AND WE ARE 3 MEMBERS AND WE USING PHP AS OUR MAIN PROGRAMMING LANGUAGE AND MY ROLE IS A PROGRAMMER/DOCUMENT ON MY THESIS

- 2017 – "PAYROLL SYSTEM OF Y'ISABEL GARMENTS AND TRADING" WE ARE 2 TEAMS AND BABY THESIS CREATED USING C# AS OUR MAIN LANGUAGE AND MY ROLE IS TO DEVELOP SYSTEM AND CREATE A DOCUMENT FOR OUR THESIS

- 2018 – "Y'ISABEL GARMENTS AND TRADING SYSTEM" WE ARE 3 TEAMS AND MY ROLE IS TO PROGRAM AND CREATE A DOCUMENT AND WE USING JAVA AS A MAIN PROGRAMMING LANGUAGE AND WE USING MYSQL FOR THE DATABASE

EDUCATION

2019-2022-Laguna State Polytechnic University Sta Cruz Campus
Sta Cruz Laguna

2017-2019-STI College Sta Cruz
Pedro Guevarra St. Sta Cruz Laguna

2013-2017-Liceo De Pila
J.P.Rizal St. Pila Laguna

SKILLS:

-Basic Web Development like PHP, HTML, CSS and JS
-I can do make a system application in .Net Framework
-Background for Android Development
-Basic Background for Microsoft Word and Microsoft Excel

EXTRA CURRICULAR ACTIVITIES

- I join Code I.T Programming Competition on August 31 2017
- I join Tagisan ng Talino on August 31 2017
- I participated the Web development 101 seminar on July 8 2020
- I join the Introduction of 3d seminar on September 2019

SEMINARS ATTENDED

- I join and completed the Webinar called "Web Development 101" A Webinar hosted by Department of Information and Communications Technology held on July 8, 2020

- I join and completed the Course "Intro to CSS" on TESDA Course online on June 1, 2020

PERSONAL BACKGROUND

I can speak Filipino Language and I can do make an application using C# and I can do back-end web development like PHP – beginner and I am willing to learn about the web designing and web development

REFERENCE:

Dorothy Escueta
Manager
Rural Bank of Magdalena
09153350615

Julie Castilio De Leon
Owner
Y'ISABEL GARMENTS AND TRADING
09153225999

Hanna Kris Fallaria
Teacher
ELIS Foreign Languages Center
09086059548

I hereby that this resume will contains information about me is true

_____
Signature over printed name

Contact

rhon2692@gmail.com

www.linkedin.com/in/anonuevo-rhoniel-llanes-35189b2a5 (LinkedIn)

Top Skills

Laravel
React.js
Next.js

Anonuevo Rhoniel Llanes

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Lumban, Calabarzon, Philippines

Experience

None
None

Management Information System Office Laguna
1 year 8 months

Computer Programmer I
November 2025 - Present (2 months)
Santa Cruz, Laguna

Laguna Farmers and Fisherfolks Registry Management System

Vet Shipping Permit System

Disaster Response Operations Monitoring and Information Communication System Laguna

Barangay Nutrition Information System

Job Order
May 2024 - November 2025 (1 year 7 months)

Education

Laguna State Polytechnic University (LSPU), Santa Cruz
Bachelor's degree, Information Technology · (March 2019 - July 2023)



EDUCATION

2008 - 2013
BS INFORMATION TECHNOLOGY
Laguna State Polytechnic University

2004 - 2008
SECONDARY EDUCATION
Pedro Guevara Memorial National High School

SKILLS

- Networking Fundamentals
- Hardware Diagnostics
- Graphic Arts
- Microsoft Office Savvy
- GIS Mapping
- Technical Support

CONTACT

09171302395
tobiasmharjun666@gmail.com
Patimbao, Santa Cruz, Laguna
www.reallygreatsite.com

MHARJUN V. TOBIAS

Graphic Artist / Technical Support

PROFILE INFO

Highly skilled professional with extensive experience in graphic design, technical support, and administrative operations. Proven ability to handle complex networking tasks and digital asset creation with precision and efficiency. Dedicated to delivering high-quality visual solutions and reliable technical troubleshooting.

EXPERIENCE

2025 - Present
CASUAL CLERK
MIS Office - MGO Santa Cruz, Laguna
Handled management information systems and administrative support for the local government unit.

2023 - 2025
CASUAL CLERK
Assessor's Office - MGO Santa Cruz, Laguna
Assisted in data management and office operations for tax assessment and land records.

2020 - 2022
GRAPHIC ARTIST / TECH SUPPORT
Mayor's Office - MGO Santa Cruz, Laguna
Provided digital creative solutions and technical maintenance for municipal operations.

2019 - 2019
ADMINISTRATIVE AIDE
PIO - MGO Santa Cruz, Laguna
Supported public information dissemination and media relations.

ACHIEVEMENTS

2022
CIVIL SERVICE PROFESSIONAL
Passed with a rating of 80.56.

2019
CIVIL SERVICE SUB-PROFESSIONAL
Passed with a rating of 82.26.

JANSSEN REYES ESPINO



Committed graduate of Bachelor of Science in Information Technology in Laguna State Polytechnic University who has experience in the field of telecommunication activation and data encoding skills that I have learned in the program and internship to elevate self-growth and to contribute to future organizations. Passionate about precision and quality of work.

ACHIEVEMENTS AND PROFESSIONAL AFFILIATIONS

- Civil Service Exam Professional Passer, March 2025
- Cisco Networking Academy Operating Systems Basics - Completer, December 2024
- eLGU Admin Training and System Setup
- System Administrator in phpMyAdmin completer
- Graduated First Honorable Mention, 2015

EXPERIENCES

- **MUNICIPALITY OF SANTA CRUZ IT STAFF**
  JULY 2025-PRESENT
- **NESTLE BRAND AMBASSADOR**
  JULY 2022 - OCTOBER 2022
- **NATIONAL ELECTION SUPPORT STAFF**
  MAY 2022
- **BARANGAY ELECTION THIRD MEMBER**
  OCTOBER 2023
- **PLDT DATA ENCODER AND SERVICE ACTIVATION**
  JANUARY 2025-JUNE 2025
- **SENATORIAL ELECTION SUPPORT STAFF**
  MAY 2025

INTERNSHIP

CENTEGRA TECHNOLOGY INC. PLDT HOME DISTRIBUTOR
Banca-Banca, Victoria, Laguna

- Took internship in the company for 600 hours as data encoder and service activation.
- Validated customer's application before passing their service order to main office.
- Calling out customers to confirm the status of their internet and voice data.
- Data activation using third party apps.
- Manual service activation.

SKILLS

- **FOUNDATIONAL KNOWLEDGE IN MICROSOFT EXCEL**
- **DATA ENTRY AND ENCODING**
- **PROFICIENT IN COMPUTER APPLICATIONS**
- **ADOPTABLE TO NEW TECHNOLOGIES**
- **BASIC UNDERSTANDING OF COMPUTER MAINTENANCE AND REPAIR**
- **DETAIL ORIENTED**
- **COMMUNICATION SKILLS**
- **TIME MANAGEMENT**
- **ENTRY-LEVEL EXPERIENCE IN COMPUTER HARDWARE SERVICES**
- **COMPUTER TROUBLESHOOTING**

EDUCATION

Laguna State Polytechnic University, 2021-2025
Bachelor of Science in Information Technology

Laguna Senior High School, 2019-2021
Senior High School
Science, Technology, Engineering, and Mathematics

Pedro Guevara Memorial National High School 2015-2019
Junior High School

Bagumbayan Elementary School, 2009-2015
Grade 1-6

CONTACT INFORMATION

0975-353-7625
ssen.jre@gmail. com
Sitio 1 Bagumbayan Santa Cruz, Laguna

CHARACTER REFERENCE

Asher H. Pasco
0925-301-6826
Education Program Supervisor
San Pedro City Laguna



Justine Jhay Lopez

ASPIRING QA ENGINEER |
WEB DEVELOPER |
IT SUPPORT TECHNICIAN

Contact

+63 976 238 5693

jhayjustinlopez@gmail.com

Santa Cruz, Laguna, 4009, Phillipines

Skills

- Leadership & Team Collaboration
- Computer Literacy (MS Office, Email, Digital Filing)
- Visual Design & Editing
- Photo & Video Editing
- Basic UI/UX & Web Content Management

Language

- English
- Filipino

Education

(2022 – 2026)
LAGUNA STATE POLYTECHNIC UNIVERSITY
Bachelor of Science in Information Technology
1st - 4th Year College

(2020 – 2022)
LAGUNA SENIOR HIGH SCHOOL
Science, Technology, Engineering, and Mathematics strand
Grade 11 - Grade 12

(2016 – 2019)
PEDRO GUEVARA MEMORIAL NATIONAL HIGH SCHOOL
Grade 7 - Grade 10

Experience

(2025 – Present)
MANAGEMENT INFORMATION SYSTEMS OFFICE
- MIS Programmer

(2024 – 2026)
MUNICIPAL SOCIAL WORKER AND DEVELOPMENT OFFICE
- Data Entry Clerk

Projects

- **IKNOWLGBC - KNOWLEDGE MANAGEMENT SYSTEM**
  -2026 (Full Stack Developer)
- **EXELENCIA ACADEMY PORTAL**
  -2024 (Full Stack Developer)
- **SHE WEAR (E-COMMERCE WEB)**
  -2024 (Programmer)
- **BARANAGAY RECORDING SYSTEM**
  -2024 (Full Stack Developer)
- **JAMAICA RECORDING SYSTEM**
  -2023 (Full Stack Developer)



JONAS T. SANTOS

Barangay. Jasaan, Santa Cruz, Laguna, Philippines, 4009
Jonassantos122113@gmail.com | 09675977572

- IT professional with three years of experience in technical support, system administration, and IT operations in the Municipal Government of Santa Cruz, Laguna. Skilled in troubleshooting hardware, software, printers, and network issues, providing IT support across multiple municipal offices, and maintaining systems to ensure efficient operations. Known for strong problem-solving abilities, adaptability, and delivering reliable technical solutions that improve productivity and service delivery.

Work Experience

- **MUNICIPAL GOVERNMENT OF SANTA CRUZ**

Management Information Systems office | MIS IT Staff **2025 – Present**

- Provide technical support for hardware, software, printers, and network issues across municipal offices.
- Install, configure, and maintain computer systems, peripherals, and user accounts.
- Perform system administration, preventive maintenance, and software updates.
- Troubleshoot network connectivity issues and support IT projects and system implementations.
- Maintain IT asset inventory, service records, and technical documentation.

Municipal Administrator's Office / Municipal ICT Office | Administrative Aide **2024 – 2025**

- Performed administrative and clerical functions to support daily office operations.
- Prepared and maintained records, reports, and official correspondence.
- Assisted with document processing, office coordination, and administrative support.

Municipal Government of Santa Cruz | DOLE - Government Intern **2023 – 2024**

- Municipal Administrator's Office / Municipal ICT Office

- **PROVINCIAL GOVERNMENT OF LAGUNA | ON-THE-JOB TRAINING** **2022**
  - Laguna Parole and Probation Office - R4A

Key responsibilities

- Provide technical and systems support for multiple LGU information systems, including the ***General Services Office Management System, Motorized Tricycle Operators Permit System***, and ***EBOSS Queueing System.***
- Troubleshoot and resolve system issues, perform database record updates and corrections, manage user accounts, and generate operational reports.
- Perform source code modifications, including layout adjustments (margins and page breaks), to improve system-generated reports and document formatting.
- Administer and maintain EBOSS Queueing System kiosks, ensuring reliable operation and resolving system-related concerns.
- Perform daily backups of LGU system databases to ensure data integrity, security, and disaster recovery readiness.
- Provide technical support for municipal offices by installing, maintaining, and troubleshooting computers, printers, network connectivity, and other IT equipment.
- Prepare and process Purchase Requests (PRs) for MIS office supplies, IT equipment, maintenance requirements, and other operational needs.
- Manage inventory and monitoring of office supplies, computer equipment, and IT tools to ensure availability and accountability.

Technical Skills & Abilities

- IT Technical Support
- Hardware Installation, Configuration & Troubleshooting
- Backup and recovery deleted files
- Software Installation & Maintenance
- Printer Installation & Troubleshooting
- Network Troubleshooting
- System Administration
- System User Account Creation and Management
- Operating System Installation & Configuration (Windows)
- Technical Documentation
- Computer System Maintenance
- Problem Solving & Troubleshooting
- Analytical Thinking
- Customer Service & End-User Support
- Communication Skills & Team Collaboration
- Time Management
- Attention to Detail
- Adaptability & Continuous Learning
- Organizational Skills
- Ability to Work Under Pressure

Education

- **TERTIARY LEVEL:** **2018 – 2022**
  Bachelor of Science in Information Technology
  Major in Service Management Program
  Laguna State Polytechnic University – Santa Cruz Campus
- **SENIOR HIGH SCHOOL:** **2016 – 2018**
  Science, Technology, Engineering and Mathematics – STEM
  AMA Computer Collage – Santa Cruz Campus

Civil Service Eligibility

- **Career Service Subprofessional Eligibility**
  Passed
  March 2026

Personal Information

- Age: 26 years old
- Birthdate: May 11, 2000
- Birthplace: Santa Cruz, Laguna
- Sex: Male
- Height: 4'11"
- Weight: 40kg
- Citizenship: Filipino
- Civil status: Single
- Religion: Roman Catholic
- Languages: English/Tagalog

Seminars & Training

• **Introduction to Networking**
Provincial Government of Laguna - Miso Training Room
April 30, 2026

• **E-LGU User's Training**
DICT- CALABARZON
April 23, 2026

• **Web Development (Laravel & MySQL) & Server Administration**
Provincial Government of Laguna - Miso Training Room
January 5 – 16, 2026

• **Data Analytics Essentials**
December 5 – 11, 2025
DICT-ITU DTC (Net. Acad. Program)

• **Gender Sensitivity and Safe Space Act of the Philippines**
LGU Santa Cruz
December 2, 2025

• **E-LGU Admin Training and System Set-Up**
DICT- CALABARZON
November 25 – 28, 2025

• **Google Workspace for Educators/LGU Employees**
DICT – CAR
October 27, 2025

• **Cybersecurity and PNPKI Awareness at LGU - Santa Cruz**
DICT - Calabarzon
August 11, 2025

• **Basic & Intermediate Geographic Information System (GIS) Training**
Uplink Integrated Solution Inc.
January 30 – February 1, 2025

• **City/Municipal-Level Training Coordination, and**
Field Visit of LGU on the Adoption of the Philippines
Standard Industrial Classification
PHILLIPPINES STATISTICS AUTHORITY
July 6, 2024

• **Computer System Servicing Training**
Provincial Government of Laguna - Miso Training Room
October 16 – 20, 2023

• **Installing and Configuring Computer System & Introduction to Computer System Servicing**
TESDA Online Program
July 2022

Character Reference

Reference available upon request.

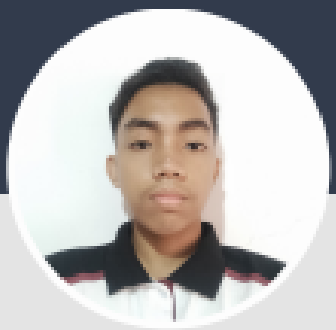
APPENDIX D

(CURRICULUM VITAE OF THE

RESEARCHERS)

JUSTIN ANGELO LUZANDE

SOFTWARE DEVELOPER



CONTACT

+63 956 640 4838

justinluzande117@gmail.com

Pila, Laguna, Philippines

SKILLS

- Software Development
- Data Analytics
- Computer Hardware
- Networking
- Leadership
- Effective Communication
- Critical Thinking
- Digital Marketing
- 3D Rendering

LANGUAGES

- English (Fluent)
- Tagalog (Fluent)

PROFILE

Driven 4th-year Bachelor of Science in Computer Science student with a strong foundation in full-stack web/mobile development, UI/UX design, applied machine learning, and robotics/IoT hardware integration. Proven project manager and technical lead across 9+ collaborative initiatives, adept at bridging the gap between hardware, system logic, and intuitive user interfaces. Award-winning robotics competitor and creator of pioneering educational platforms.

WORK EXPERIENCE

Creotech 2022 - 2023

IT INTERN

- Standardized and restructured messy cross-functional datasets in Excel and Power BI enabling the analytics team to build reliable, error-free operational dashboards.
- Collaborated with senior analysts to audit and harmonize discrepancies across multiple data sources, ensuring 100% compliance with internal data governance standards.
- Cleaned, validated, and preprocessed records, resolving missing values and inconsistencies to improve internal reporting accuracy.

Rockies Fitness 2025 - 2029

Facility Manager

- Directed day-to-day front desk operations, facilitating smooth check-ins and delivering high-touch customer support.
- Oversee market research to identify emerging trends, customer needs, and competitor strategies.

UandWorld 2025 - 2026

Marketing Intern

- Produced and published high-engagement digital content across TikTok, Instagram, and Facebook, driving brand visibility and growing the audience.
- Monitor and maintain brand consistency across all marketing channels and materials.
- Developed creative assets and promotional copy for e-commerce campaigns, directly contributing to an increase in sales conversions.

EDUCATION

Bachelor of Computer Science 2023 - 2027

Laguna State Polytechnic University

Reserved Officers Training Corps 2023 - 2024

Laguna State Polytechnic University

Humanities and Social Sciences 2021 - 2023

San Antonio de Padua College

KEAN GABRIEL E. SALVAHAN

AI Enthusiast | Machine Learning | Data Scientist

Nagcarlan, Laguna | +63 920 581 5366 | keangabriel101@gmail.com | https://kenji-dev.vercel.app

PROFILE

Intelligent systems specialist with a background in developing machine learning models, computer vision systems, and data management pipelines. Experienced in building full-stack AI solutions and using cloud platforms to deploy and scale data-driven projects that solve real-world problems.

EDUCATION

Bachelor of Science in Computer Science - Major in Intelligent Systems — **August 2023 – June 2027**

- Laguna State Polytechnic University - Santa Cruz, Laguna

SKILLS

- **Programming & Backend:** PHP, Python, JavaScript, SQL, HTML5, CSS3
- **AI/ML:** PyTorch, Scikit-learn, NLP, RAG, Gen.AI, AI Agents, Computer Vision
- **Web & Databases:** Laravel, Tailwind CSS, Inertia.js, PostgreSQL, MySQL
- **Data Analysis & Visualization:** Pandas, Matplotlib, Power BI

PROJECTS

SafeScape V2: AI-Driven Fire Safety and Information Dissemination

- Developed and deployed a live, responsive web application for fire prevention education built using PHP Laravel, MySQL, and Tailwind CSS.

Rockies Fitness Tracker App

- Designed and developed a cross-platform mobile fitness tracking application focused on personal workout logging and goal progress monitoring.

CERTIFICATIONS

- **EPCPS:** HANDS ON FIGMA UI/UX DESIGN WORKSHOP — **November 2025**
- **Google:** Agent Fundamentals — **March 2026**
- **HP:** Agile Project Management — **March 2026**
- **HP:** Data Science & Analytics — **March 2026**
- **Google:** Introduction to Generative AI — **March 2026**
- **ITTP (TOPCIT):** TOPCIT Level 2: Competent — **August 2026**
- **TESDA:** Microsoft AI Course: Azure AI Fundamentals — **August 2026**

SEMINARS & TRAININGS

- AI-Driven Software Development: From Wireframe to App - with a Focus on Security and Assurance organized by Computer Science Society Organization.

ACHIEVEMENTS

- **Robotics - Battle Bots Champion** — **2026**



PRINCE GENEL R. UMALI

Data Scientist | Software Developer

09691082388 genelumali@gmail.com https://github.com/UmaliPrinceGenel

ABOUT ME

Solutions-driven Computer Science student specializing in Intelligent Systems. Passionate about bridging full-stack web development with artificial intelligence to build modern, accessible, and scalable applications.

EDUCATION

2023 - 2027 **Laguna State Polytechnic University (LSPU), Santa Cruz, Laguna**
Bachelor of Science in Computer Science - Major In Intelligent Systems

PROJECTS

SafeScape V2: An-Driven Fire Safety and Information Dissemination

Developed an AI-driven fire safety education portal using Laravel 11, React, and PostgreSQL, featuring a Gemini API RAG pipeline and Web Speech API for interactive, accessible learning.

Fitness Tracker App

Built a mobile fitness tracker backed by Supabase Edge Functions, integrating the Gemini API to deliver dynamic, personalized workout plans and progress tracking.

LSPUMART - E-commerce App

Architected a secure, highly responsive e-commerce marketplace for the LSPU community using Dart, Flutter framework, Firebase Database and Supabase, featuring comprehensive product management and cart systems.

SKILLS

Programming Languages: Python, PHP, SQL, Java, C#, Dart

Frameworks & Libraries: Laravel, React, Inertia.js, Flutter, Scikit-learn, Pandas, Flask

AI & Machine Learning: Gemini API, NLP

Tools & Databases: PostgreSQL, Version Control (Git), Power BI, MS Excel, Supabase, Firebase

CERTIFICATIONS & SEMINARS

TOPCIT (Test of Practical Competency in IT) – Level 2

Institute for Information & communications Technology Planning & Evaluation (IITP) |July, 2026

The Role of Cloud in Artificial Intelligence and Big Data

Ethel Programming Computer Programming Services | October 2025

CompTIA Network Security for Beginners & Professionals

Ethel Programming Computer Programming Services |
September 2025